

الكترونيات 1

ملخص مادة الميد + الفاينل !!

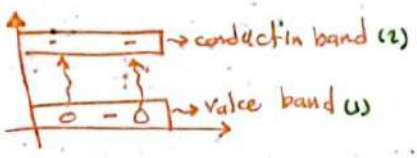
MECHATRONICS
SPECTRUM
TEAM

ch 1 :-

* Materials (المواد) consists (تتكون) form atoms..
* valance electrons : Number of $\bar{e}$ in valanc orbit
* → conductors (موصلات) : Grop → 1, 2, 3
  → semi condaters (أشباه الموصلات): Grop → 4 → Si , Ge, Ga
  → Isolaters (العوازل) : Grop → 5, 6, 7

* بسواعدنا - بنبيها

* Coahent Baunds (رابطة التساهمية) : piece of pure crystal
* holes and free $\bar{e}$ →



① تنتقل الـ $\bar{e}$ من إلى holes فتترك مكانها فجوة تسمى holes
② يأتي الـ $\bar{e}$ آخر ويتقوقع

current carries :- ① holes
② free $\bar{e}$

Energy :
1. The farther an $\bar{e}$ is form the nuchens the higher is the energy state. (ابتعد)
2. any $\bar{e}$ that has heftits parent! That has higher energy

* Doping : adding impurity to pure si or Ge (شائبة "جوهرية")
* Doping "إضافة"
  → n-type : adding impurity atomes that five ne "free $\bar{e}$" ("increase the num of conduction $\bar{e}$")
  5-th-Group "Pentavalent" "donor"
  free $\bar{e}$: Majority current carriers , holes :- Minority current carries
  → P-type : adding impurity atomes the 3 ne "free $\bar{e}$"
  3-th Grop "Trivalent" "accept"
  free $\bar{e}$: Minority current carries , holes : Majority curnt carriers

semicondactor materials have a :-
negativetempratur coefficient

Spectrum Team
الممسوحة ضوئيا بـ CamScanner
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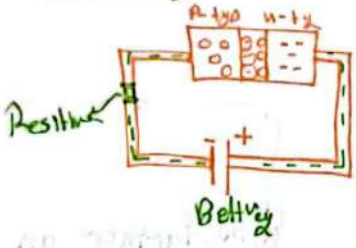
Semiconductor Diode :-

تشغيل الـ "Diode"

1) No Applied Bias " عندما يكون لا يوجد جهد خارجي مطبق I=0 "
2) Reverse Bias
3) Forward Bias

0.9

" The connect the n-type of the positive side of the Battery of connect the P-type of the negative side of the Battery "



* لو كان الـ Diode عبارة عن - switch

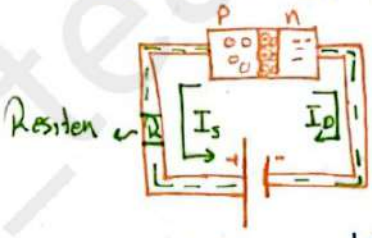
Depletion Region will increase ∴ open

* I ≃ 0 قريب من صفر لكن ليست صفر لأنه يوجد تيار بسيط جداً

* "$V_D < V_{BR}$" هيك أمورنا بتكون بخير الدايود ما بيتدمر

"otherwise damaged diode" غير

" The canned the n-type the negative side of the Battry of connect the P-type of the positive side of the Battery "



* لو كان الـ Diode عبارة عن - switch

Depletion Region will increase ∴ closed

* $V_{Battery} > V_{Diode}$

ideal → $V_D = 0$

depletion region :- The region of uncovered positions and negative ion's

" سنضحك لك الحياة عندما تجد شخصاً ما تتضحك به "

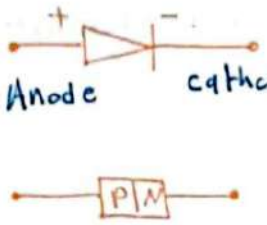
ابتسم :)

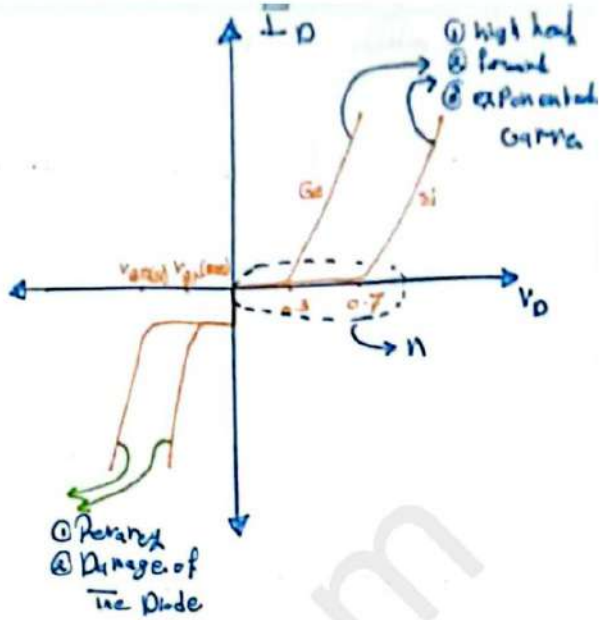
الممسوحة ضوئيا بـ CamScanner

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Symbol of Diode :-

رمز الدايود





note :-

RRT → is very small

"تأثيرات درجة حرارة على المتحنى"

↳

اذا زادت درجة حرارة في Forward :-

decrest ← Depletion ① Region

V_D ← نزاح لليسار V_{BR}

② عكس متتبع

Diode Two Types :-

1) not ideal → مثال مستويات للمقاومة يمكن حسابها

2) ideal → $V_D = IR$

$R = \frac{V}{I}$

"Resitance levels" مستويات المقاومة

1) Dc or static Resitance "R_D" :-

في حالة تأخذ نقطة وتبين I_D و V_D إلها، ممكن بطبيق R_D وأنا أوجد V_D ، I_D

2) Ac or Dynamice Resitance "R_d" :-

في حالة تأخذ نقطة وحماس عندها إنقطة ثم تأخذ نقطتين من أطراف الحماس ونقسم فرق الجهد على فرق التيار

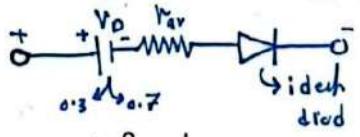
3) Avargy Resitance :-

$$r_{avg} = \frac{V_2 - V_1}{I_f - I_1}$$

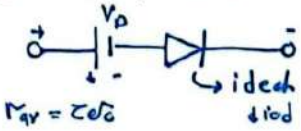
"equivalent Circuits of the Diode"

الدوائر المكافئة للدايود

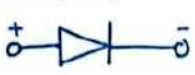
1 Piece-wise linear equivalent :-



2 simplified :-



3 ideal :-



$r_{av} = 0\Omega$

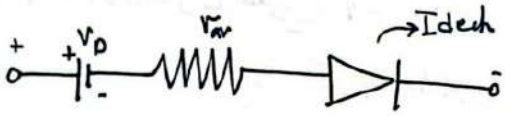
$V_D = 0\Omega$

الممسوحة ضوئيا بـ CamScanner

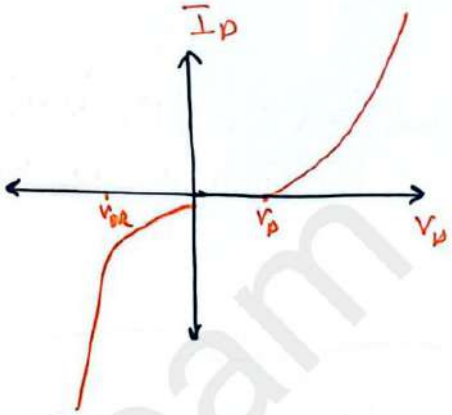
Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

الرسمات بتحون الدوائر المكافئة للديود :-

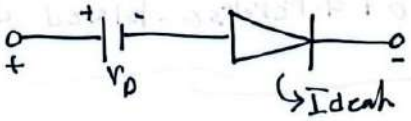
1 Piece-wise linear equation :-



V_D = value not fixed

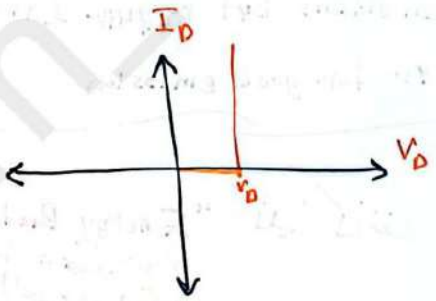


2 simplified :-

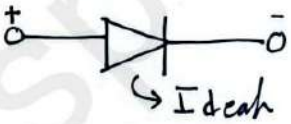


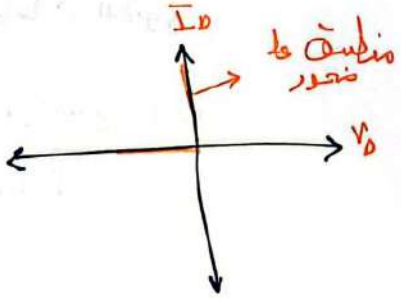
r_{av} = zero

V_D = fixed value



3 Ideal :-





ضغوط الحفاظ

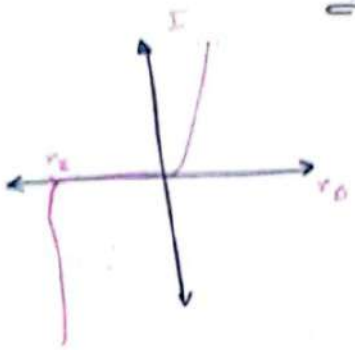
* بساعدنا نبسيها

الممسوحة ضوئيا بـ CamScanner
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ZENER DIODES :-

رسمة





* The location of the Zener region can controlled by? varying the doping levels

increase the doping will the decrease the voltage "Zener Potential

"Si" is the pefered material in the manufacture of Zener diode"

* light-ek emitting Diodes :-
is a diod that gives off visible or invis-ible "infrared"

* some of this energy is given off in the form of heat and some in the form of photons.

note :-

① Si and Ga are not used in the construction of "LED" degrees on the other hand

② Diodes constructed of GaAs emit light in the infrared "invisible" zone during the recombination at the p-n junction

"البدايات للجميع لكن الثبات للمداهبين"

spectrum_team

الممسوحة ضوئيا بـ CamScanner

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* for a forward-biased diode, as temperature is increased the forward current increased for a given value forward voltage.
ازا العطاني في حجب لتجهد

※

note's :

* في حالة الانحياز عكسي : حيث يمر تيار صغير جداً ويعرف بإسم تيار التسريب العكسي يكون قليلا مقارنة بالتيار الامامي لذلك يزداد قليلاً مع ارتفاع درجة الحرارة

* light emiting diode "led" emit light under forward bias-conditions but requir 2v to 4v for good emission

1 تيار تسريب العكسي :
is reverse leakage current
There is a small amount of current across the barrier of a reverse-biased diode

لون الضوء يتعلق بفجوة الطاقة "Energy Bandgap" كلما كانت فجوة طاقة أكبر كان طول الموجة أقصر
wavelength is shorter

لا يمكن استخدام LED لتوليد مصدر تيار متماسك للضوء
led cannot be used to generate coherent source of light

الممسوحة ضوئيا بـ CamScanner
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* whereas "si" at resulted primarily in heat dissipation (المحاولة الأولى) (حرارة) (تبريد)
on recombination (لعادة تركيب)

* The wavelength and frequency of light of a specific color (طول الموجة) (تردد) (الضوء) (محدد) (اللون)
are directly related to the energy band gap of the materials

* one of the major concerns when using an "LED" (رئيسي) (مخاوف)
is the "reverse-bias breakdown voltage", which is typically between
3 ~ 5 V

* The best semi-conductor that is used "LED" is P
GaAs

* تطبيقات عل الزينر :-
1) Voltage Regulation (أفضل تتبع)
2) Voltage Reference
3) protection over-voltage

طريقة التوصيل :-
توصيل "الكاثود" مع جهد الموجب
و "الأنود" مع جهد السالب

"اللهم ان كنت قد قصرت فيما قدمت من عمل أو ملفات
فاغفر لي وما محني وان وجدت خيراً فيه خيراً ونفعاً فاجعل
دعاءك لي بركة وفقني اذا ما تبت وتوفني واجعل عملي
خالصاً لوجهك الكريم يا أرحم الراحمين"

الممسوحة ضوئيا بـ CamScanner
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Note's :-
The term bias in E usually means :-
The value of Dc voltage the divice to opearate properly

There is a smalth amount of current across the barrier of a reverse-biased diode. This current is called:
Reverse leakage current

junction : حمام تاي
controlled : تحكم

انتهى الشبتر ت"

"بسواعدنا نبنيها"

تم تلخيص الشبتر الاول كامل

اللهم لك الحمد كما ينبغي
لجلال وجهك وعظيم سلطانك، اللهم اني
اسالك أن تنفعني بما كتبت وأن تجعله
خالصا لوجهك الكريم وأن تسهل لي فهمه
واستذكارة وقت الحاجه، انك على كل شيء قدير

هدف جاريه على روح والدي "ابو مهيب"

الممسوحة ضوئيا بـ CamScanner
Scanned with CamScanner
CamScanner بـ ممسوحة ضوئيا

بسم الله الرحمن الرحيم

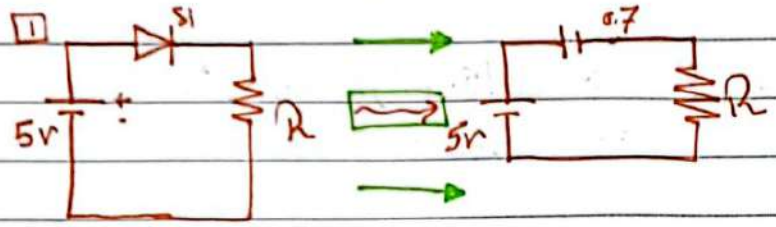
ch 2 :~

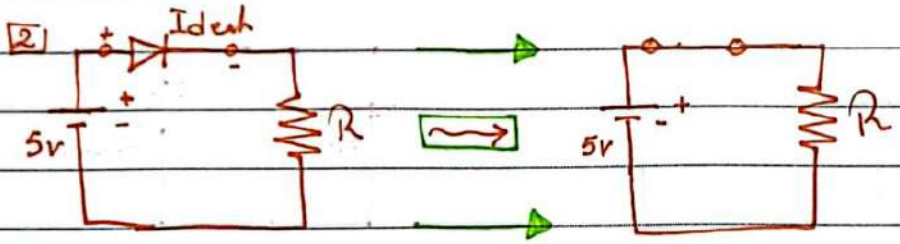
* Diode الدايود: عبارة عن [1] "انحياز أمامي" forward
[2] البعد الطبيعي لدايود "أي" عند "V_T" الجهد العتبة

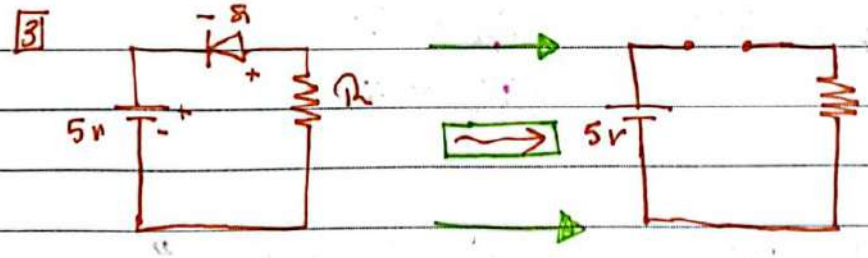
"الإصرار على تحقيق الآمال صعبٌ قرآني؛ في سورة الكهف ﴿ لَا أَبْرَحُ حَتَّىٰ أَبْلُغَ ﴾"

V_T → Si → 0.7 v
→ Ge → 0.3 v
→ Ideal → 0 V

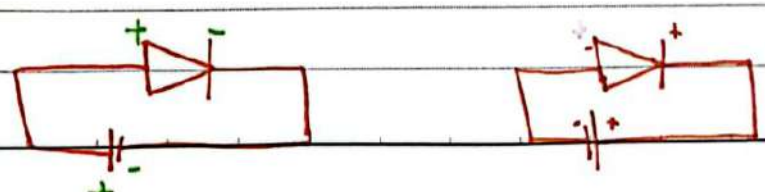
Ex :~







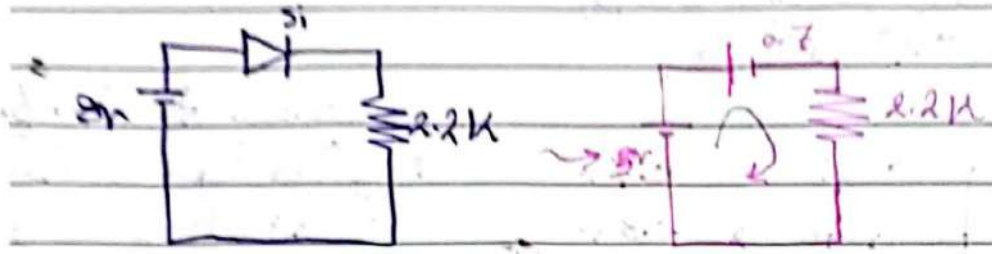
note's :~



"forward" "reverse"

الممسوحة ضوئيا بـ CamScanner
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Ex 8~ find The V_D, I_D, V_R



$V_D = 0.7$

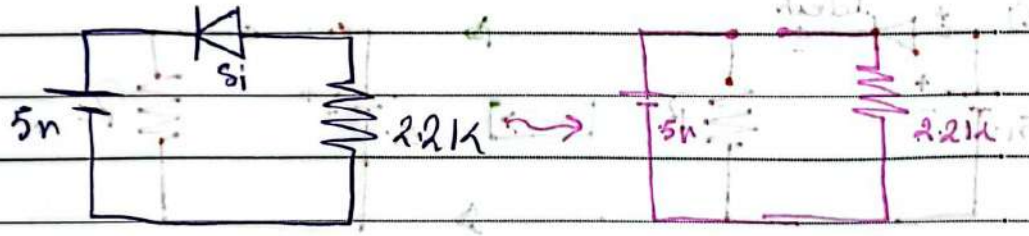
$I_D = ?$ ⇝ $-8 + 0.7 + I(2.2K) = 0$

$I_D = 1.95 \cdot 10^{-3}$

$V_R = I R$ ⇝ $(1.95 \cdot 10^{-3}) \cdot (2.2 \cdot 10^{3})$

$= 4.3 V$

Ex 9~ find The V_{DQ}, I_{DQ}, V_R



$V_D = 0$

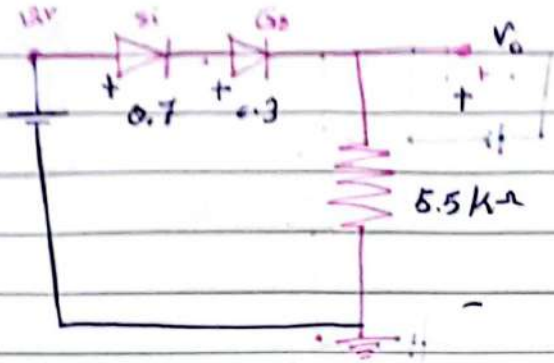
$I_R = 0$ $\quad V_R = V$

$V_D = 5V$

"وما نيل الطلاب بالتمني ولكن يؤخذ الكتاب على رضاه"

الممسوحة ضوئيا بـ CamScanner

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* Ex8- Find the V_{D_1}, V_{D_2}, I_D, V_R, V_o !!

$V_{D_1} = 0.7$

$V_{D_2} = 0.3$

$I_D = ?$ ⇝ $-12 + 0.7 + 0.3 + I_D \cdot 5.5k = 0$

$\boxed{I_D = 2mA}$

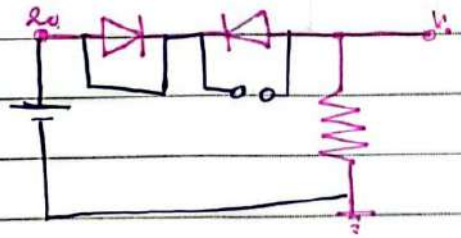
$V_R = IR$

$\boxed{V_R = 11V}$ ~~//~~ $V_o = V_R = 11V$

open ckt ⇝ $I = 0$ $V_D = ?$

short // ⇝ $I = ?$ $V_D = 0$

* Ex8- find the I_D, V_D, V_{D_2}, V_R, V_o ... ?



$I = 0$

$V_{D_1} = 0$

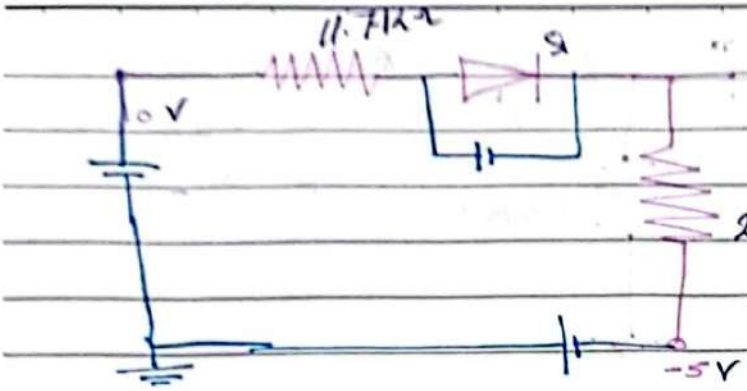
$V_{D_2} = 20$

$V_o = 0$

$V_R = 0$

الممسوحة ضوئيا بـ CamScanner

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تم المسح ضوئيا بـ CamScanner



ex3- find the I_D, V_{R_1}, V_{R_2}, V_o

sol:-

$$-10 + 4.7k \cdot I + 0.7 + 2.2k I - 5 = zero$$

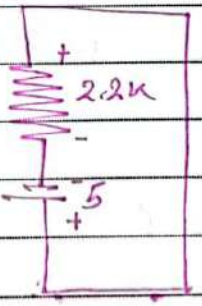
$$6.9 I - 14.3 = zero$$

$$\boxed{I = 2.072\ mA}$$

$$V_R = I R = 9.738.4\ V$$

$$V_{R_2} = 4.558.4\ V$$

$V_o \rightarrow$



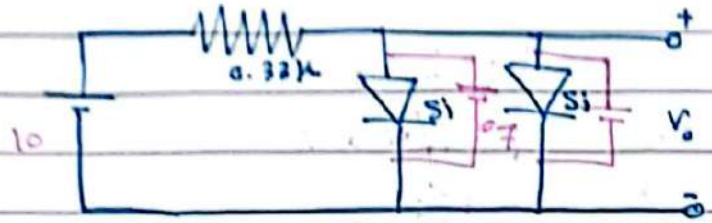
$$5 - 2.2k + V = zero$$

$$\boxed{V_o = 0.442\ V}$$

الممسوحة ضوئيا بـ CamScanner

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Ex: find the V_o, I_D ?



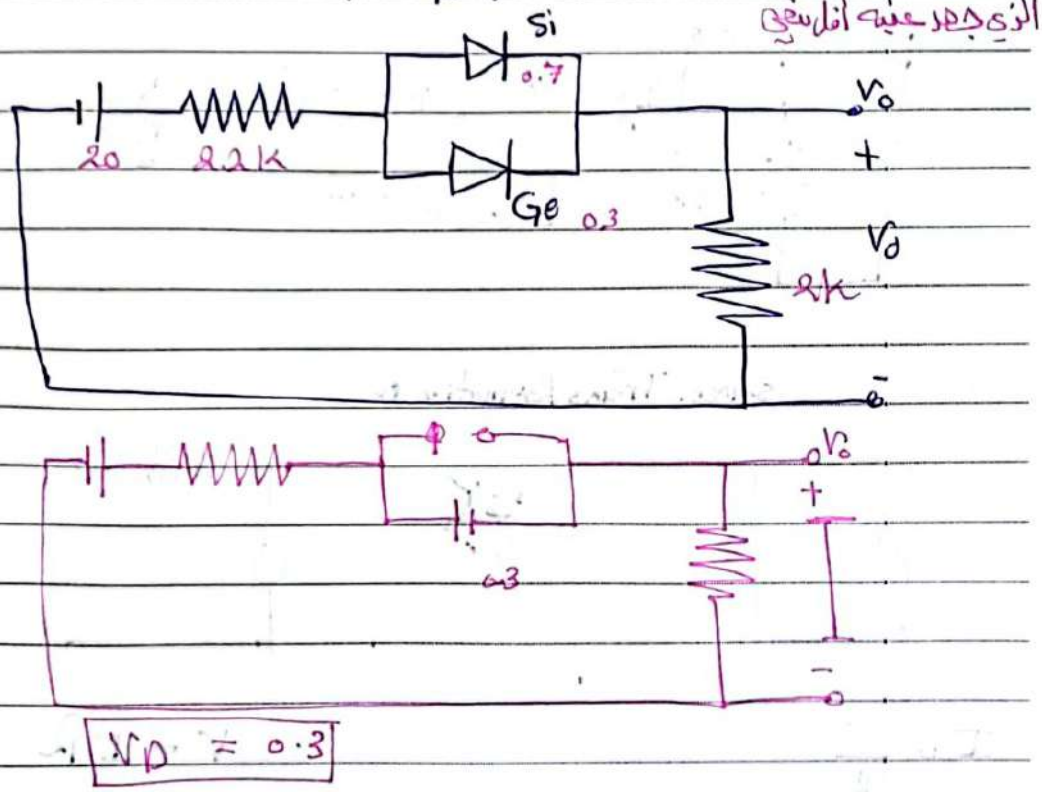
$V_o = 0.7$

$-10 + 0.33I + 0.7 = zero$

$I_D = 0.028\ A$

Ex: find the I_D, V_D ??

الذي جهد عتبته اقل يشتغل



$\therefore V_D = 0.3$

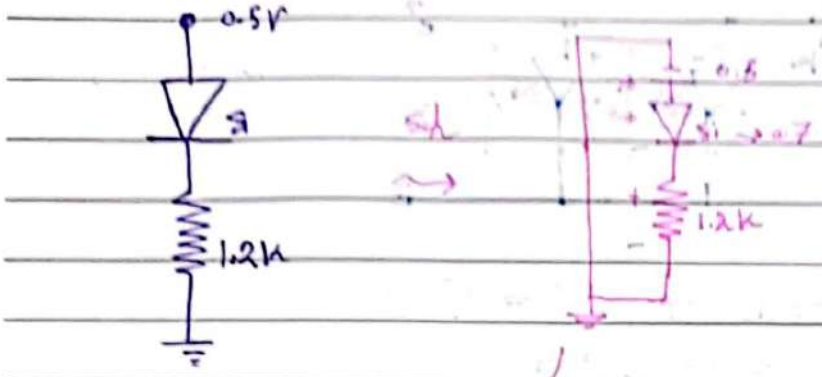
$-20 + 2.2 \cdot 10^3 I + 2 \cdot 10^3 I = zero$

$I = 4.76\ m$

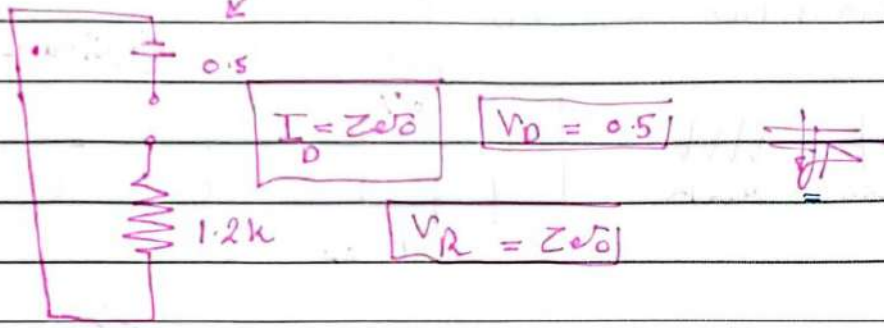
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex8- find The I_D . V_D . V_R



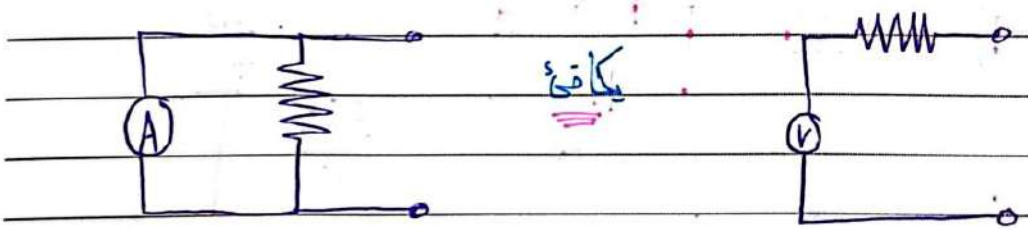
$V_D = 0.$ $\quad 0.5 < 0.7 \rightsquigarrow 0.5$



$I_D = 2mA$ $\quad V_D = 0.5$

$V_R = 2mA$

Note 8~ source Transformation 8~



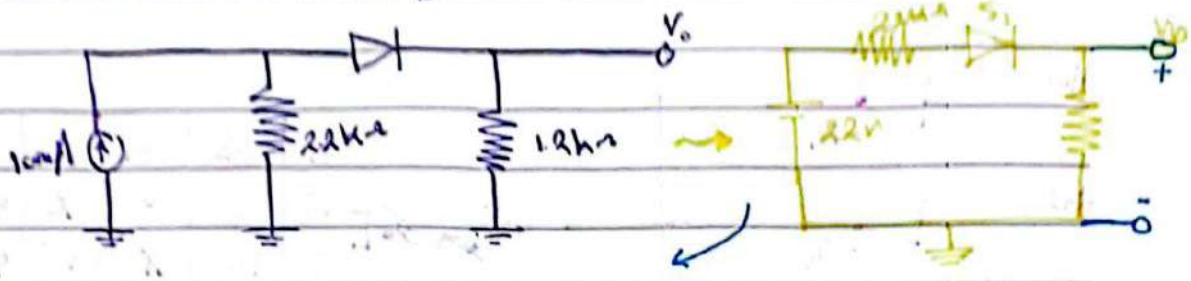
$I = \frac{V}{R}$ $\qquad V = IR$

قال الله تعالى: ﴿ واتقوا يوماً ترجعون فيه إلى الله ﴾

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex 8~ find the V_o and I_D



$$-22 + 2.2kI + 0.7 + 1.2kI = 20$$

$$I_D = 6.26 \cdot 10^{-3}$$

$$V_o = 7.517$$

And - or - Gate 8~

A	B	A or B
0	0	0
0	1	1
1	0	1
1	1	1

"or Gate"

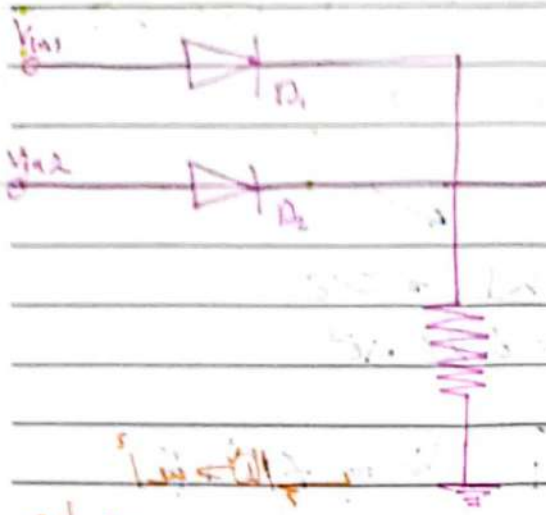
A	B	A and B
0	0	0
0	1	0
1	0	0
1	1	1

"and Gate"

الممسوحة ضوئيا بـ CamScanner

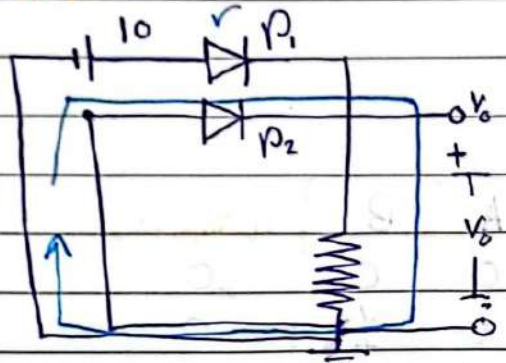
Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

ex- Find the D_1, D_2, V_o



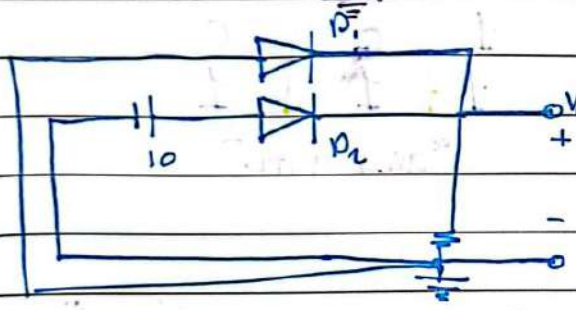
V_1	V_2	D_1	D_2	V_o
10	0	✓	X	9.3V
0	10	X	✓	9.3V
10	10	✓	✓	9.3V
0	0	X	X	0V

Sol:-



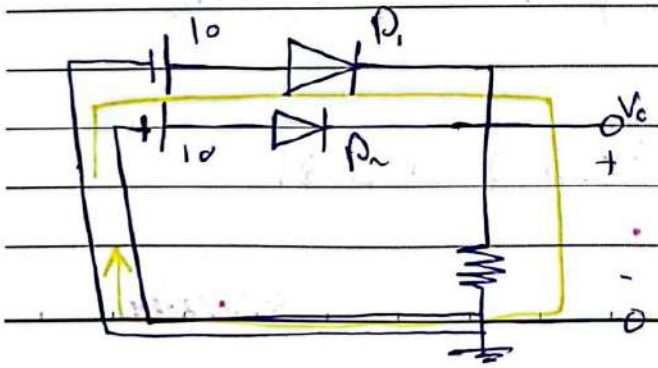
$$-10 + 0.7 + V_o = 0V$$

$$V_o = 9.3V$$



$$-10 + 0.7 + V_o = 0V$$

$$V_o = 9.3V$$

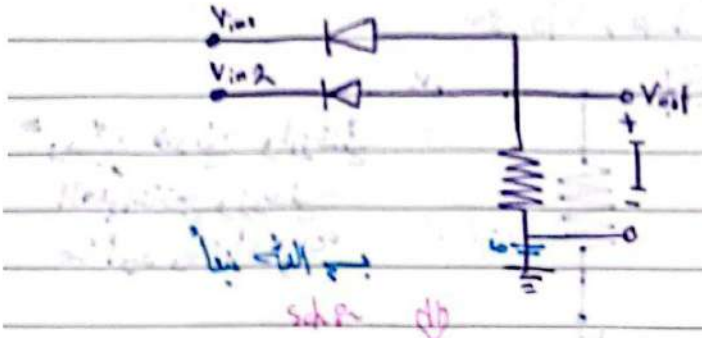


$$-10 + 0.7 + V_o = 0V$$

$$V_o = 9.3V$$

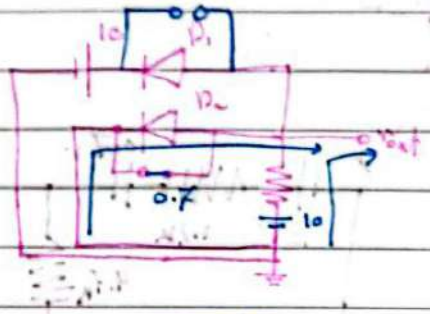
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

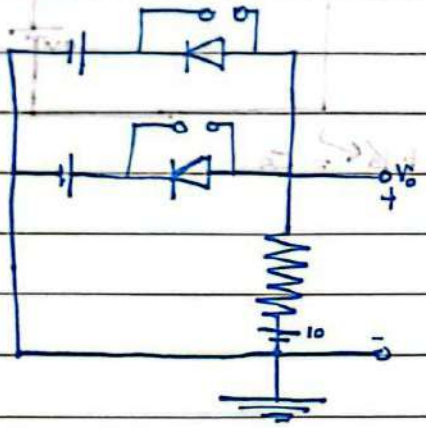


V_1	V_2	D_1	D_2	V_o
10	0	X	✓	0.7
0	10	✓	X	0.7
10	10	X	X	10v
0	0	✓	✓	0.7

: and "



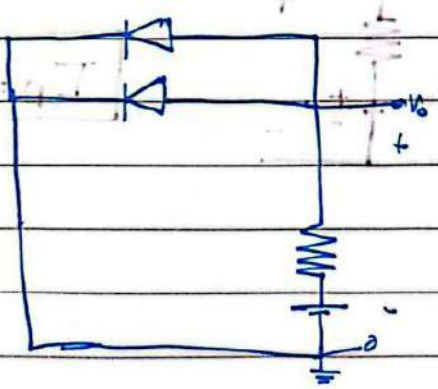
$0.7 = V_{out}$



$I = \frac{10}{2}$

$-10 + 2I = V_o$

$V_o = 10$

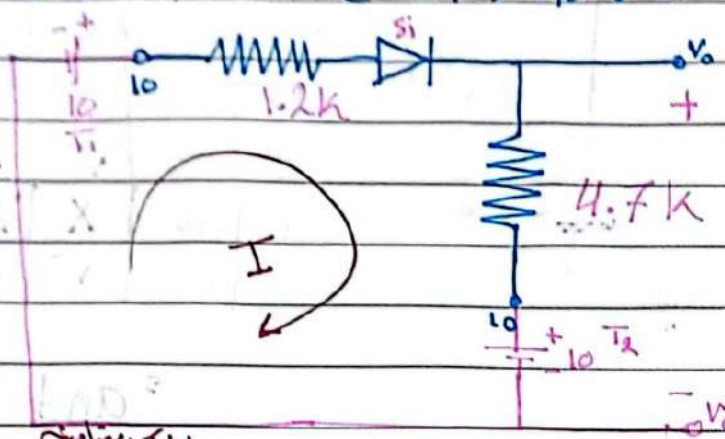


$V_o = 10.7$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex:- find The I_D, V_D :-



ربي هدني وارزقني العلم النافع وزدني ما أريد وما أهواه

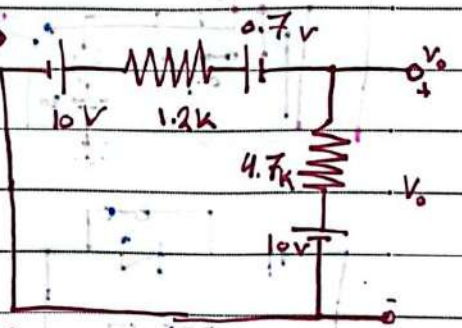
سايكون مشابه

طريقة تحديد الجهد الذي بادئ الديود !؟

① بكون الديود Default شغال الجهد بيلي تغلو T_0.

② بطرح قيمة البار

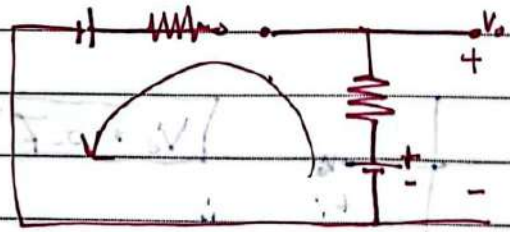
③ اذا الطرح في البار مال فرايدل ان اتجاه تيار عكس بار فنهمله



$$-10 + 1.2kI + 0.7 + 4.7kI + 10 = 0$$

$$I = 0.18 \cdot 10^{-4}$$

قيمة موجبة اتجاه التيار



$V_o = 10v$

$I_D = 0$

④ بطرح الجهدين

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

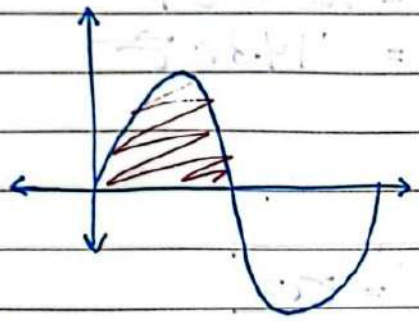
Rectification :-

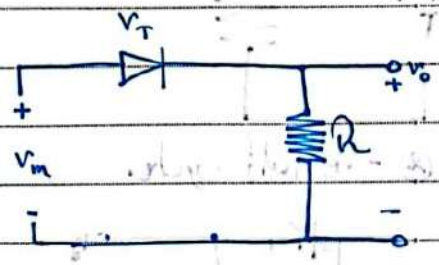
Ac $\xrightarrow{\text{dest}}$ Dc

- half wave Rectifir
- full wave Rectifir
  - Bridge
  - center tapped

"أمي هي الأمان وهي سلام"

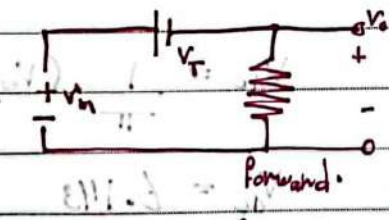
1- half wave Rectifir :-





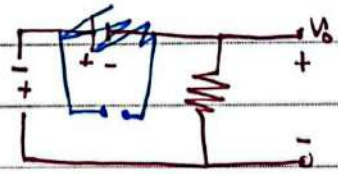
sch :-

① +ve half cycles

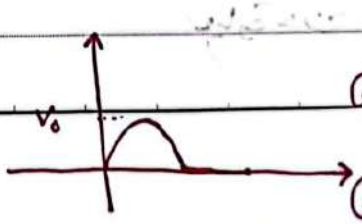


$V_o = V_m - V_T$

② - ve half cycle



$V_o = zero$



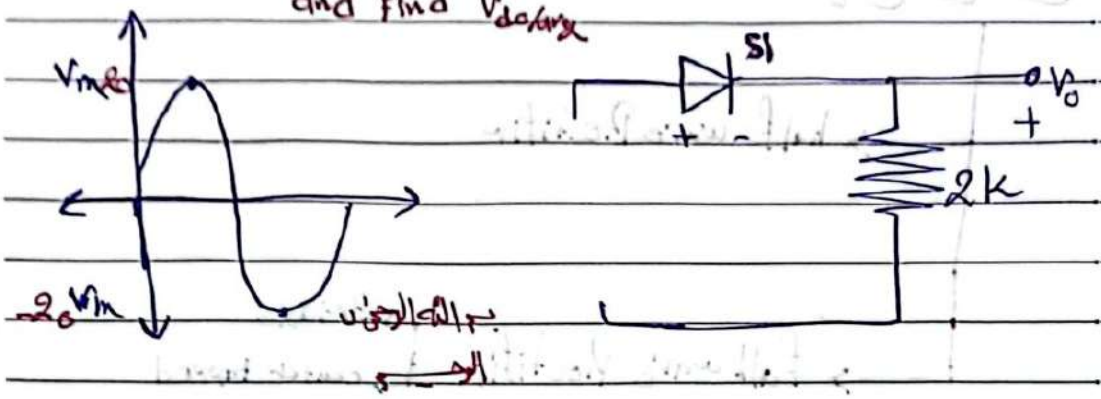
① $V_{dc/avg} = \frac{1}{\pi} \cdot V_o$

② $f_{in} = f_{out}$

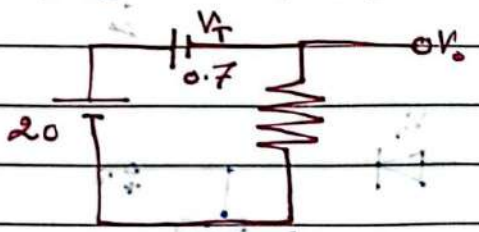
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner ممسوحة ضوئيا

Ex 8- Find and Drawn output signal &
and find V_{dc} value



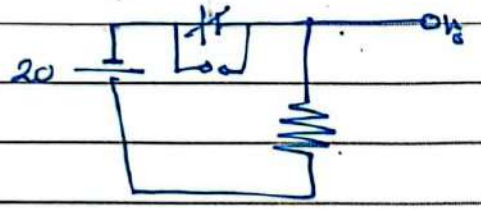
sol 8- ① +ve half cycle



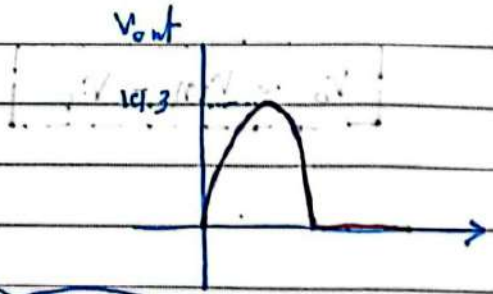
$V_o = 20 - 0.7$

$V_o = 19.3V$

② -ve half cycle



$V_o = Zero$



$V_{dc} = \frac{1}{\pi} \cdot (V_o)$

$V_{dc} = 6.143$

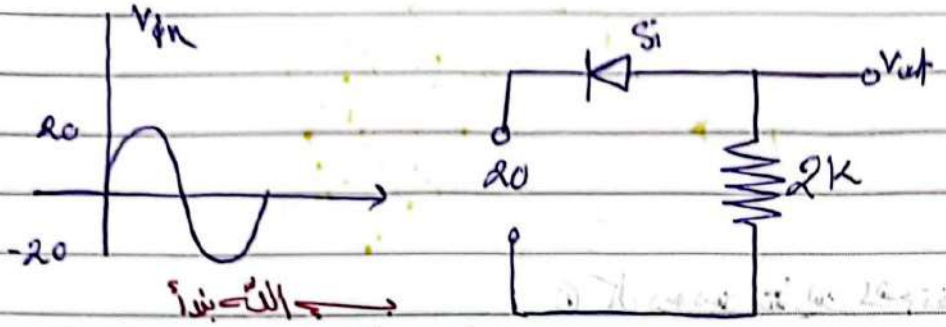
note-

"الفولتية في خلال ح.ت تكون
مساوية حت PIV"

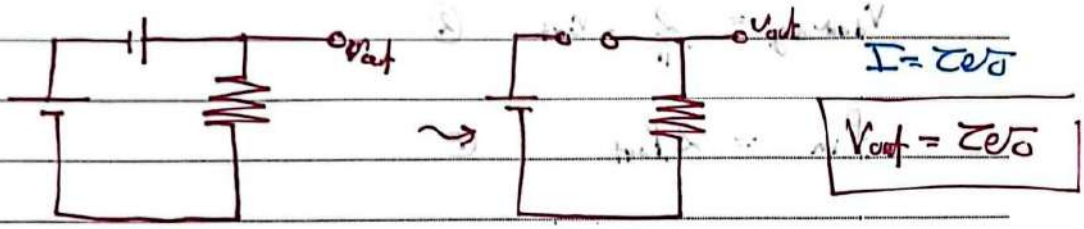
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner

Ex2 : find and sketch the output signal ov and find $V_{dc/avg}$?

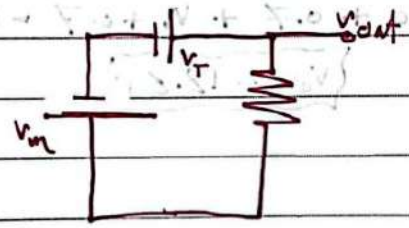


sch :- ① +ve half cycle



inverse

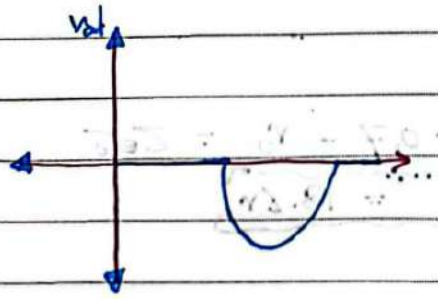
② -ve half cycle



$V_{out} = V_{in} - V_T$

$= -20 + 0.7$

$V_{out} = -19.3\,V$



$V_{dc} = \frac{1}{\pi} \cdot V_{out}$

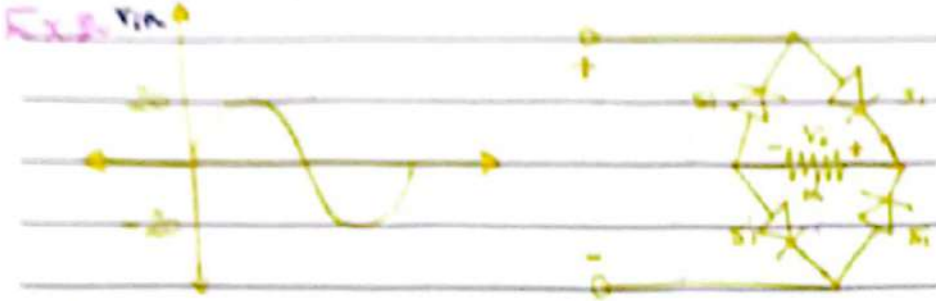
$= \frac{1}{\pi} \cdot 19.3$

$V_{dc} = 6.143$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

[2] Full wave rectifier

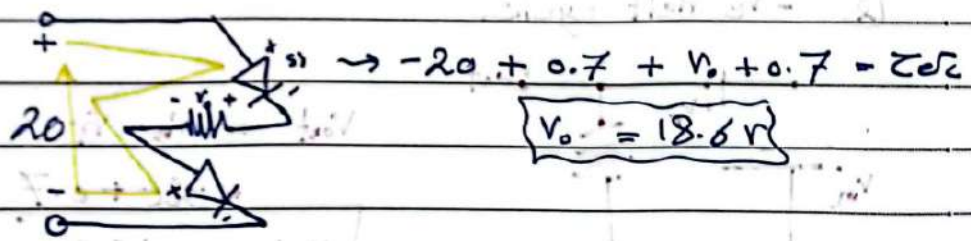


not :- كل ديودين متقابلين يشتغلوا
في نفس الوقت

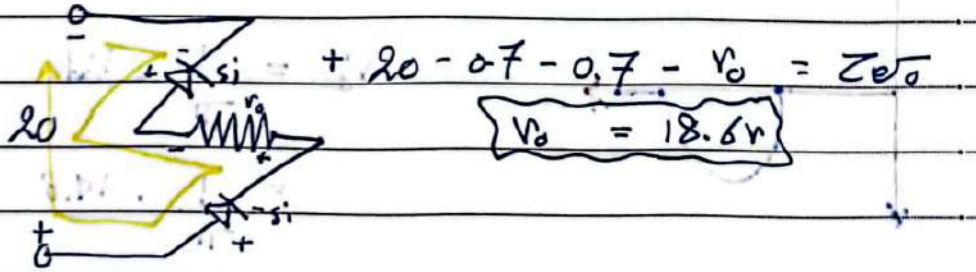
$$V_{dc\ avg} = \frac{2}{\pi} \cdot V_{out} \quad (2)$$

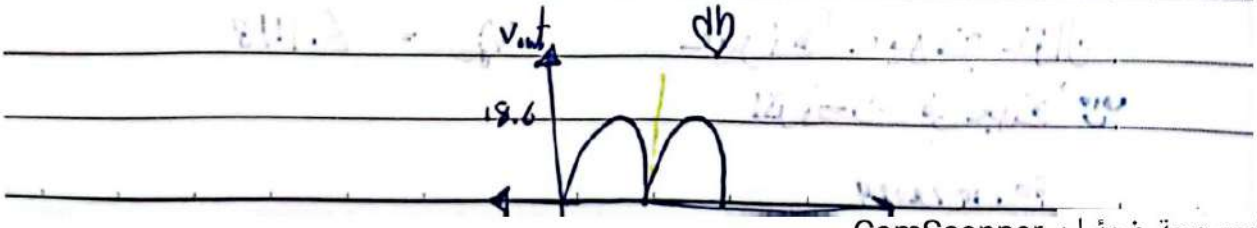
$$f_{in} = 2 f_{out} \quad (3)$$

sol :- +ve half cycle :-



-ve half cycle :-

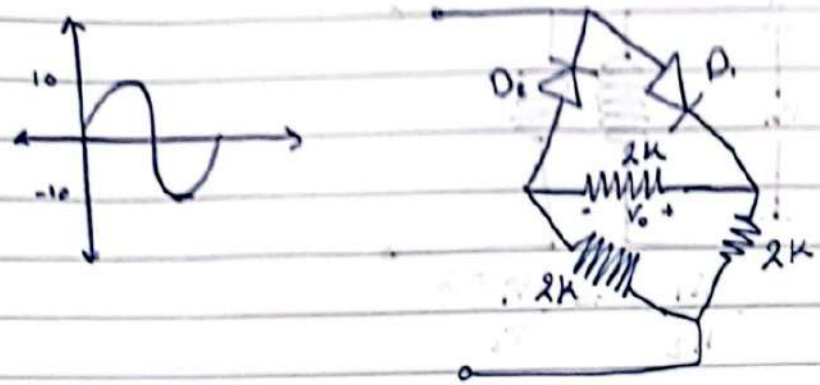




الممسوحة ضوئيا بـ CamScanner

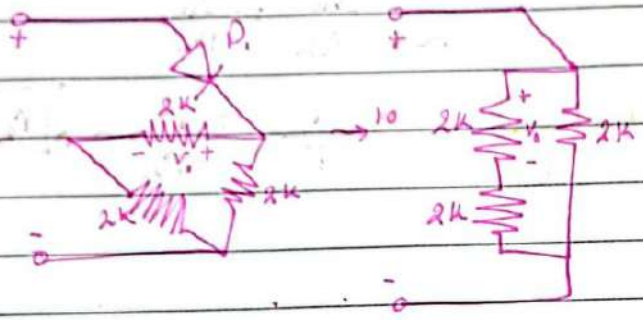
Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

Ex 8- find of Draw The output signal "D1 & D2 Ideal"



sch 8-

+ Ve



$R_{eq} = \frac{4000}{3}$

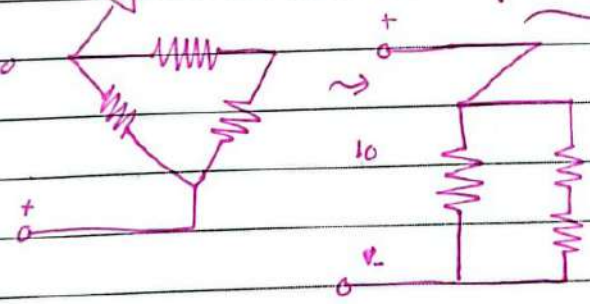
$V = I R$

$I = \frac{3}{400}$

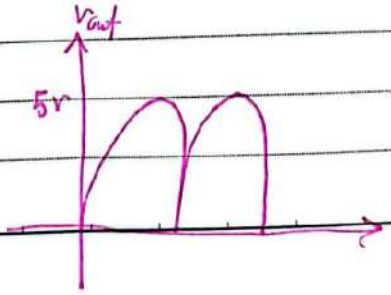
$I = \frac{10}{4 \cdot 10^3} \rightarrow \frac{1}{400}$

$V_o = 5V$

- Ve



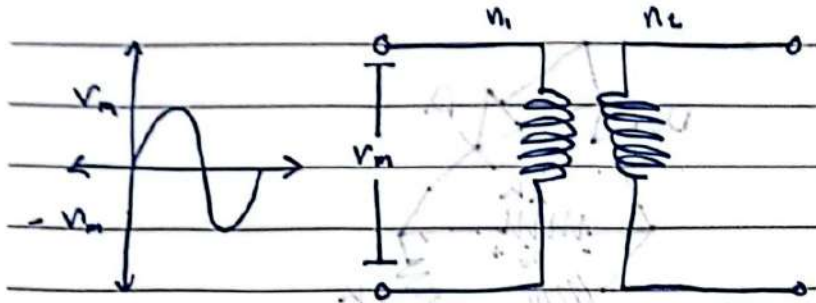
$V_o = 5V$



الممسوحة ضوئيا بـ CamScanner

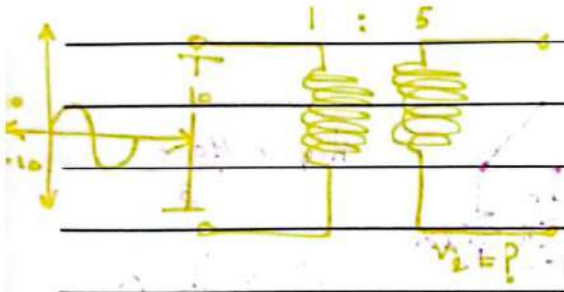
Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

الحول



$$\frac{n_1}{n_2} = \frac{v_1 \to v_m}{v_2}$$

Ex



$$\frac{n_1}{n_2} = \frac{n_1}{n_2}$$

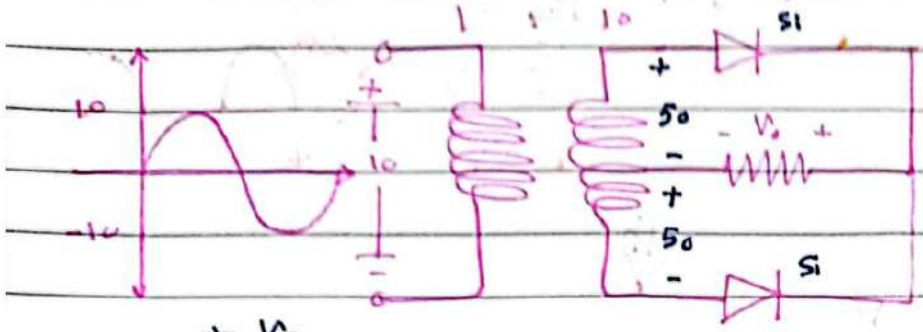
$$\frac{1}{5} = \frac{10}{v_2} \rightarrow \boxed{v_2 = 50v}$$

الممسوحة ضوئيا بـ CamScanner

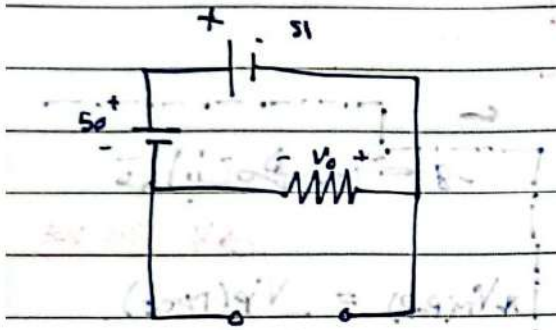
Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

[B] Center tapped transformer :-

Ex: Find & Draw the output signal :-



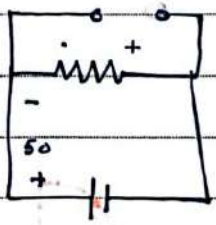
$$V_2 = \frac{n_2 \cdot V_1}{n_1} \rightarrow 100 \rightsquigarrow V_2 = \frac{100}{2} \rightarrow 50$$



$$-50 + 0.7 + V_o = 0$$

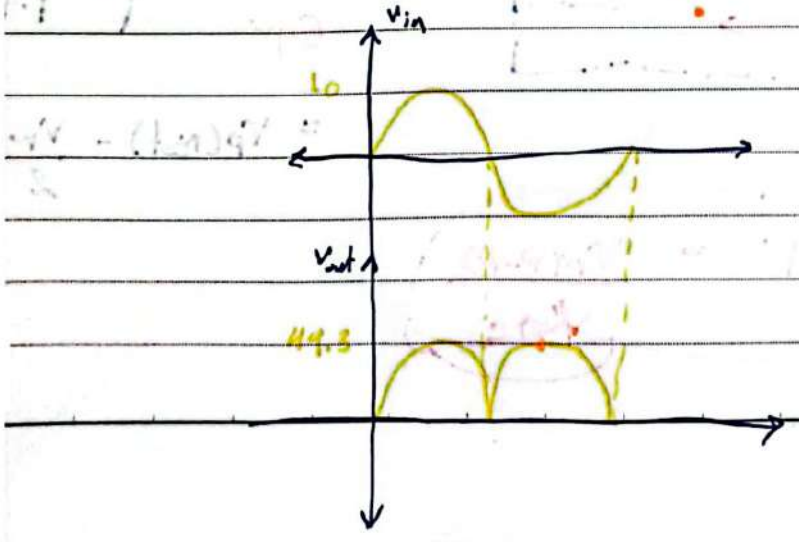
$$\boxed{V_o = 49.3\,V}$$

- V_{in}



$$50 - V_o - 0.7 = 0$$

$$\boxed{V_o = 49.3\,V}$$

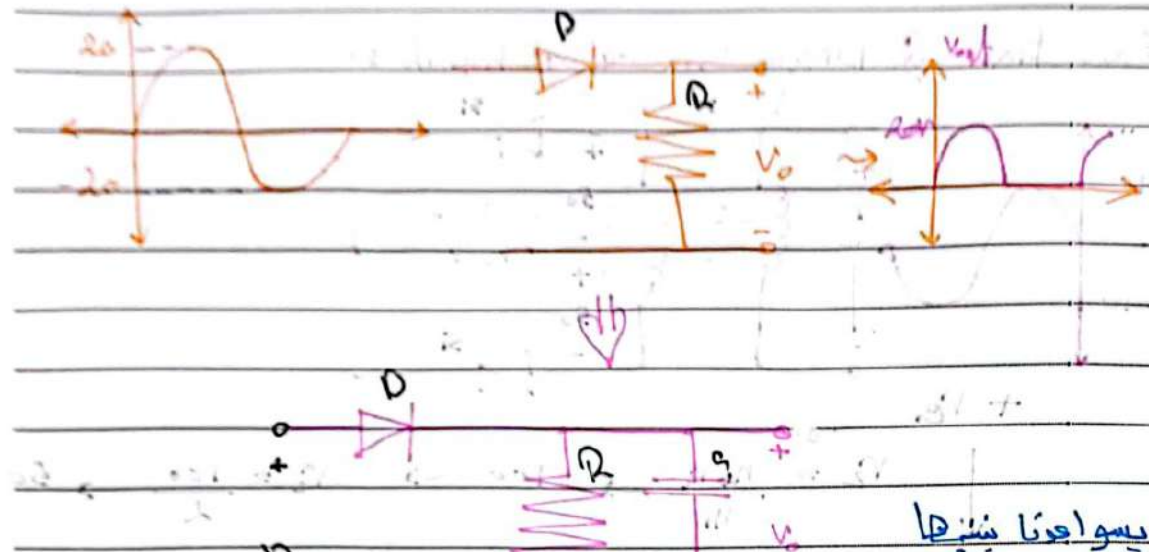


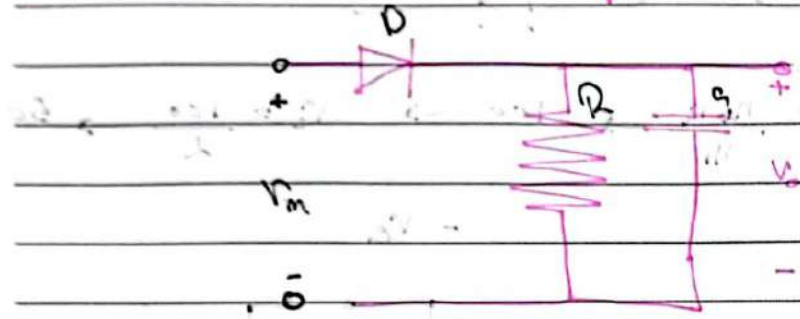
سلاح الهندسة
كتائب المقاوم

الممسوحة ضوئيا بـ CamScanner

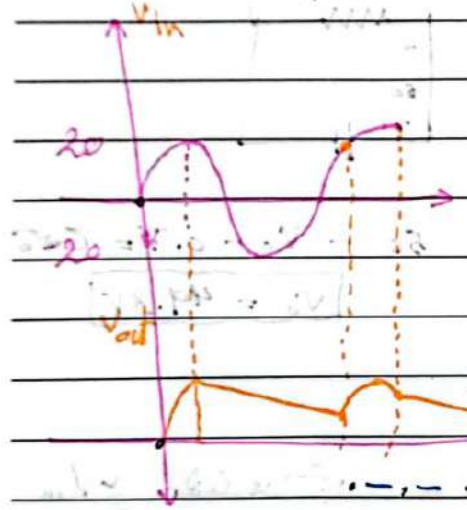
Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Filter's 8-





* يسواوا بنيها



قوانينها ؟-8

V_o (يعني موجة)

$$V_{r(p-p)} = \frac{V_{p(rect)}}{f_p \cdot R_L \cdot C}$$

$$V_{DC} = V_{p(rect)} \cdot \left(1 - \frac{1}{f_p \cdot R_L \cdot C}\right)$$

OR

$$= V_{p(rect)} - \frac{V_r}{2}$$

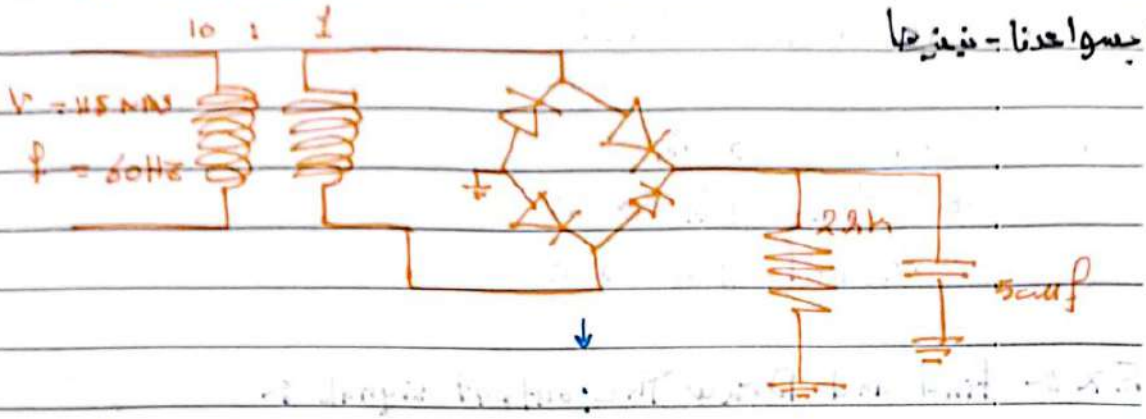
Ripple factor $= \frac{V_{r(p-p)}}{V_{DC}}$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex 8- find the ripple factor for the circuit
* All diode is si ??

بسواعدنا - نبنيها //



Sol:-

$$\frac{N_1}{N_2} = \frac{V_1}{V_2} \Rightarrow \frac{10}{1} = \frac{162.63}{V_2}$$

$$V_2 = 16.263\ V$$

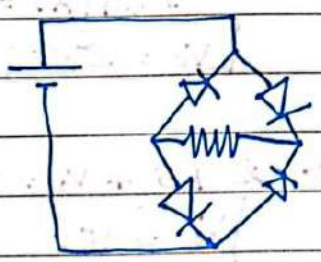
$V_{rms} \rightarrow m$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_m = V_{rms} \cdot \sqrt{2}$$

$$V_m = 162.63\ V$$

$$V_o = V_m - 2V_D$$
$$= 16.263 - 2(0.7)$$
$$V_o = 14.863 = V_{(rect)}$$



$$V_{R.P} = \frac{14.863}{120 \cdot 2.2 \cdot 10^{-3} \cdot 50 \cdot 10^{-6}}$$

$$= 1.125\ V$$

$f = 2f_{out}$

$f = 2 \cdot 60$

$= 120$

$$V_{DC} = 14.863 - \frac{1.125}{2}$$

$$= 14.3005\ V$$

$$r = \frac{V_{R.P}}{V_{DC}}$$

$$= \frac{1.125}{14.3005}$$

$$= 0.078$$

$$= 7.8\ \%$$

أسعى ..

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بيقى تعرف ال

clippers "limiter" دوائر القطع

وجود مقاومة وديود "البطارية" D و

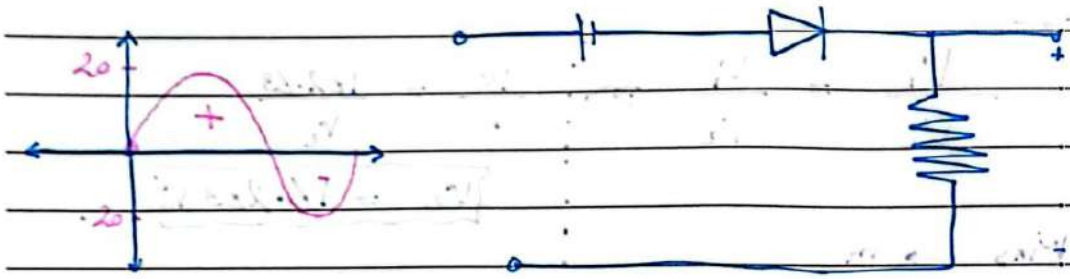
كيفية حل هذه الأسئلة ؟؟

① S.C ⟶ $V_o = 0$

② +Ve half cycle ⟶ V_o

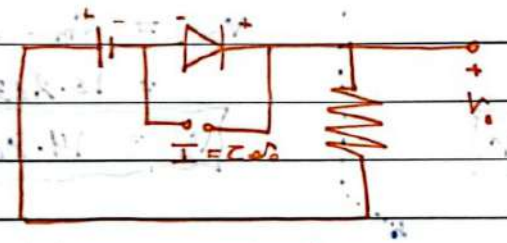
③ -Ve half cycle ⟶ V_o

Ex:- find and Draw The output signal :-



Sol:-

① S.C

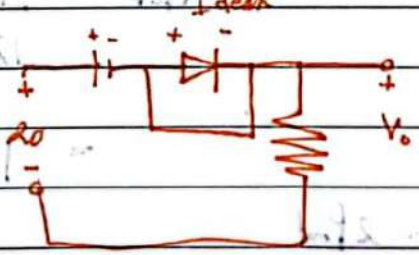


$V_o = V_R = IR$

$= (0)(R) = 0$

$V_o = 0$

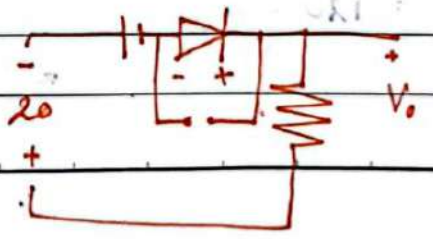
② +Ve



$-20 + 5 + V_o = 0$

$V_o = 15\ V$

③ - Ve

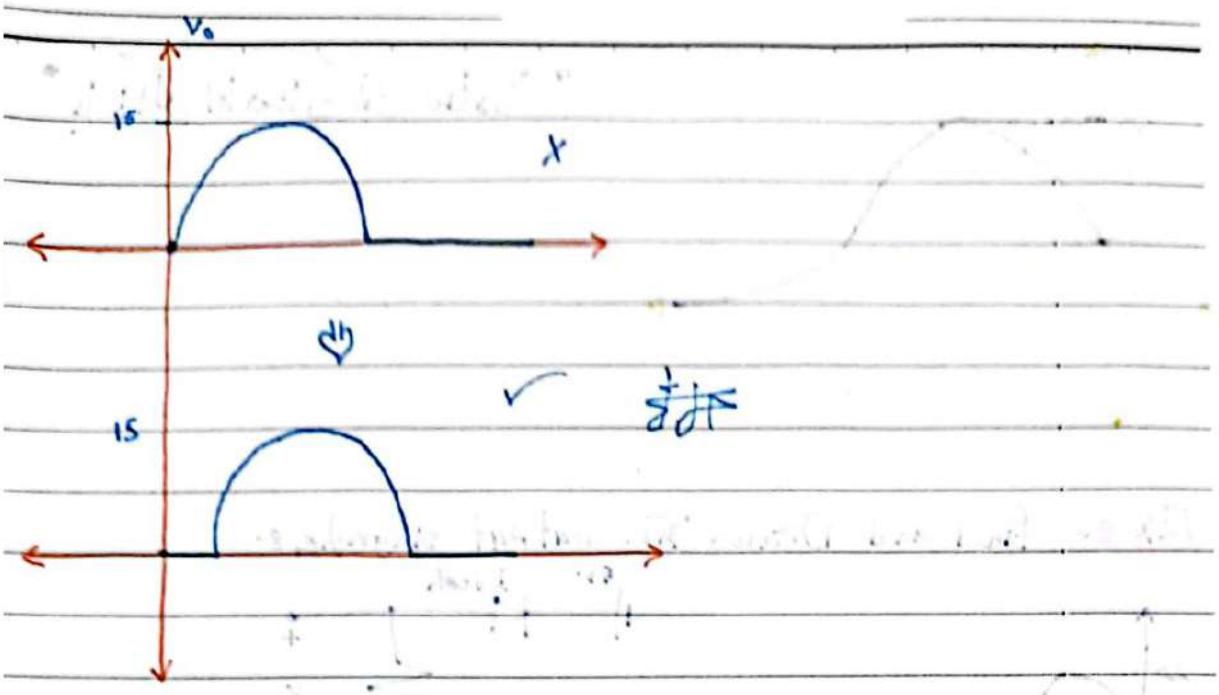


$V_o = V_R = 0$

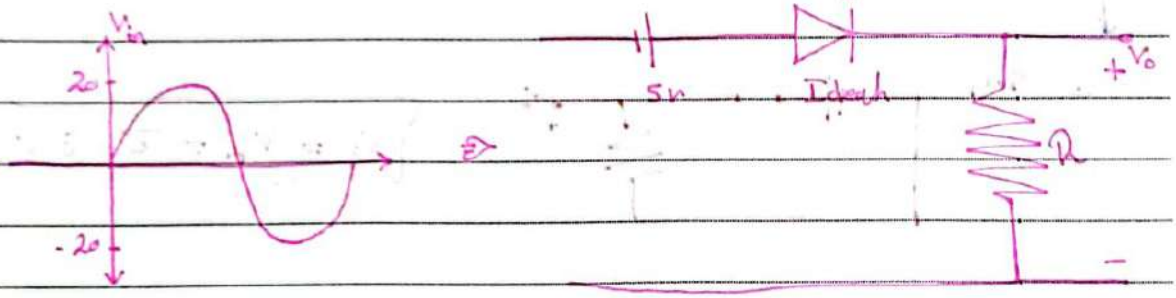
$V_o = 0$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

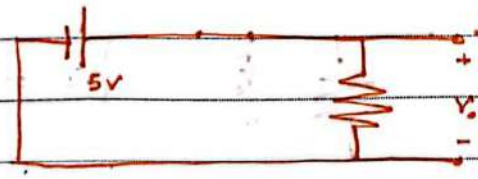


Ex:- Find and Draw the output signal



Sol:-

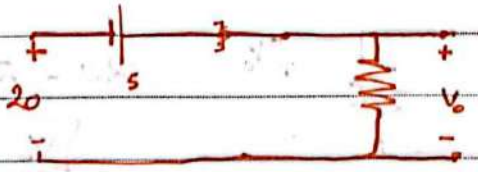
① S.C



$-5 + V_o = zero$

$V_o = 5\,v$

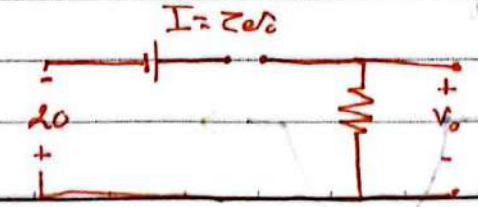
② + Ve



$-20 - 5 + V_o = zero$

$V_o = 25v$

③ - Ve

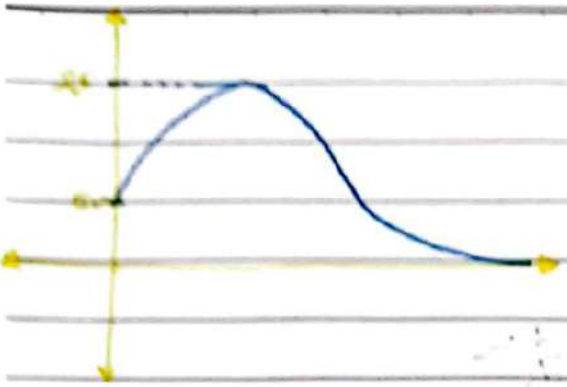


$V_o = zero$

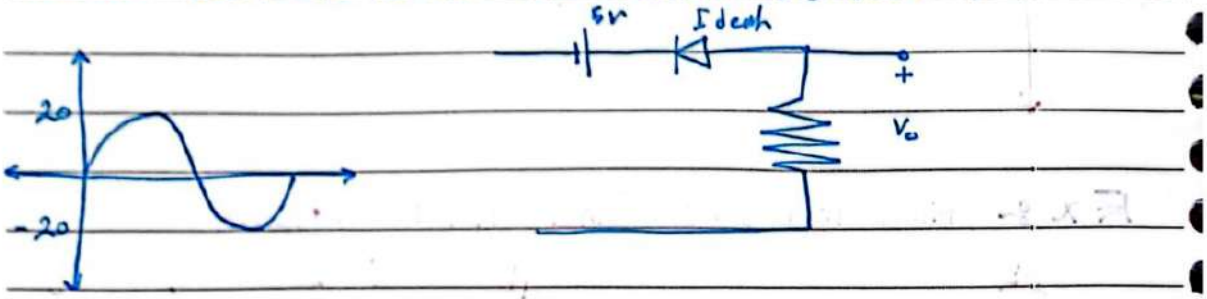
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

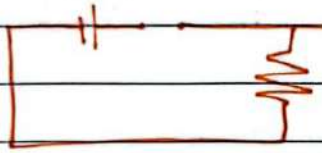
"يا ذاكر الاصحاب كن مصاحبا"



Ex:- find and Draw the output singnals:-



① S.S



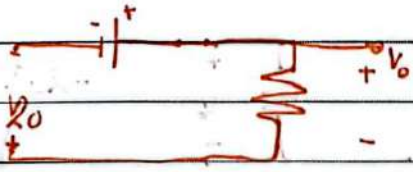
$V_o = V_R = 20 \cdot 5\ V$

② + Ve



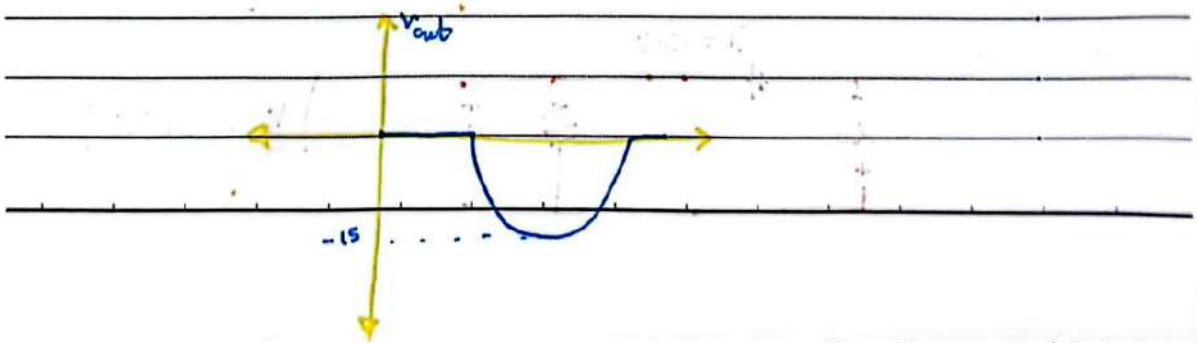
$V_o = 20 \cdot 5$

③ - Ve



$20 - 5 + V_o = 20 \cdot 5$

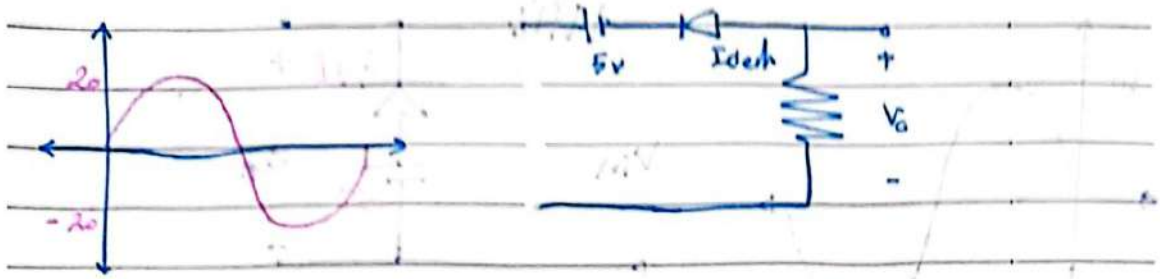
$V_o = -15$



الممسوحة ضوئيا بـ CamScanner

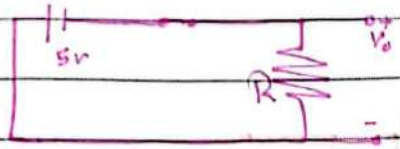
Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex: find and Draw the voltage output PP



Sol:-

① S.C



$5 + V_o = 2e0$

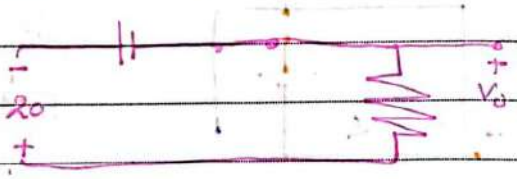
$V_o = -5$

② $+V_a$



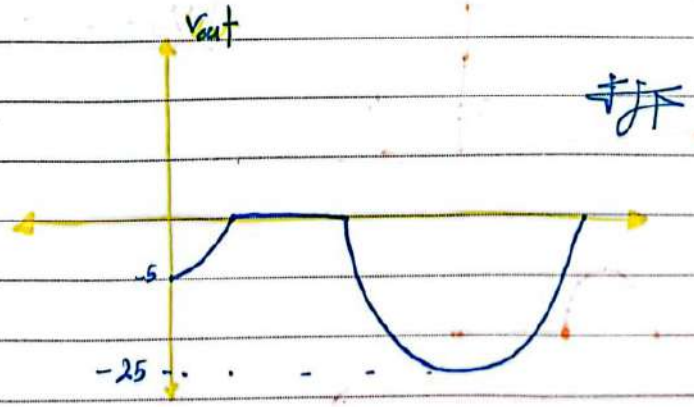
$V_o = 2e0$

③ $-V_a$



$20 + 5 + V_o = 2et$

$V_o = -25\ V$

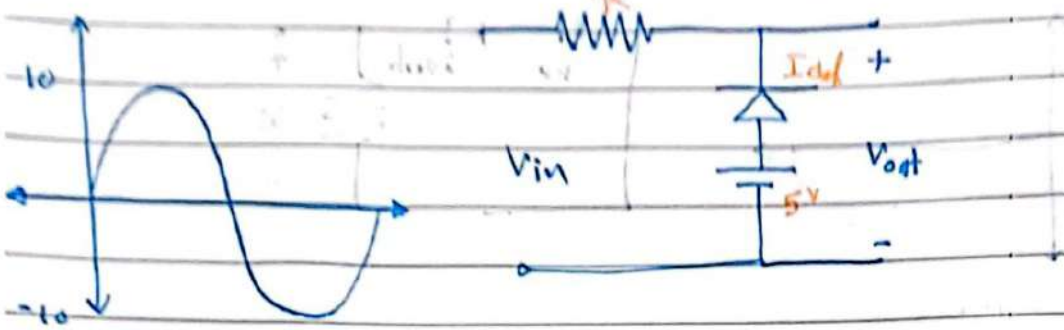


"القول تحول المعارف"

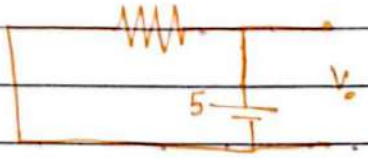
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

Ex :- find and Draw the out put voltage a Br

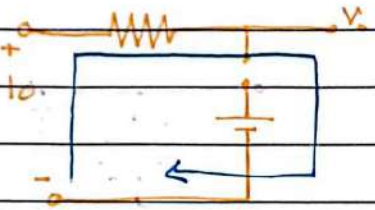


1 S.G



$V_o = 5V$ dc

2 + Ve

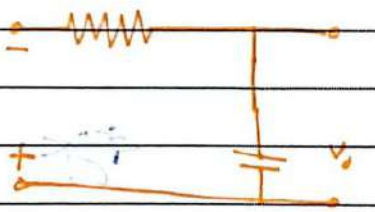


$\Sigma = 2\mu_0$

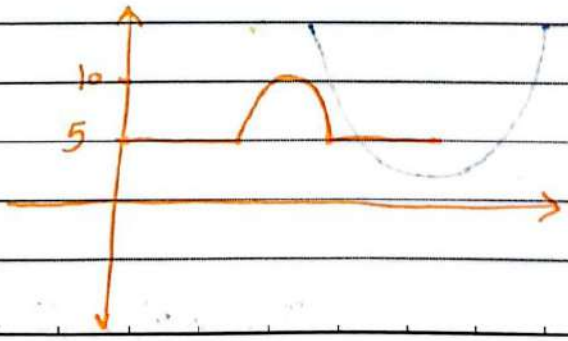
$-10 + V_R + V_o = 2\pi$

$V_o = 10V$ Ac

3 - Ve



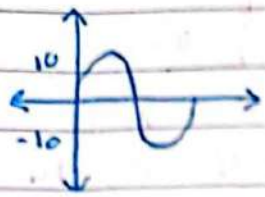
$\therefore V_o = 5V$ dc

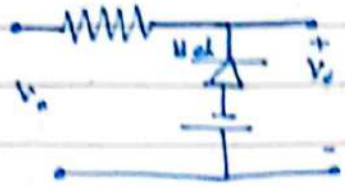


الممسوحة ضوئيا بـ CamScanner

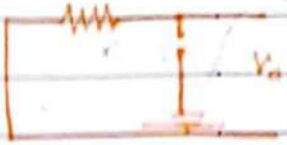
Scanned with CamScanner ب CamScanner الممسوحة ضوئيا

Ex2 Sol



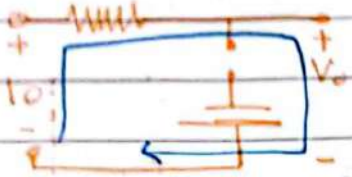


① D.C



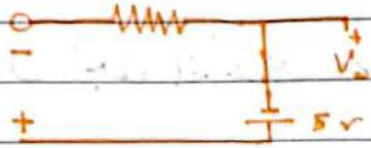
$V_o = Zero$

② +Ve

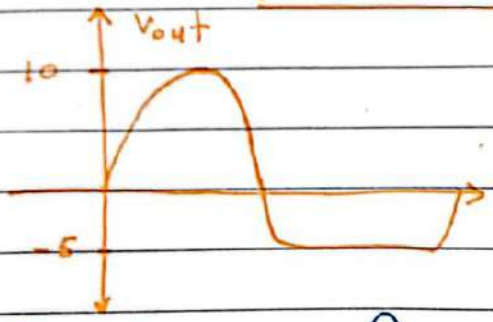


$V_o = 10\,V$ Ac

③ -Ve

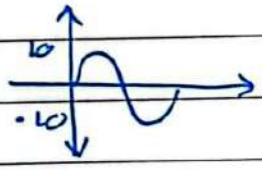


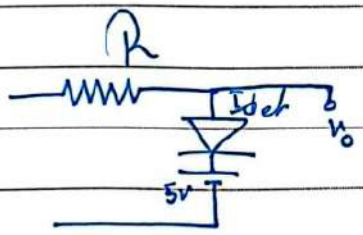
$V_o = -5V$ dc



"لسنا حالمين
بما يكفي
لكننا نخاف الله"

Ex3 Sol



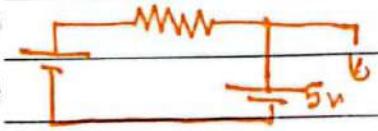


① D.C



$V_o = Zero$

② +Ve

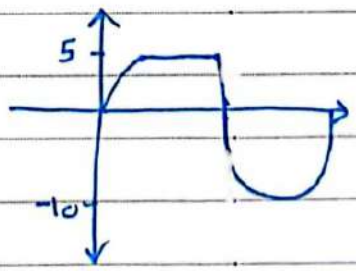


$V_o = 5V$ dc

③ -Ve



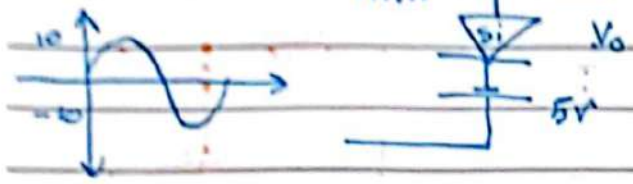
$V_o = -10V$ Ac



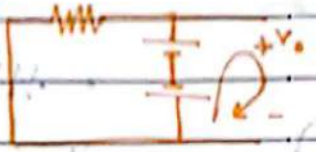
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex 8~



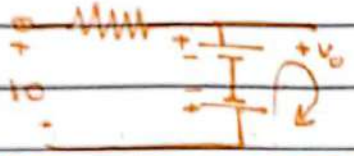
① S.C



$V_o = 5 - 0.7$

$V_o = -4.3V$ dc

② $+V_c$

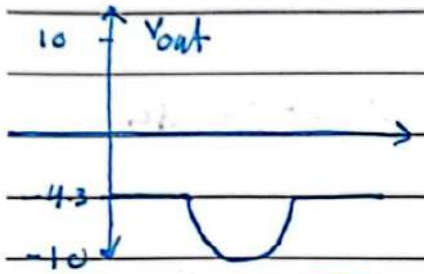


$V_o = -4.3V$ dc

③ $-V_c$



$V_o = -10V$ AC



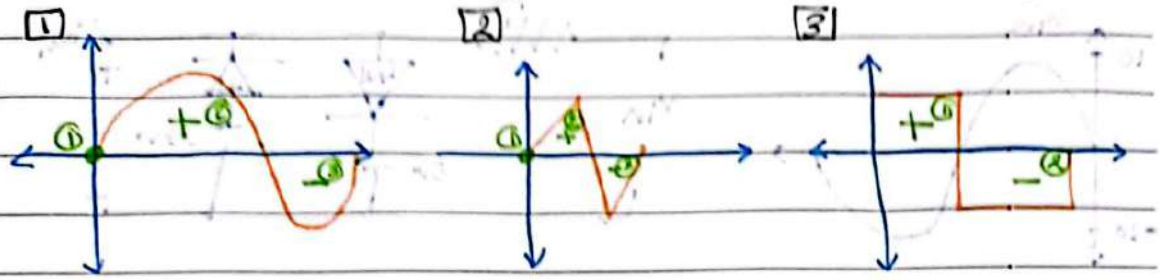
"طلاب من اجل الأفضل"

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

ch "2"

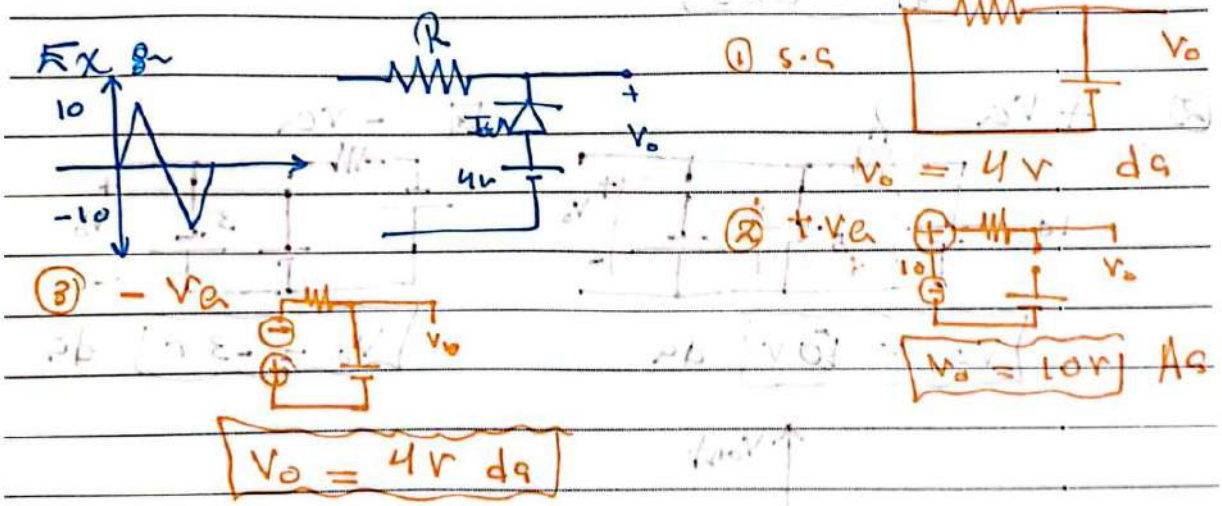
أنواع الموجات :-

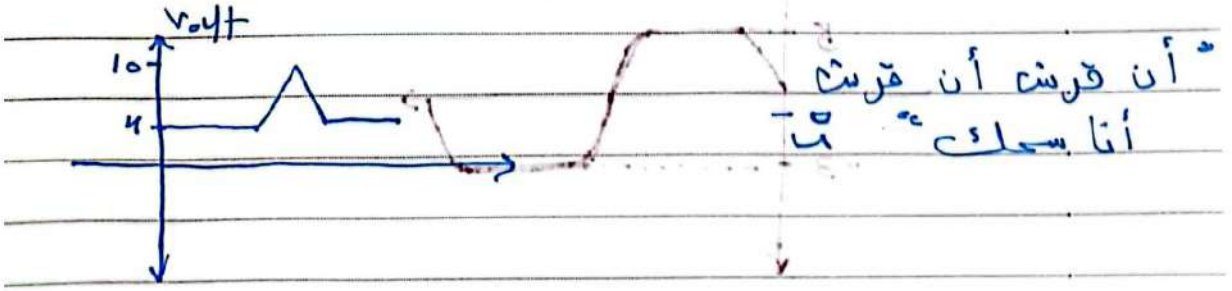


sunsoidal wave triangle wave square

* "الحل نفس في حالة sunsoidal و triangle ولكن الحل يختلف في حالة مختلف في حالة square باقي الأول "

* "موجة out تأخذ نفس شكل موجة inp"



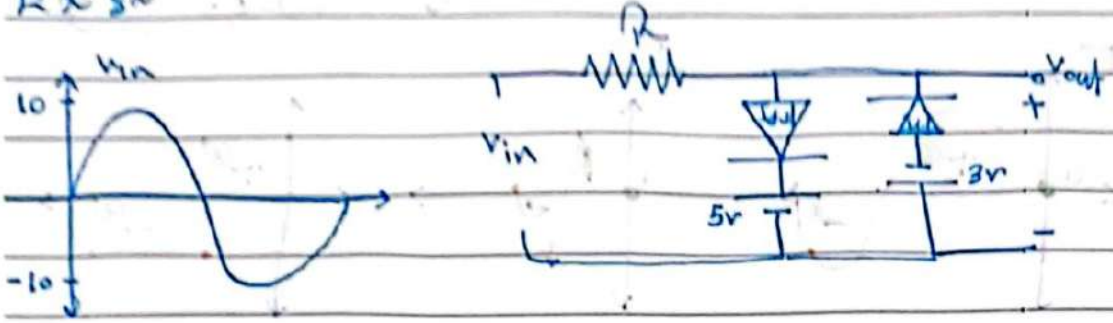


" أن قرش أن قرش
أنا سمك " ت

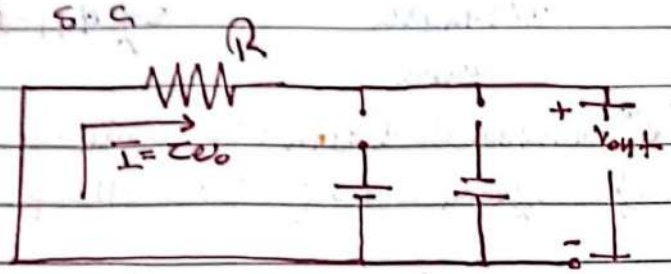
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

Ex 8~

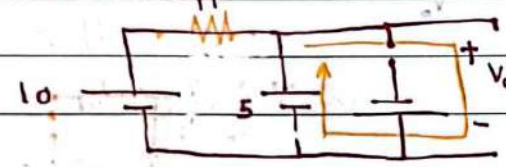


① 0.9



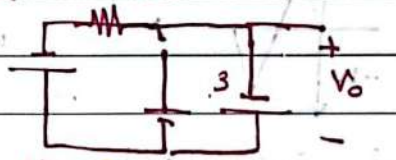
$V_{out} = Zero$

② $+V_a$

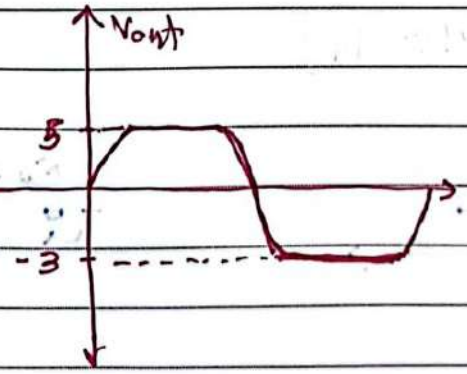


$V_o = 5\ V$ dq

③ $-V_a$



$V_o = -3\ V$ dq



يا حسرة عين ودماي حراحنا الحنيك

وعالم يلي بشاف وسمح حسو بحوجنا سوي

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

"اللهم اني أعوذ بك أن أكون قد اخطأت في هذا العمل عن جهل أو غفلة، فإن كان فيه خير فوفقني وان كان فيه زلل فأصلحني وآتني الصواب وتجاوز عني ان اخطأت
اللهم اني أعوذ بك من شيطان وشركه واستغفرك لكل خطأ وزلل"

"أتمنى منكم دعوة خالصة من قلوبكم لي تفرج بها كربي وتزيل همي"

* spectrum

* بسواعدنا - نبنيها
* هي - لله

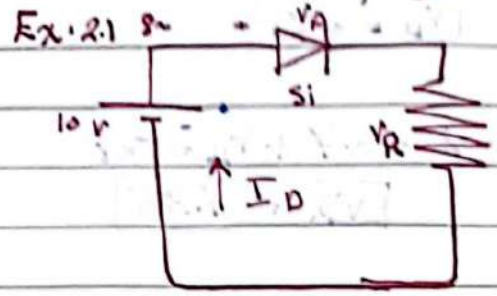
* خدمتكم - عبادة - تقرب - بها - الى الله

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ب CamScanner

حل الأمثلة الموجودة بسلايدات دكتور نابت "ch2"

Ex 2.1



Find The: V_R, I_D, V_R

$V_D = 0.7\ V$

$I_D = \frac{V_R}{R} =$

$-10 + 0.7 + V_R = Zero$

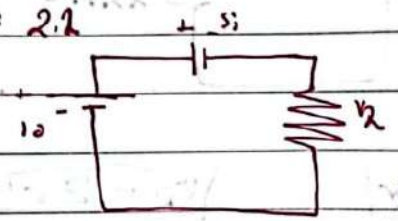
$V_R = 9.3\ v$

$I R = 9.3$

$I \cdot 0.5 \cdot 10^{3} = 9.3$

$I = 0.0186 \rightarrow 18.6\ mA$

Ex : 2.2



$V_D = 0.7\ v$

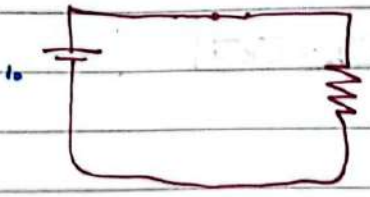
$-10 + 0.7 + V_R = Zero$

$V_R = 9.3\ v$

$I_D R = 9.3$

$I_D = \frac{9.3}{0.5\ k} \rightarrow 18.6\ mA$

Ex : 2.3 I dash



$V_D = Zero$

$-10 + 0 + V_R = Zero$

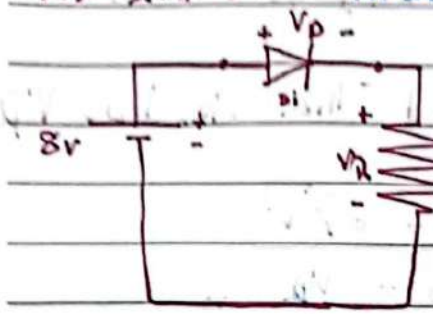
$V_R = 10\ V$

$I = \frac{10}{0.5 \cdot 10^{3}} = 20\ mA$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex 2.4 :- forward



$V_D = 0.7v$

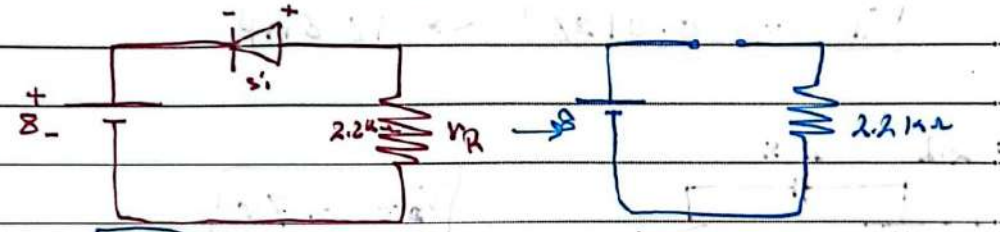
$-8 + 0.7 + V_R = 0$

$V_R = 7.3\ v$

$I_D R = 7.3$

$I_D = \frac{7.3}{2.2 \cdot 10^3} \rightsquigarrow 3.32\ mA$

Ex 2.5 :- reversed

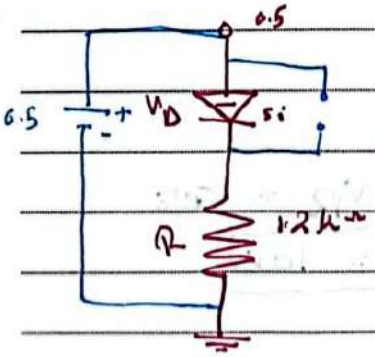


$I = 0$

$V_R = 0$

$V_D = 8v$

Ex 2.6 :-



$I = 0$

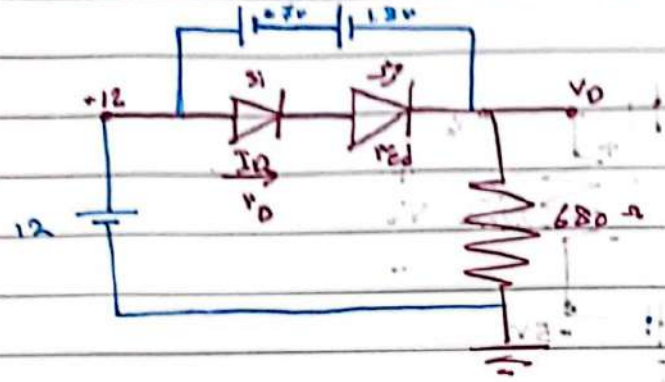
$V_D = 0.5v$

$V_R = 0$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex: 2.7 find the V_D, I_D



$-12 + 0.7 + 1.3 + V_R = 0$

$V_R = 9.5\,V$

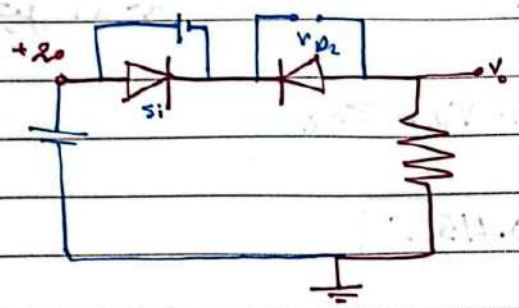
$I_R = 9.5\,V$

$I \cdot 680 = 9.5\,V$

$I = 13.9\,mA$

$V_o = V_R = 9.5\,V$

Ex: 2.8 , find I_D, V_{D_2}, V_o



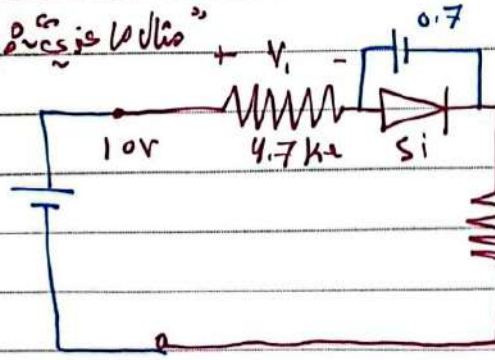
$-20 + 0.7 + V_{D_2} + V_R = 0$

$V_{D_2} = 19.3\,V$

$V_o = V_R = 0$

$I_D = 0$

Ex



$-10 + V_1 + 0.7 + V_2 = 0$

$I(4.7 + 2.2) = 9.3$

$I = 1.34\,mA$

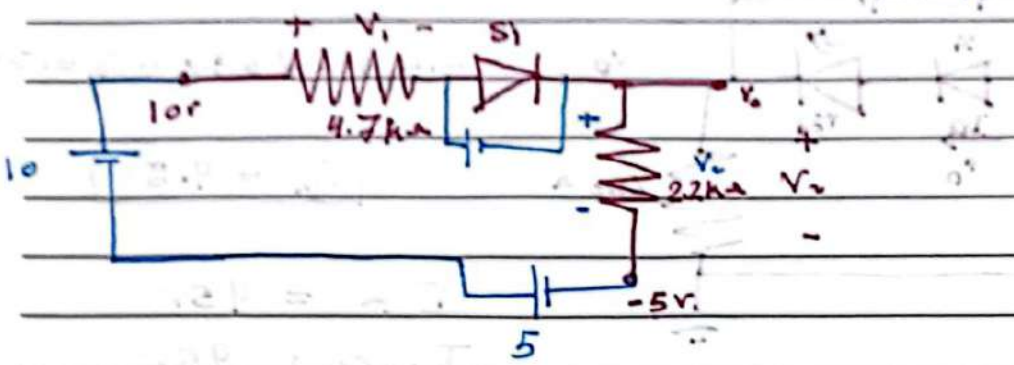
$V_1 = IR = 6.29\,V$

$V_o = V_2 = 2.95\,V$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex 2.9



$$-10 + I(4.7\cdot10^3 + 2.2\cdot10^3) + 0.7 - 5 = 2\text{eso}$$

$$\boxed{I = 2.07\ mH}$$

$$\boxed{V_1 = IR = 9.74\ V}$$ "الجزء آخر"

$$V_2 \rightsquigarrow \quad V_2 + 5 + V_R = 2\text{eso}$$

$$\boxed{V_2 = -0.45\ V}$$

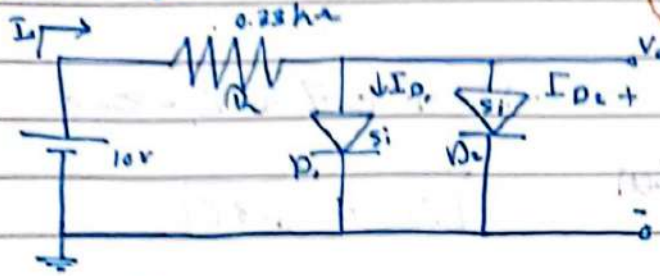
$$V_2 = 2.2\cdot10^3 \cdot 2.07\ m$$

$$\boxed{V_2 = 4.55\ V}$$

الممسوحة ضوئيا بـ CamScanner

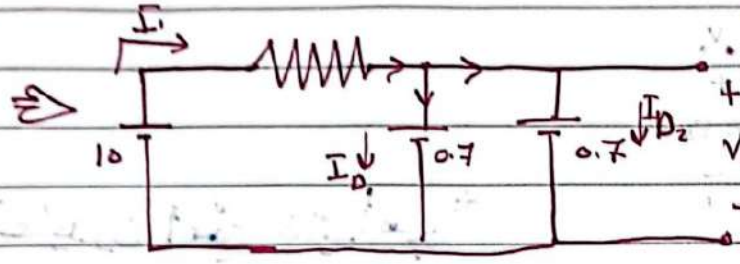
Scanned with CamScanner بـ CamScanner

Ex2-10



"اذا كان الديودان متطابقين تماماً" "بحيث لا يوجد فرق الجهد" "صفر في البار الكلي على جانبي الديودين"

$$I_D = \frac{I_{total}}{2}$$



$V_o = 0.7\,V$

$$-10 + IR + 0.7 = 0$$

$$I = \frac{10 - 0.7}{0.33 \cdot 10^3} \rightarrow \boxed{28.1\,mA}$$

$$I_1 = I_{D_1} + I_{D_2}$$

$$28.18 = 2\,I_{D_1} \rightarrow \boxed{I_{D_1} = 14.09\,mA}$$

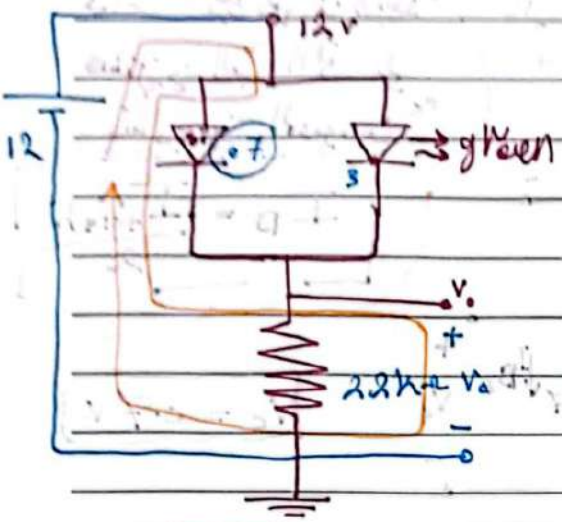
$$I_D = \frac{I_{total}}{2}$$

"و تنسى موعدك مع النوم في ميناء الليل"

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ليس عضواً CamScanner

Ex 8- 2.12



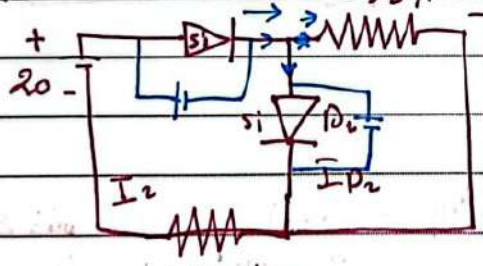
$-12 + 0.7 + V_o = 0$

$V_o = 11.3V$

"عايدون أمتي لانياسي"

note's :- اذا كان الديود ت موصلين على توازي لكن كل واحد منهم ديود مختلف ناخذ جهد الجنب الاغلى

Ex 8- 2.13



Find the I_1, I_2 I_D ?

$-20 + 0.7 + 0.7 + V_R = 0$

$V_R = 18.6V$

$I_R = 18.6V$

$I = \frac{18.6}{5.6 \cdot 10^3}$

$I_2 = 3.32mA$

$V_{R_1} = 0.7$

$I_R = 0.7$

$I_1 = 2.12 \rightarrow 0.212mA$

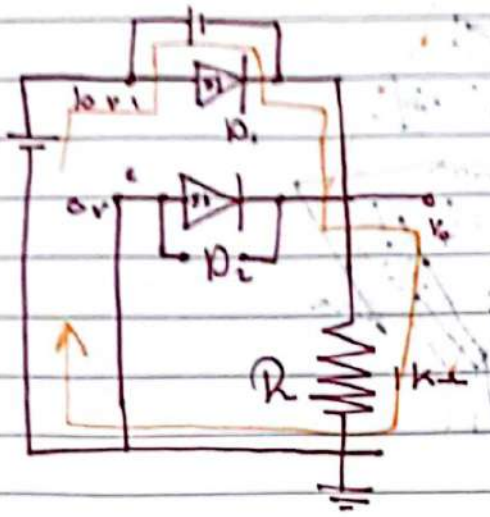
$I_2 = I_D + I_1$

$I_D = 3.11mA$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex 2.14 : Determin V_o



$-10 + 0.7 + V_o = 0$

$V_o = 9.3V$

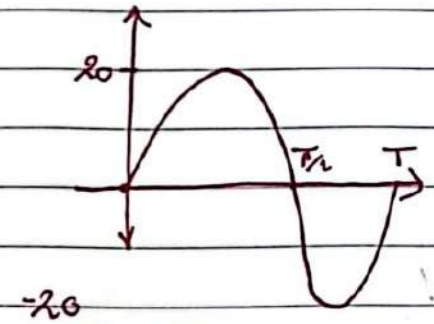
$I = \frac{V_o}{R}$

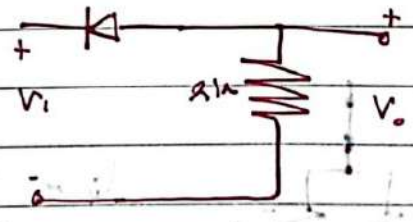
$I = \frac{9.3}{1k} \rightarrow 9.3mA$

Ex : 2.16 .

① sketch output v_o and Detemine The dc hevle of The output for The network

②





① $+v_e$

$V_o = 0$

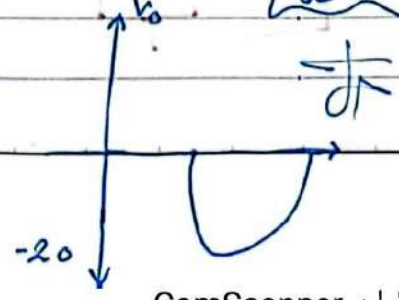
② $-v_e$

$-20 + V_R + 0 = 0$

$V_o = -20$

$V_{dc} = \frac{1}{T} V_{out}$

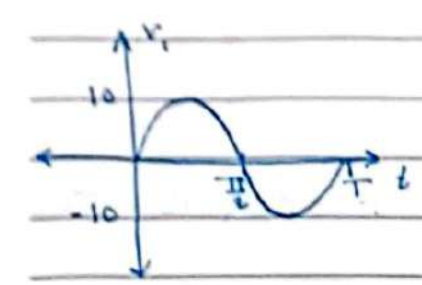
$V_{dc} = -6.36V$

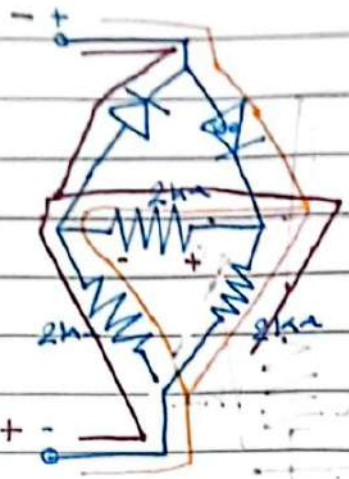


الممسوحة ضوئيا بـ CamScanner

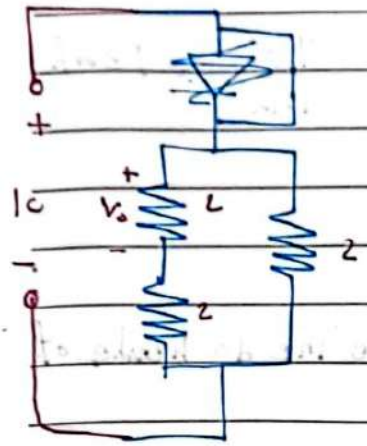
Scanned with CamScanner
الممسوحة ضوئيا بـ CamScanner

Ex 2.17 Sol: Find output waveform









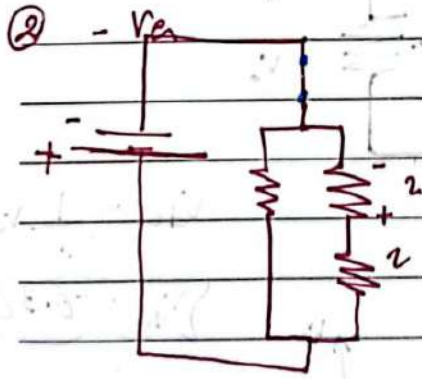
$V_o = IR$

$10 = I \cdot 4$

$I = \frac{10\ A}{4 \cdot 10^3}$

$V_o = I \cdot R$

$= \frac{10}{2 \cdot 4 \cdot 10^3} \cdot 4 \cdot 10^3 = \boxed{5\ V}$



$V_{ii} = IR$

$-10 = I \cdot -4$

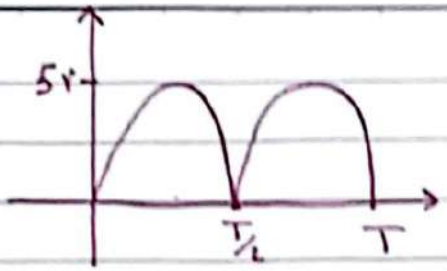
$I = \frac{10}{4 \cdot 10^3}$

$V_o = I R$

$= \frac{10}{2 \cdot 4 \cdot 10^3} \cdot 4 \cdot 10^3 = \boxed{5\ V}$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

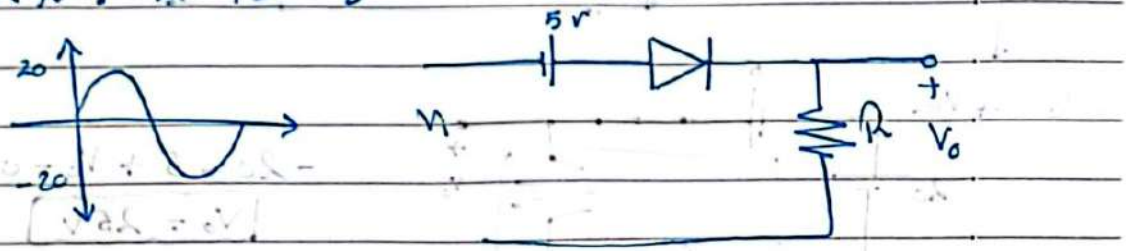


$$V_{dc} = \frac{2}{\pi} \cdot V_{max}$$

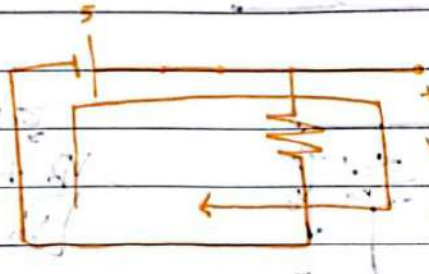
$$= 3.18\,V$$

قدي، ثم دمي، ثم، دمي، ثم دي فاستجاب
الله له اللهم هذا النجور لنا جميعا"

Ex : 2.18 P~



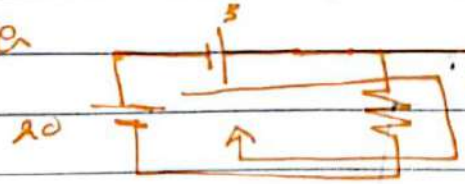
① S.C



$$-5 + V_o = zero$$

$$V_o = 5V$$

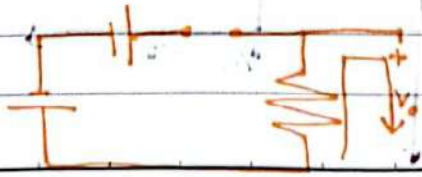
② + Ve



$$-5 - 20 + V_o = 0$$

$$V_o = 25V$$

③ - Ve



$$V_R + V_o = zero$$

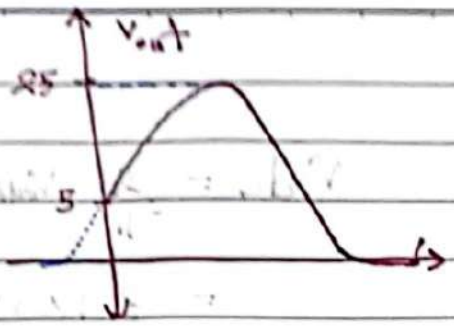
$$zero + V_o = 0$$

$$V_o = 0$$

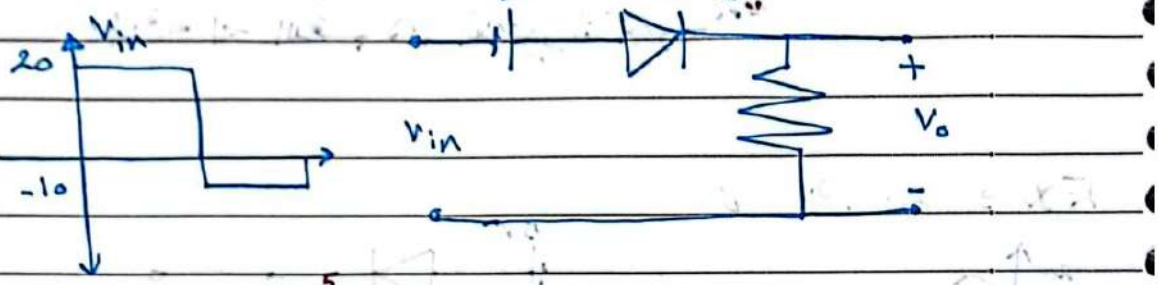
ينتج ⇒

الممسوحة ضوئيا بـ CamScanner

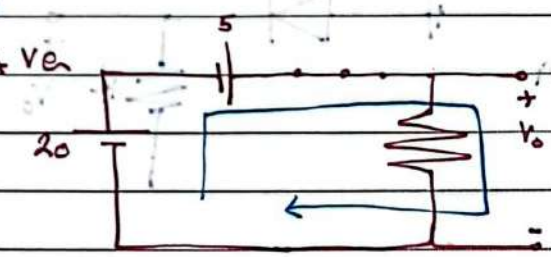
Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner



Ex 8~ 2.19 8-



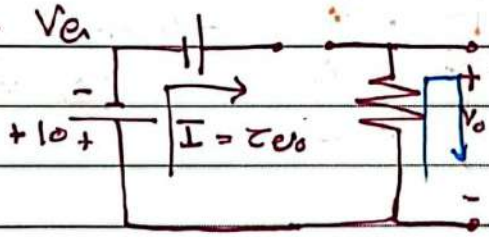
① $+V_{in}$



$-20 - 5 + V_o = 0$

$V_o = 25V$

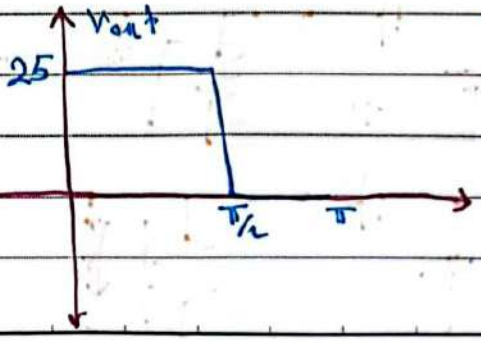
② $-V_{in}$



I_{ES}

$V_R + V_o = I_{ES}$

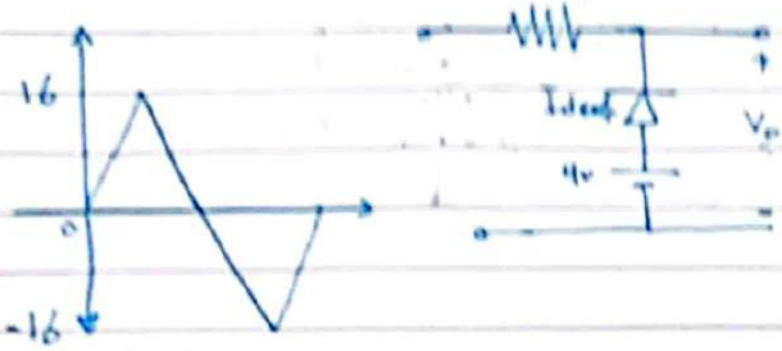
$V_o = I_{ES}$



الممسوحة ضوئيا بـ CamScanner

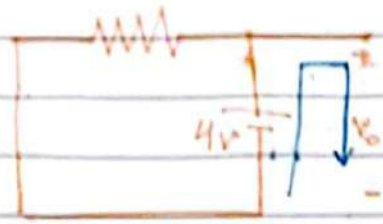
Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

ex 8 2.20 8-



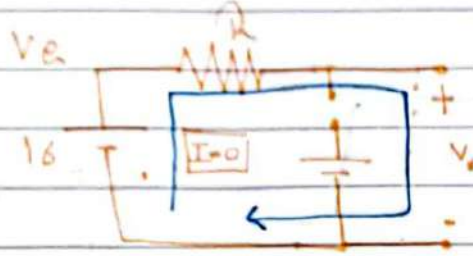
sol 8-

① D.C.



$V_o = 4V$ dc

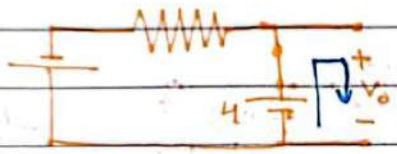
② + Ve



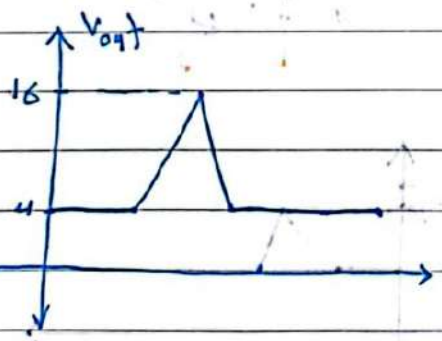
$-16 + V_R + V_o = 0$

$V_o = 16V$ Ac

③ - Ve



$V_o = 4V$ dc

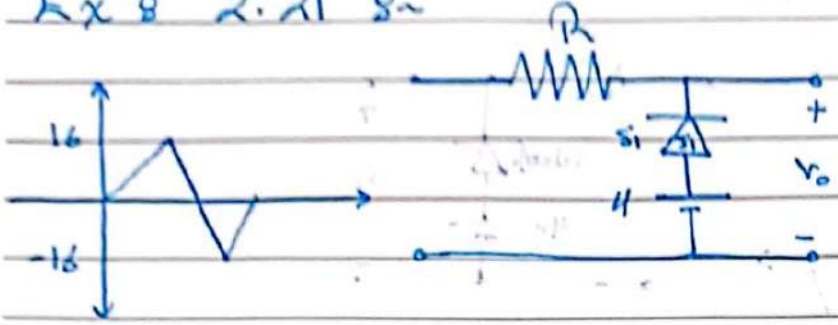


"كن محسنا"

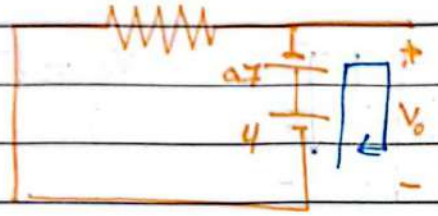
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex 2.21 :-



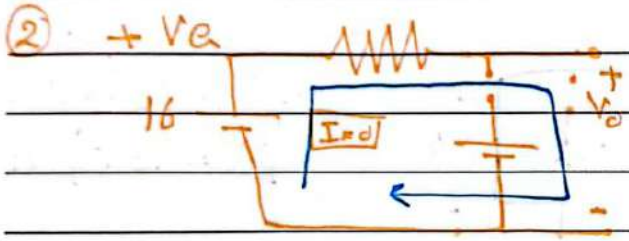
① S.C



$-4 + 0.7 + V_o = zero$

$V_o = 3.3\ V$ ds

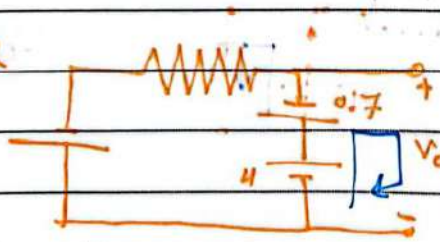
② $+V_c$



$-16 + V_R + V_o = zero$

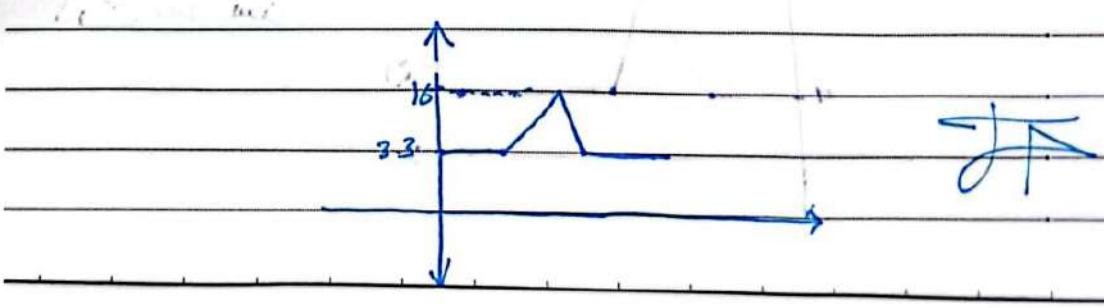
$V_o = 16\ V$ As

③ $-V_c$



$-4 + 0.7 + V_o = 0$

$V_o = 3.3\ V$ ds

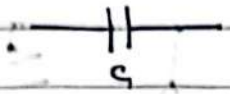


الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

clamper :-

2024/11/14

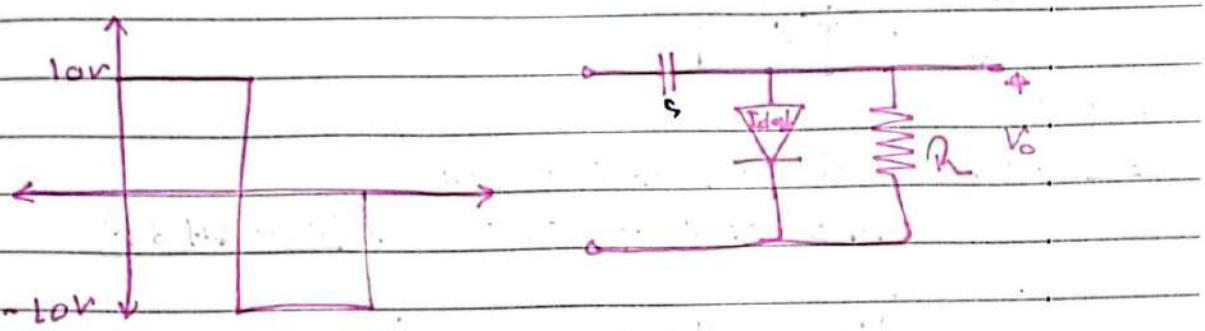


كيفية الحل عملية ؟؟ ① نجد الجزء الذي يجعل الدود "forward" حتى يوجد (V_c and V_o)

② نهمل الجزء المتبقي

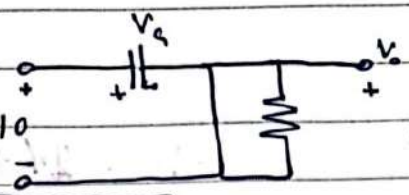
③ نوجد V_o في حالة الموجة الحقيقية والمثالية

Ex :- Find and Draw the output Figure :-



الحل :-

+Ve

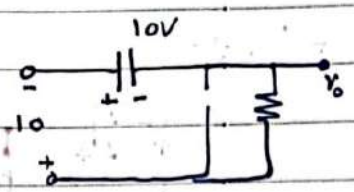


$V_o = 20V$

$-10 + V_c + \cdots - 20V$

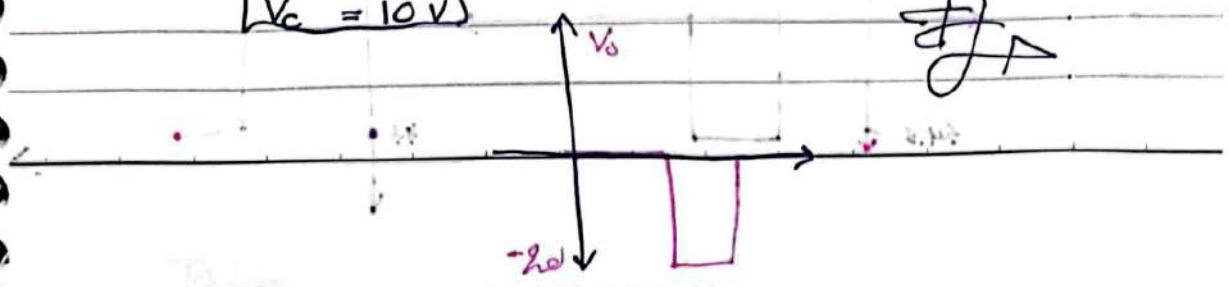
$V_c = 10V$

-Ve



$V_o + 10 + 10 = 20V$

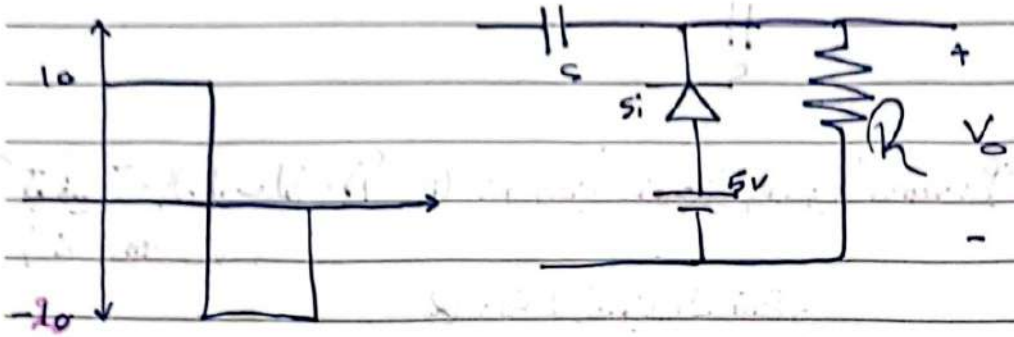
$V_o = -20V$



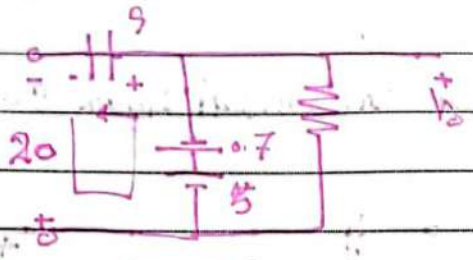
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex: Find and Draw the output signal



Sol/ sch.



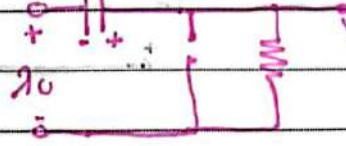
-ve sch.

$-20 + 5 + 0.7 + V_c = 20$

$+14.3 + V_c = 20 \rightarrow \boxed{V_c = 24.3\,V}$

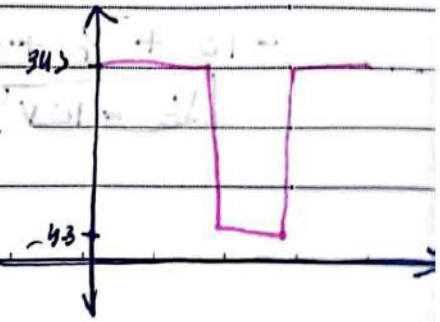
$\boxed{V_o = 4.3\,V}$

+ve sch.



$-10 - 24.3 + V_o = 20$

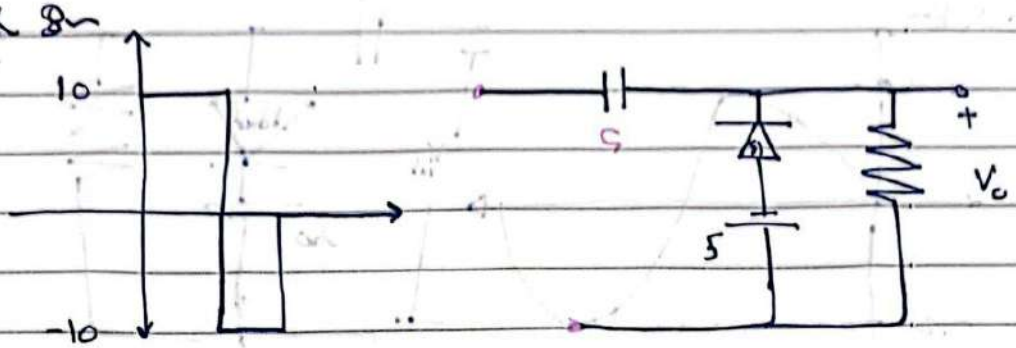
$\boxed{V_o = 34.3\,V}$



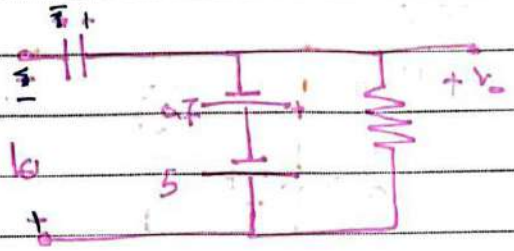
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex 8~



- Ve 8~

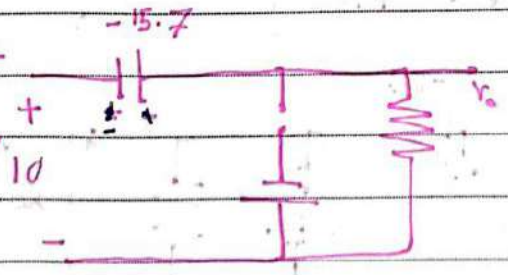


$-10 + 5 + 0.7 + V_c = 2e5$

$V_c = +4.3$

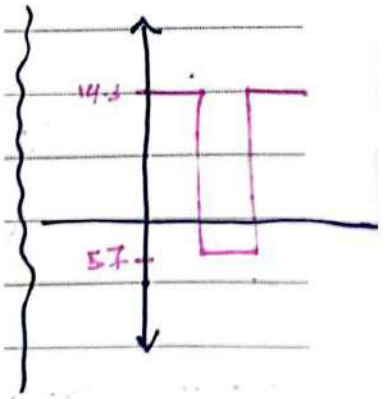
$V_o = -5.7$

+ Ve 8~



$-10 - 15.7 + V_o = 2e$

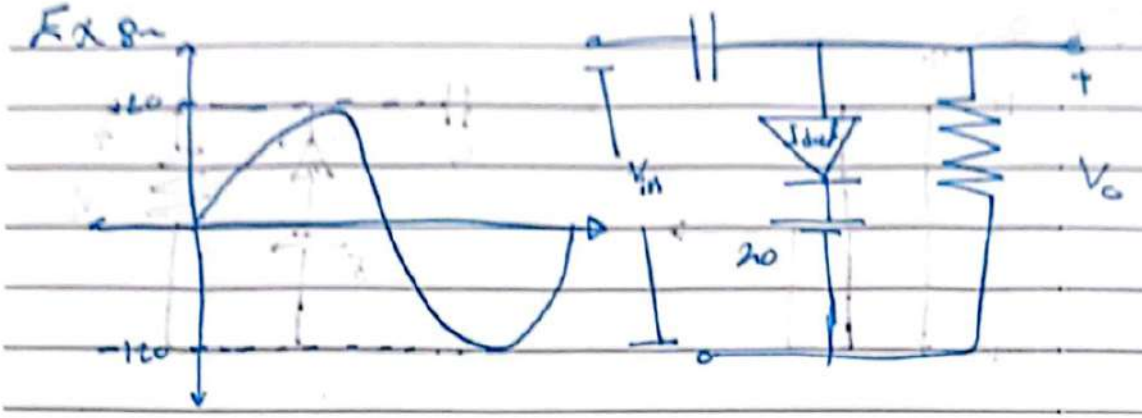
$V_o = +14.3$



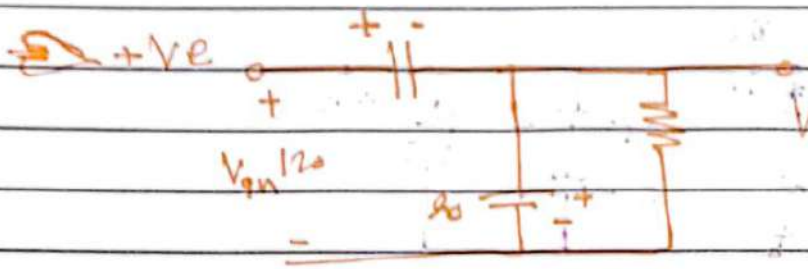
الممسوحة ضوئيا بـ mScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex 8-



+ve

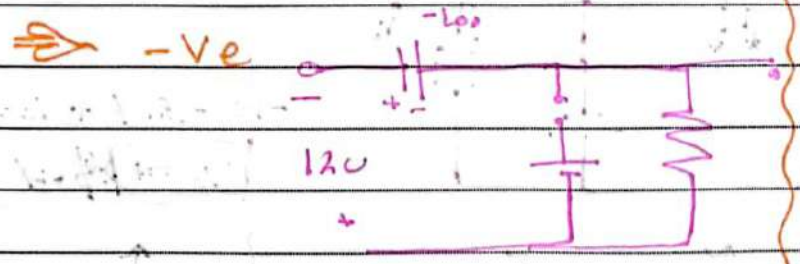


$V_o = 20V$

$-120 + V_c + 20 = 2e5_0$

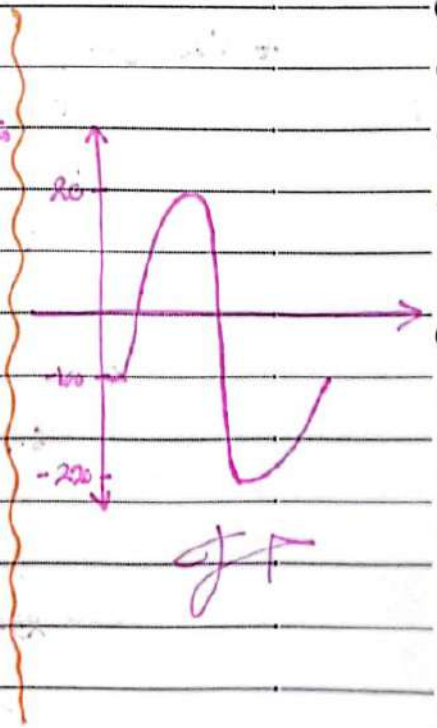
$V_c = +100$

-ve

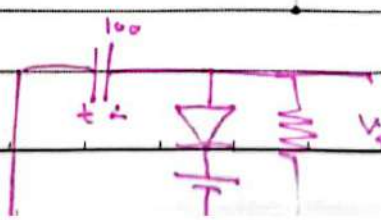


$V_o + 120 + 100 = 2e5_0$

$V_o = -220$



S.C



الممسوحة ضوئيا بـ CamScanner

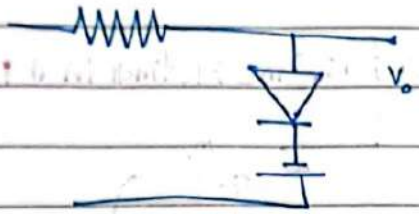
Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Design

1 clipper

* بسولعدنا - بنبيها
* spectrum

توازي



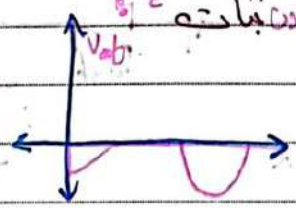
توالي



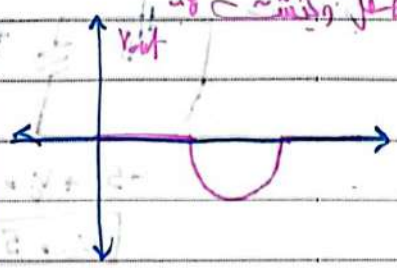
كيف اعرف الدائرة توالي ولا توازي ؟؟

توالي :-

1 اذا بلش موجة الاوت بقيمة الموجة اي فيه غير شغل دبنربة واذا الموجة بتطلع مباشرة "بدون ثبات"

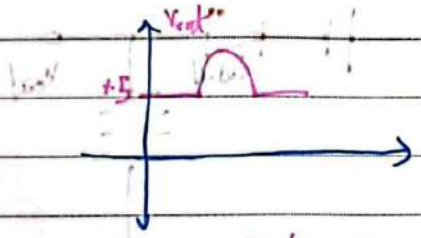


2 في حالات بدأن الموجة من الصفر وبتشتغ

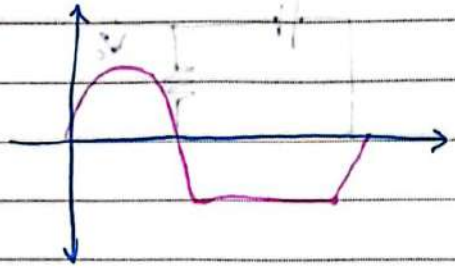


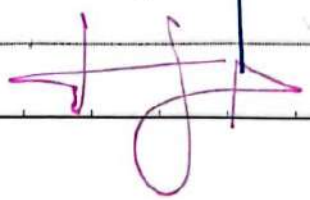
توازي :-

1 اذا بلش الموجة اي قيمة غير صفر وبتشتغ وبعدها بتغيرت



2 اذا الموجة بتطلع





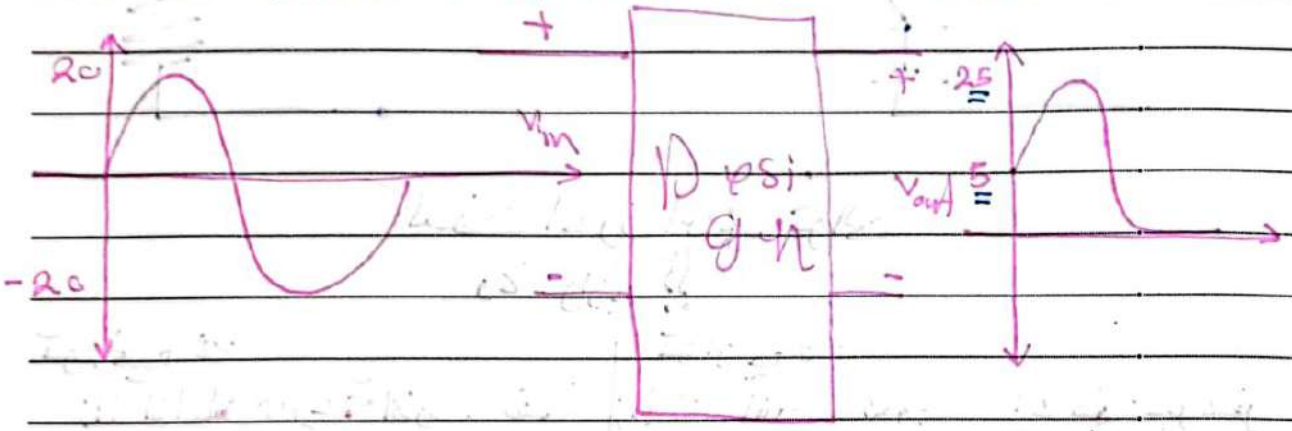
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

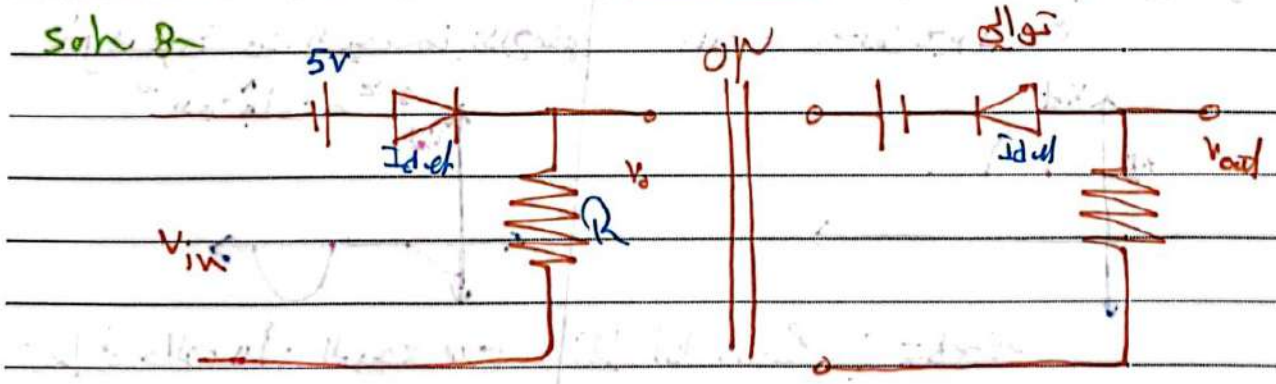
ملاحظة هامة:

① اذا كانت بداية الموجة فوق Diode reverse
② " " " " تحت Diode forward

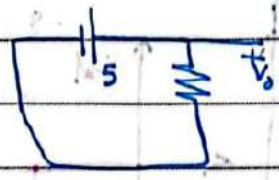
Ex: Design a clipper to perform the function in circuit

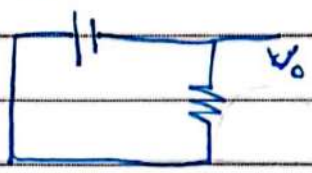


Sol:



S.C





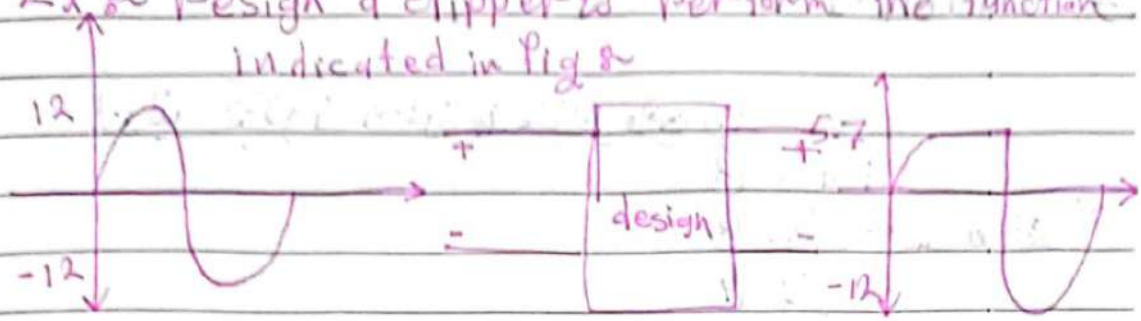
$-5 + v_o - 5$

$v_o = 5$ ✓

الممسوحة ضوئيا بـ CamScanner

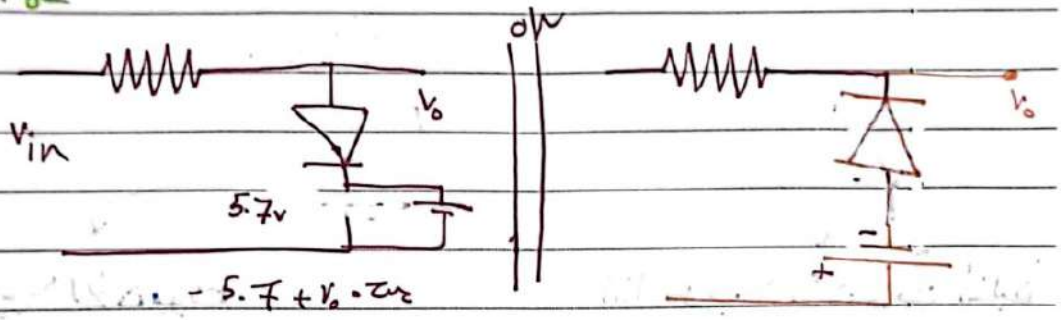
Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

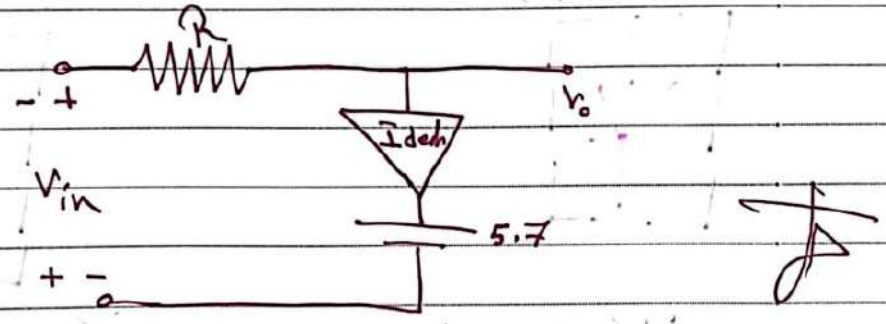
Ex 8~ Design a clipper to perform the function indicated in fig 8~



here, توازي

Sch 8~





من انت؟

- القوي الذي تورثت الحياة أمر
ايساميتي الباكية

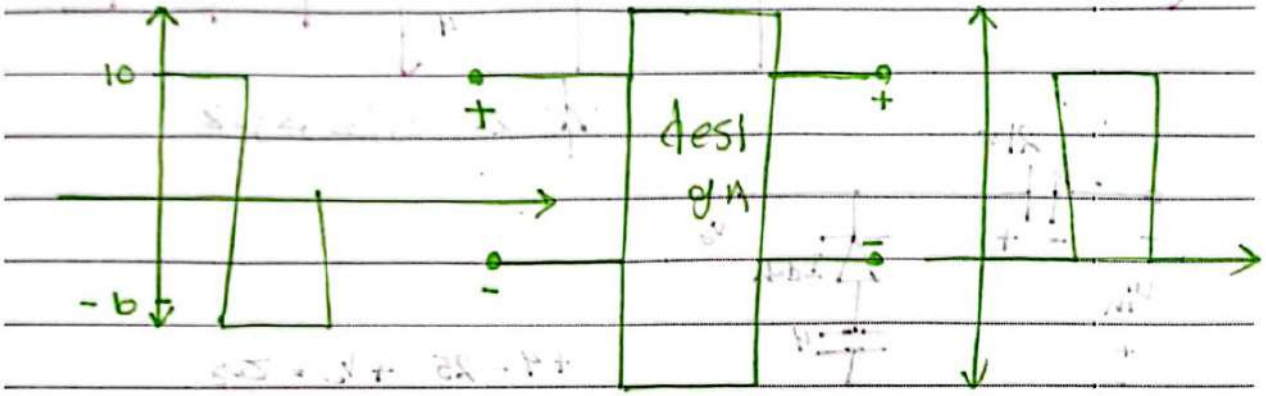
الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner

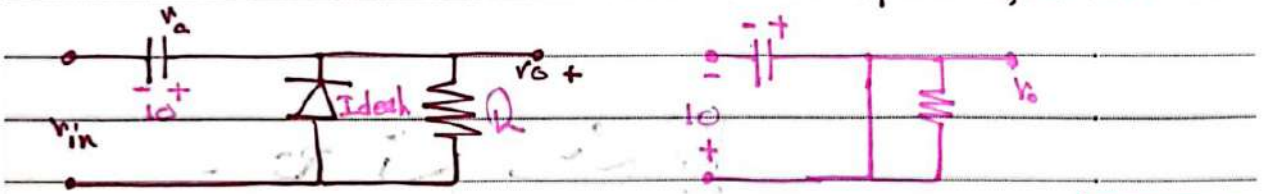
③ قيمة البطارية هي قيمة أقل قيمة من موجة الاوتبوت لما نوخذلها, طلت

* قبل الحواسح بطلعه من خلال الاوب

Ex:- Design a clamper to perform the function in fig 8~



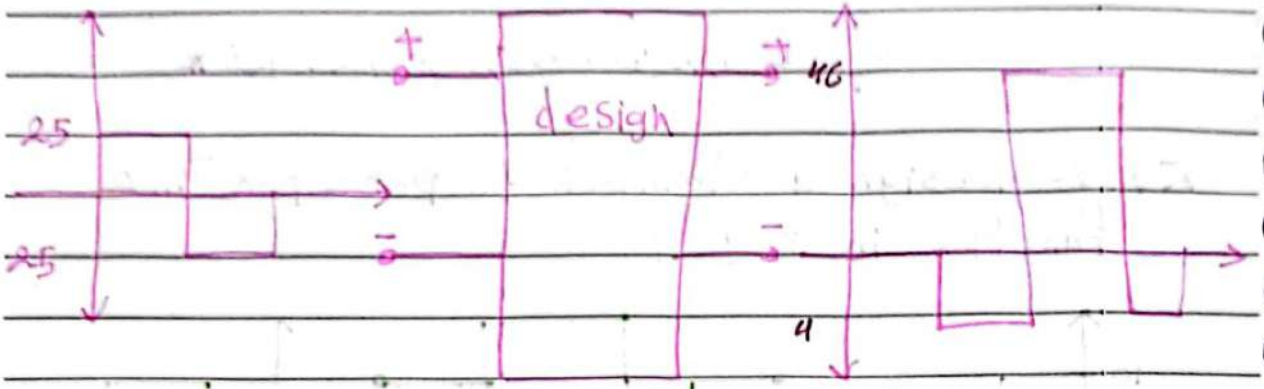
كايوجد بطارية



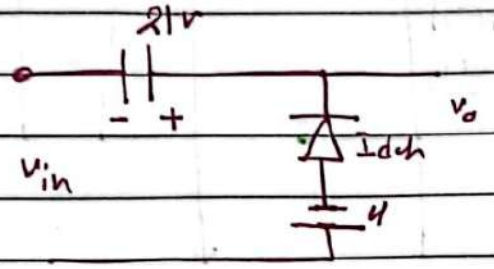
$-10 + V_c = 2V_0$ $\boxed{V_c = 10}$

الدعوة لوالدي بالرحمة

سؤال
Ex 8~ Design a clamper to perform the
function in fig 8~



بوجد بطارية ← +



$+4 - 25 + V_c = 0$

$V_c = 21$

- تعتبر ؟!
- كثيراً.

الممسوحة ضوئيا بـ CamScanner
Scanned with CamScanner بلغني تحويل

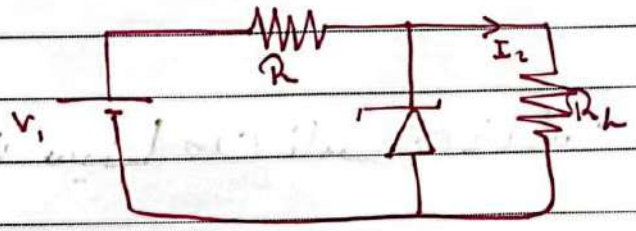
Zener diodes

- forward: يشتغل زيو زي الديود العادي
- reverse: يشتغل كـ منظم جهد

السؤال بالامتحان:

- I_z , V_z اوجد
  - 3 دوائر
  - حسب دفعتهم
- Design

[II] fixed V_i and fixed R_L



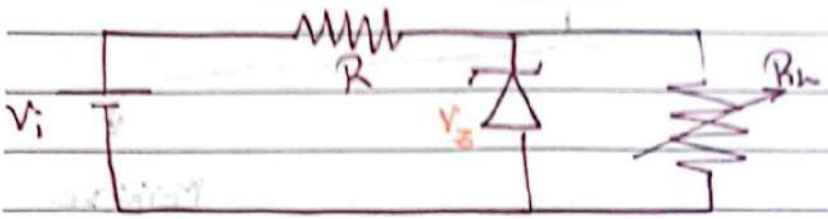
① open نحذف الزينر ونحسب V_{R_L}

- $V_{R_L} > V_z$ ، $V_{R_L} = V_z$
- $V_{R_L} < V_z$ open

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بيقى عندها

[2] fixed V_i and var R_L



* $R_{L\,min} = \frac{V_z \cdot R}{V_i - V_z}$

- $I_R = \frac{V_i - V_z}{R}$

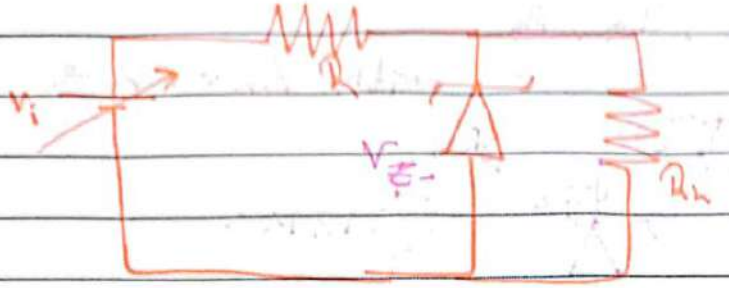
- $I_{L\,max} = I_R - I_{z\,min}$

* $R_{L\,max} = \frac{V_z}{I_{L\,max}}$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

[3] var V_i and fixed R_L



* $V_{in(min)} = \frac{V_Z (R + R_L)}{R_L}$

* $I_{R(max)} = I_{Zm} + I_{R_L}$

* $V_{in(max)} = (I_{R(max)} * R) + V_Z$

"أمي ولن ملأ الحشيب ~ عوارضي
سأ ذل حلماً عندها أترلك"

الممسوحة ضوئيا بـ CamScanner

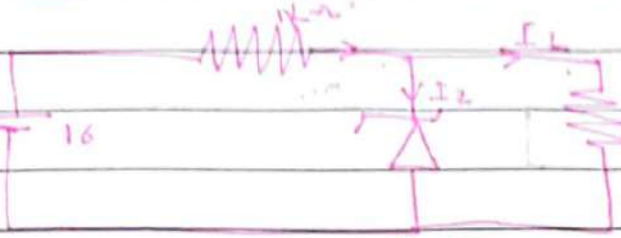
Scanned with CamScanner بـ CamScanner

Ex :-

find [1] V_R , V_{RL} , I_Z , P_Z

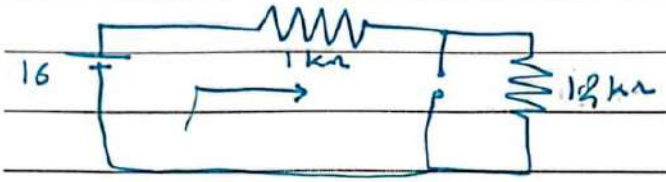
[2] " " " " , with $R_L = 3k\Omega$

P1)



Soln :- ① ① حالة

② D.C ← زينر



$$-16 + I(1+1.2)k = 0$$

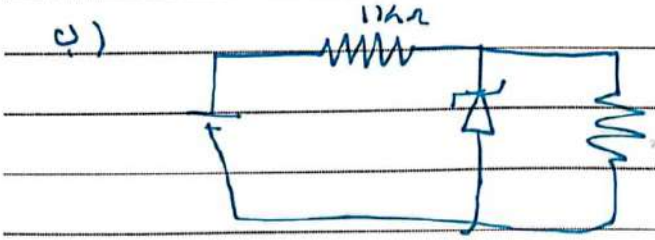
$$I = 7.27\,mA \quad , \quad \boxed{V_R = 8.72\,V}$$

$V_R < V_Z \rightarrow$ off

$\therefore$ $\boxed{V_{RL} = 8.72}$, $\boxed{V_R = 7.27\,V}$

$\boxed{I_Z = 0}$, $\boxed{P_Z = IV = 0}$

ب)



① ① حالة

② Test

$$I = \frac{16}{4k} \rightarrow 4mA$$

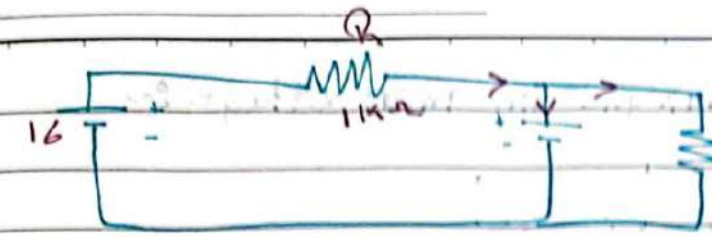
$V_{RL} = 12\,V$

$V_{RL} > V_Z$

$\therefore$ $\boxed{V_{RL} = V_Z = 10\,V}$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner



$V_L = I R$

$10 = 3 I$

$I = 3.33 mA$

$V_{RL} = 10V$

$I_L + I_Z = I_R$

$I_Z = 6 - I_L$

$= 6 - 3.33$

$I_Z = 2.667 mA$

$-16 + V_R + 10 = 0$

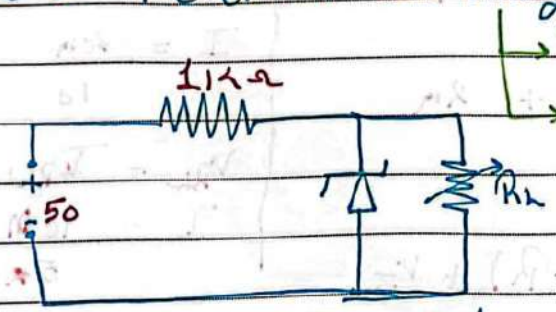
$V_R = 6V$

$I_R = 6mA$



* Spectrom ♡

Ex :- Determin the rang of R_L



$I_{Zm} = 32mA$

$V_Z = 10V$

$R_{L min} = \frac{V_Z \cdot R}{V_{in} - V_Z}$

$= \frac{10 \cdot 1 \cdot 10^3}{50 - 10}$

$R_{L min} = 250\Omega$

$R_{L(max)} = \frac{V_Z}{I_{L(min)}}$

$= \frac{10}{8m} = 1250\Omega$

$I_{R(R)} = \frac{50 - 10}{1 \cdot 10^3}$

$I_R = 0.04A$

$I_{L_{min}} = I_R - I_{Zm}$

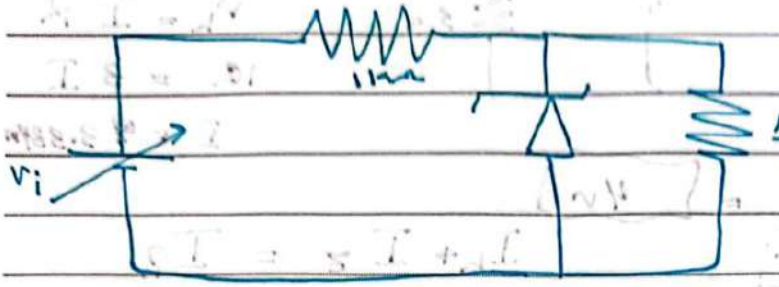
$= 0.04 - 0.032$

$8m$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

Ex :- A) Find the max and min input voltage :-



$V_Z = 10V$

$P_{Zm} = 30\ mW$

$V_{in(min)} = \dfrac{V_Z (R + R_L)}{R_L}$

$= \dfrac{10 \cdot (1k + 5k)}{5k} \quad \therefore = 12V$

$I_{R(max)} = I_{Zm} + I_{RL}$

$= 3m + 2m$

$= 5m$

$V_{max} = (I_{R(max)} \cdot R) + V_Z$

$= (5m \cdot 1k) + 10$

$= 15\ V$

$P_Z = I V$

$I = \dfrac{30m}{10} = 3m$

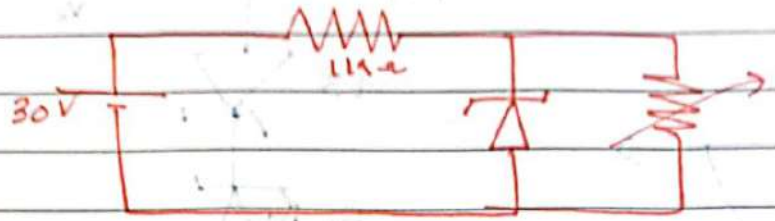
$V_{RL} = I_{RL} \cdot R$

$I = \dfrac{10}{5k} = 2m$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

[B] Find if $V_i = 30V$ The min and max. load "R"



$V_z = 10V$

$P_m = 30mW$

$R_{min} = \frac{V_z}{I_R}(R)$ ... $I_R = \frac{V_i - V_z}{R}$

$= \frac{V_i - V_z}{500\Omega}$ ... $= 20mA$

$I_{RL_{min}} = I_R - I_{Z_m}$

$= 20m - 3m$

$= 17mA$

$P = IV$

$I = \frac{30m}{10} \rightarrow 3m$

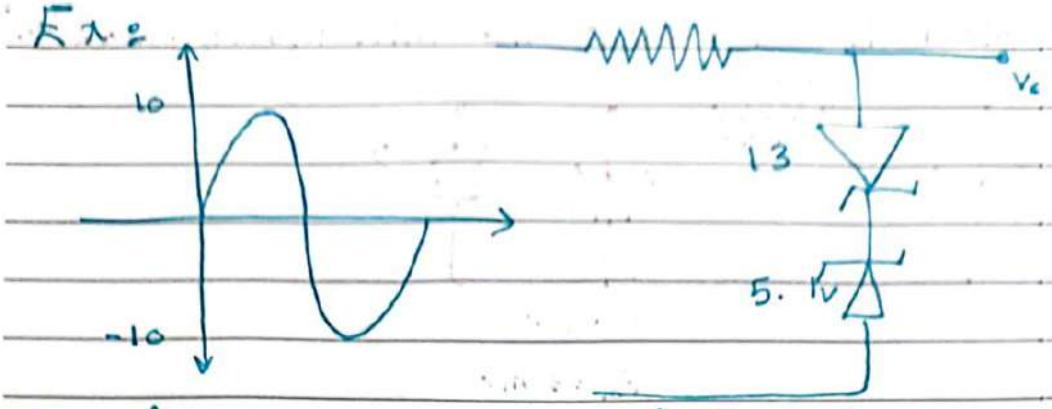
$R_{max} = \frac{V_z}{I_{min}}$

$= \frac{10}{17m} = 588.23\Omega$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner بـ CamScanner الممسوحة ضوئيا

Ex:



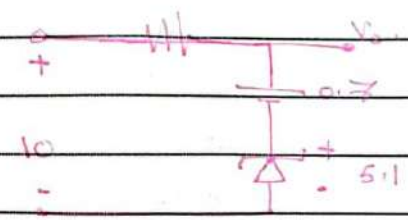
Ex: if the zener voltage of forward is (0.7 V)

Find ① V_o at +ve half cycle

② V_o at -ve " "

Sch 8~

① +ve

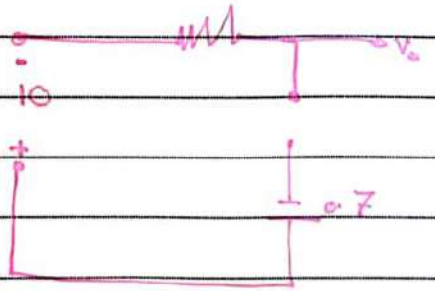


$10 > 5.1$

$-5.1 - 0.7 + V_o = 0$

$V_o = 5.8V$

②



$V_o = -10V$

الممسوحة ضوئيا بـ CamScanner

Scanned with CamScanner ممسوحة ضوئيا بـ CamScanner

الحمد لله هيك خلصنا بشبتر مادة الحيد
كاملة

هذا عمل مقدم من قربة spectrometer على والدي

اذا لفيت في هذا الملخص الخ خير ما تنسى
الدعاء لوالدي بالرحمة

اخوكم ♡

الممسوحة ضوئيا بـ CamScanner

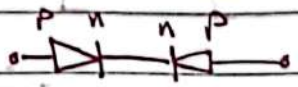
Scanned with CamScanner ب CamScanner

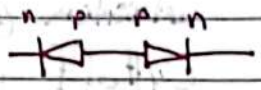
الفصل الثالث الترانزستور

Ch 3 :- Transistor

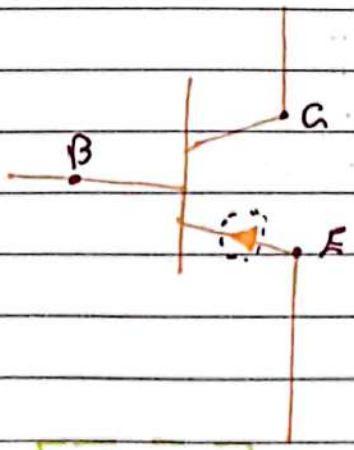
- BJT
- FET

BJT:
- (2) PNP
- (1) NPN

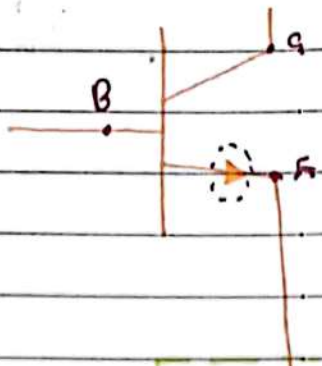




- (1) Emitter
- (2) Base
- (3) Collector



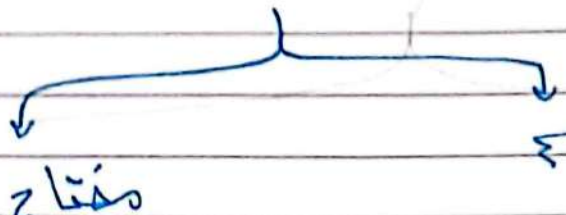
PNP



NPN

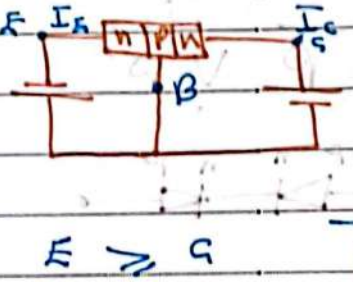
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تم المسح ضوئيا بـ CamScanner

Transistor application :



مناطق عمل الترانزستور

Active ①
saturation ②
Cut off ③



$E \geq C$

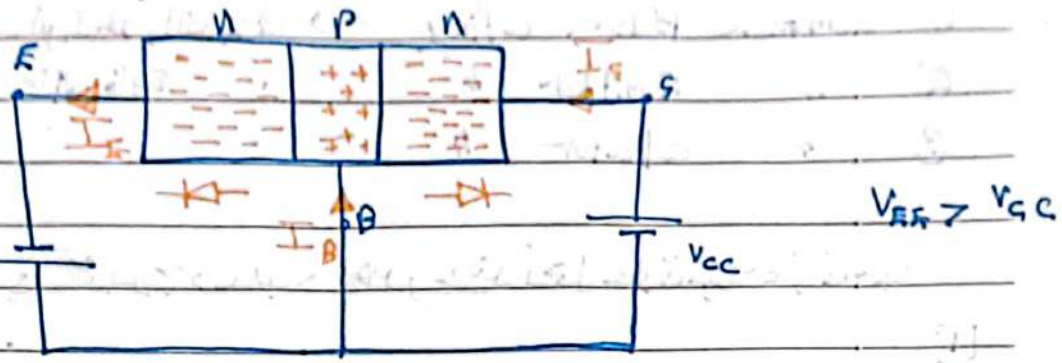
	Active	saturation	Cut off
BE	forward	forward	reverse
BC	reverse	forward	reverse

مضخم — مفتاح

"يا مقلب القلوب ثبت قلبي
على دينك وطاعتك"

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تم المسح ضوئيا بـ CamScanner

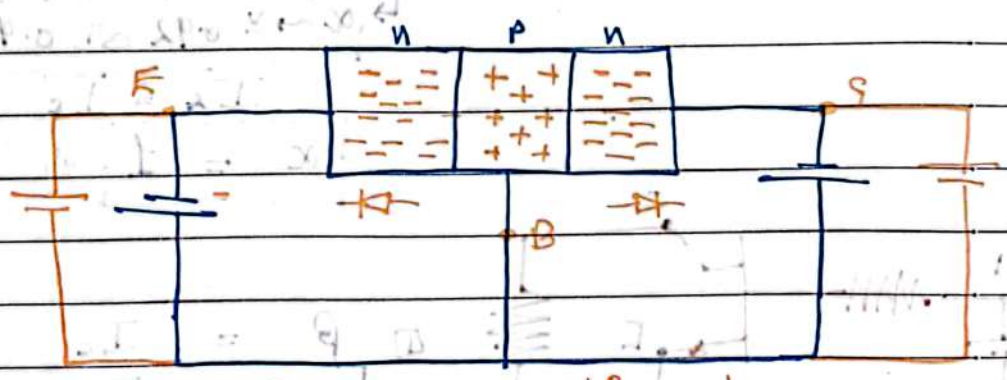
[1] Active region :-



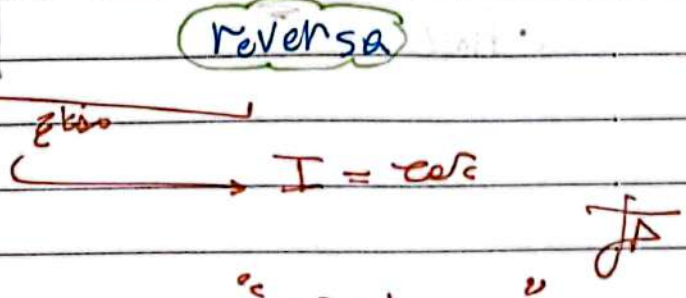
$I_E = I_B + I_C$; $\alpha \rightarrow 10^{-6}$, $\beta \rightarrow 10^{-3}$

$I_E \cong I_C$

[2] saturation region :-



[3] catt off — Fever — reversa



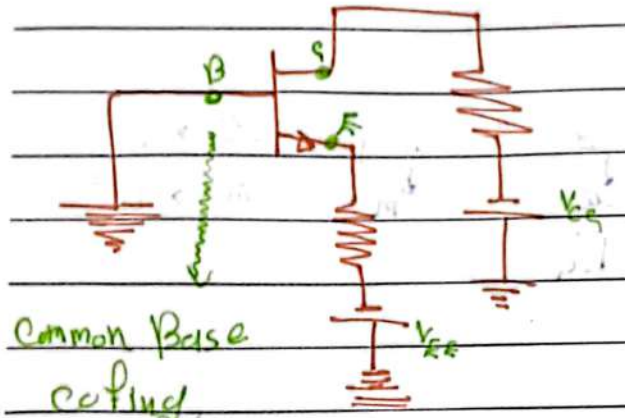
"spectrum"

التوصيلات هي:

① Common Base coling ← ③ يكبر الجهد والقدرة
② " Emitter " ← يكبر التيار فقط
③ " Collector "

الخط الذي يوجد عليه مصدر جهد نعمل على تسميته على نسمي الخط

[1]



Common Base coling

$$[1]\ \alpha = \frac{I_C}{I_E}$$

$$I_E = I_B + I_C$$

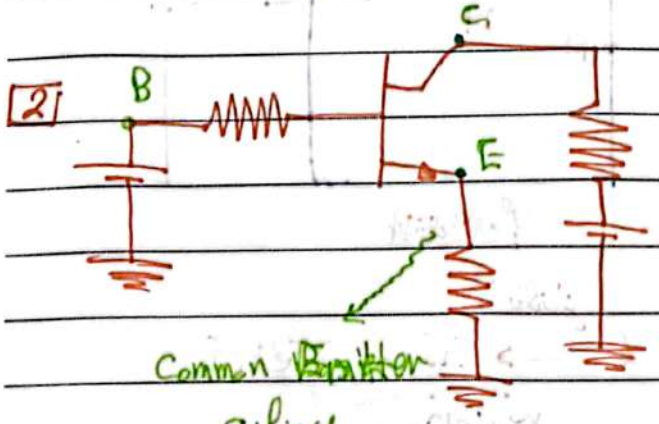
$$I_E \cong I_C$$

$$\alpha \sim \cong 0.92 \text{ إلى } 0.99$$

$$I_E = I_C$$

$$\alpha = 1$$

[2]



Common Emitter coling

$$[1]\ \beta = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$\Rightarrow 50 \leq \beta \leq 400$$

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تم المسح ضوئيا بـ CamScanner

3 Common collector

1 X

2 بكرة التيار

العلاقة بين α و β

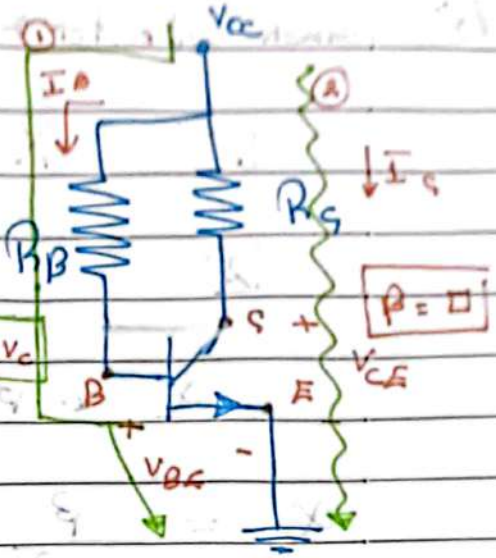
$$\alpha = \frac{\beta}{1+\beta} \quad , \quad \beta = \frac{\alpha}{1+\alpha}$$

كل ما تم شرحه اذا اللان لا يفيدني الامتحان
فقط للعلم ومعرفة ان الحالة الطبيعية مايحتي بالتفاصيل

"الجاهد"

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تم المسح ضوئيا بـ CamScanner

[1] fixed Bias :-



notes:-

[1] $V_{CE} = V_C - V_E \rightsquigarrow V_{CE} = V_C$

$V_{BE} = V_B - V_E \rightsquigarrow V_{BE} = V_B$

$V_{BC} = V_B - V_C \rightsquigarrow V_{BC} = V_B - V_C$

[2] $V_E = 0$

[3]

Si $\rightarrow V_{BE} \rightarrow 0.7$

Ge $\rightarrow V_{BE} \rightarrow 0.3$

خطوات حل ؟

I_B [1]

I_C [2]

V_{CE} [3]

[4] $I_{C(sat)} =$ يخلي V_{CE} تصفر

loop (1) "kvl"

$-V_{cc} + I_B R_B + V_{BE} = 0$ (0.7)

$$I = \frac{V_{cc} - 0.7}{R_B}$$

$$I_C = \beta I_B$$

loop (2) "kvl"

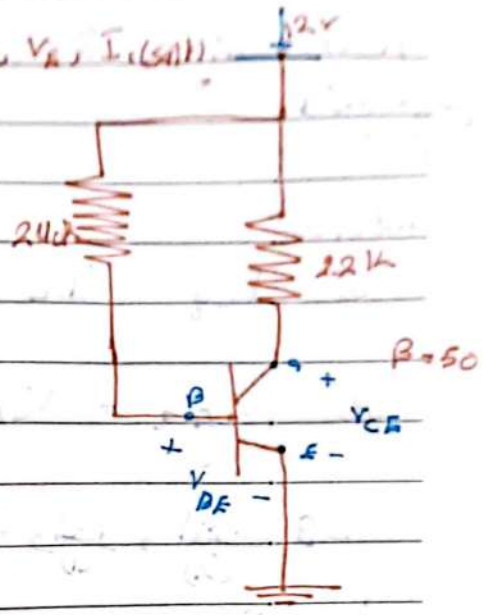
$-V_{cc} + I_C R_C + V_{CE} = 0$

$$V_{CE} = -I_C R_C + V_{cc}$$

الله يحفظني ويسرلي أموري

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تم المسح ضوئيا بـ CamScanner

Ex2 find I_B, I_C, V_{CE}, V_B, V_C, V_E, $I_{C(sat)}$.



[1] $-12 + 240k \cdot I_B + V_{BE} = 0$ (0.7, zero)

$$I_B = \frac{12 - 0.7}{240k}$$

$$I_B = 47.08\,\mu A$$

[2] $I_C = \beta I_B$

$I_C = (50)(47.08\mu)$

$I_C = 2.35\,m$

[3] $-12 + 2.2(2.35m) + V_{CE} = 0$ V_{CE}

$V_{CE} = 12 - (2.2)(2.35m) = 6.83V$

[4] $V_{BE} = V_B \rightarrow V_B = 0.7V$

[5] $V_{CE} = V_C \rightarrow V_C = 6.83V$

[6] $V_E = 0$ (zero)

[7] $I_{C(sat)} \Rightarrow$ V_{CE} (zero)

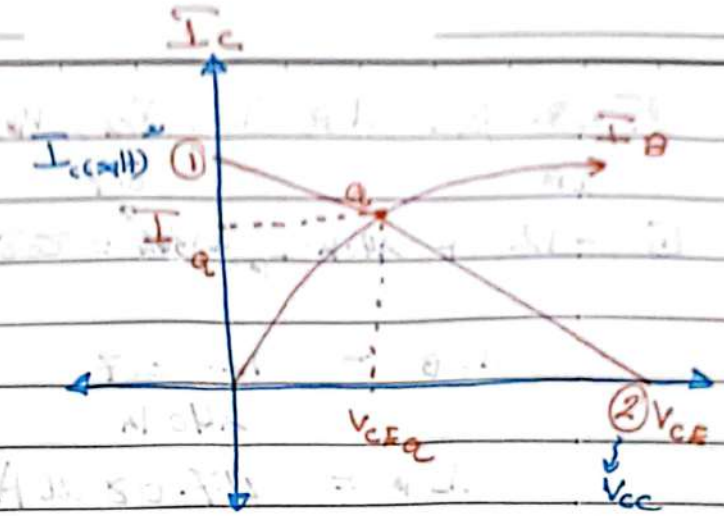
$\Rightarrow -V_{CC} + I_B R_C + V_{CE} = 0$

$$I_C = \frac{V_{CC}}{R_C} = \frac{12}{2.2k} \Rightarrow 5454\,\mu A$$

NEVER Give UP

spectrum.

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CamScanner بلغني تعرف



hadoj hints

notes:

① $I_{CQ} = I_C$

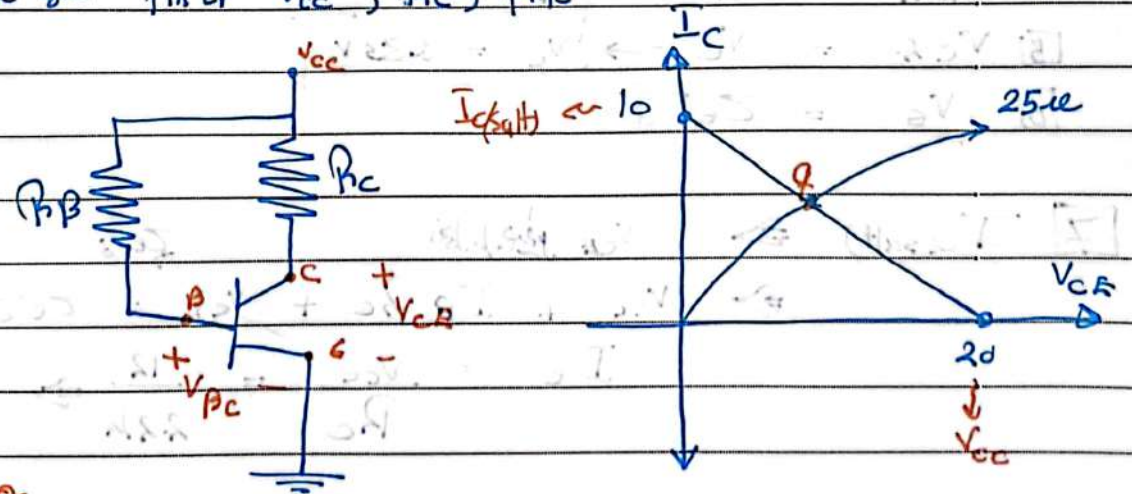
② $V_{CEQ} = V_{CE}$

يحادفي واحد و انتين من (1) (2)

نجيب ① بدل $V_{CE}=0$ نحل على جدل محور x

نجيب ② بدل $I_C=0$ نحل على جدل محور y

Ex: find V_{cc}, R_C, R_B



sol:

① $V_{cc} = 20\ V$

② $-20 + R_B(25u) + 0.7 = 0$

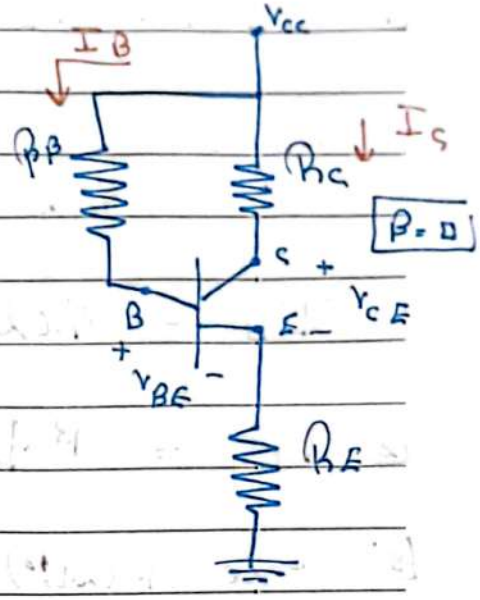
$R_B = 772\ K\Omega$

Scanned with CamScanner

[3] $-20 + R_C(10.10^{-3}) = 2\text{volt}$

$$R_C = 2\,k\Omega$$

[2] self Bias



not 3

[1] $V_E \neq 2\text{volt}$

[2] $I_E = I_C$

خطوات الحل

I_B [1]

I_C [2]

V_{CE} [3]

"KVL"

I_B

$$-V_{CC} + I_B R_B + V_{BE} + V_E = 2\text{volt}$$

$$-V_{CC} + I_B R_B + 0.7 + R_E \cdot \beta I_B = 2\text{volt}$$

$$I_B = \frac{V_{CC} - 0.7}{R_B + \beta R_E}$$

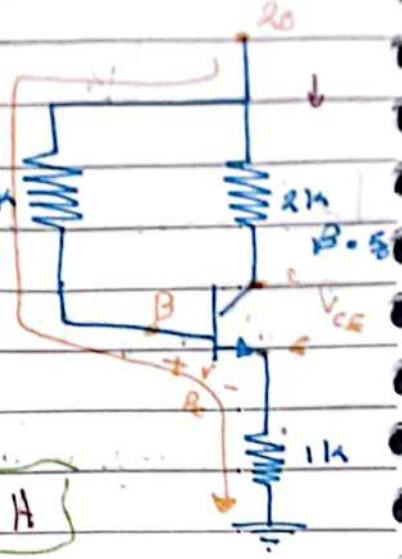
$$I_C = \beta I_B$$

$$-V_{CC} + I_C R_C + V_{CE} + R_E I_E = 2\text{volt}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

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Find The $I_B, I_C, V_{CE}, V_B, V_E$ Saddle



I_B : ?

$\beta . I_B$

① $-20 + 430.10^3 I_B + V_{BE} + 1.10^3 I_E = 0$ (0.7)

$$I_B = \frac{20 - 0.7}{430.10^3 + 10^3 . 50}$$

$I_B = 4.02 . 10^{-5} \rightsquigarrow$ 40.21 μA

② $I_C = \beta . I_B \rightsquigarrow$ 2.01 A

③ $-20 + (2.10^{+3})(I_C) + V_{CE} + 1 I_E = 0$ ($I_C = I_E$)

$V_{CE} = 20 - (2.10^{-3})(2.01) + 1(2.01) = 0$

$V_{CE} = 13.98$ V

$V_{BE} = V_B - V_E$

$0.7 = 2.714 - V_E$

$V_E = 2.01$ V

$-20 + (430)(10^3)(40.2.10^{-6}) + V_B = 0$

$V_B = 2.714$

$V_{CE} = V_C - V_E$

$13.98 = V_C - 2.01$

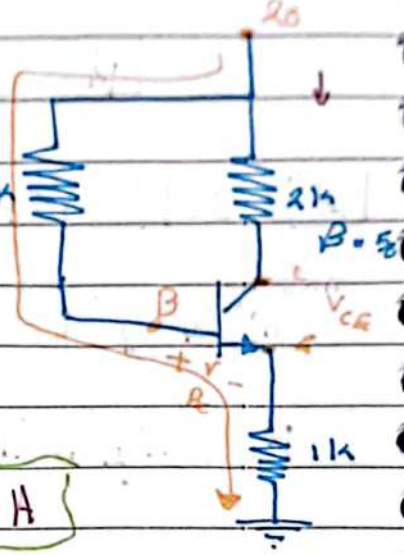
$V_C = 15.99$ V

$-20 + 2.10^3 . I_C + 10^3 I_C = 0$

$I_{C\,sat} = 6.66$ mA

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CamScanner بلطف تحويل

Find The $I_B, I_C, V_{CE}, V_B, V_E$ Sadly



I_B : ?

βI_B

① $-20 + 430 \cdot 10^3 I_B + V_{BE} + 1 \cdot 10^3 I_C = 0$ (0.7)

$$I_B = \frac{20 - 0.7}{430 \cdot 10^3 + 10^3 \cdot 50}$$

$I_B = 4.02 \cdot 10^{-5} \rightarrow$ 40.21 μA

② $I_C = \beta \cdot I_B \rightarrow$ 2.01 A

③ $-20 + (2 \cdot 10^3)(I_C) + V_{CE} + 1 I_E = 0$

$V_{CE} = 20 - (2 \cdot 10^{-3})(2.01) - 1(2.01) = 0$

$V_{CE} = 13.98$ V

$V_{BE} = V_B - V_E$ (0.7) $= 2.714 - V_E$

$V_E = 2.01$ V

$-20 + (430)(10^3)(40.2 \cdot 10^{-6}) + V_B = 0$

$V_B = 2.714$

$V_{CE} = V_C - V_E$

$13.98 = V_C - 2.01$

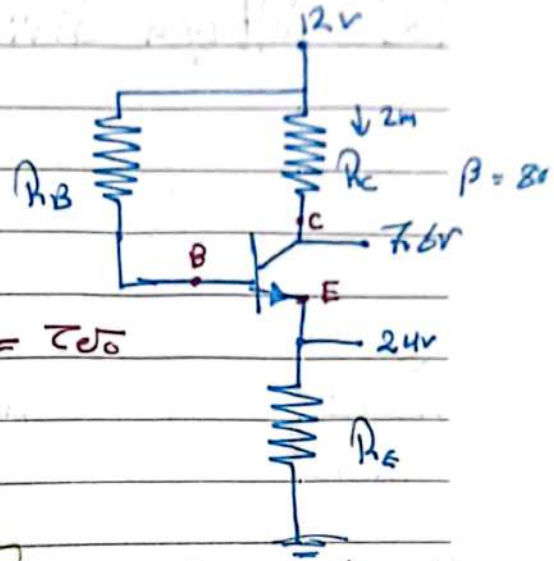
$V_C = 15.99$ V

$-20 + 2 \cdot 10^3 I_C + 10^3 I_C = 0$

$I_{C\,Sadly} = 6.66$ mA

Scanned with CamScanner

Ex: find the V_B, R_C, R_E, R_B



$V_{CE} = V_C - V_E$

$= 7.6 - 2.4$

$V_{CE} = 5.2\,V$

$-12 + 2\cdot10^{-3} R_C + 7.6 = 2\sigma_0$

$R_C = \frac{12 - 7.6}{2\cdot10^{-3}}$

$R_C = 2.2\,k\Omega$

$V_{BE} = V_B - V_E$

$0.7 = V_B - 2.4$

$V_B = 3.1\,V$

$I_C = \beta I_B$

$2\cdot10^{-3} = 80 \cdot I_B \rightsquigarrow I_B = 25\mu A$

$-12 + 25\mu \cdot R_B + 3.1 = 2\sigma_0$ $\quad V_E = 1$

$R_B = 356\,k\Omega$

$I_C = 2mA$

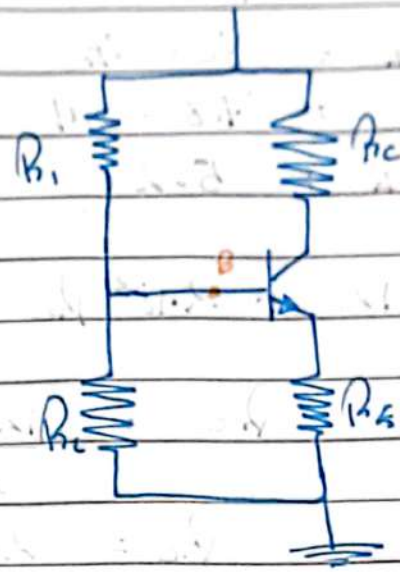
$-12 + (2\cdot10^{-3})(2.2k) + 5.2 + R_E I_E = 2\sigma_0$

$R_E = 1.2\,k\Omega$

"لا تفرح بسعة غيرك،
فإنك لا تدري ما يجدث
الزمان بك"

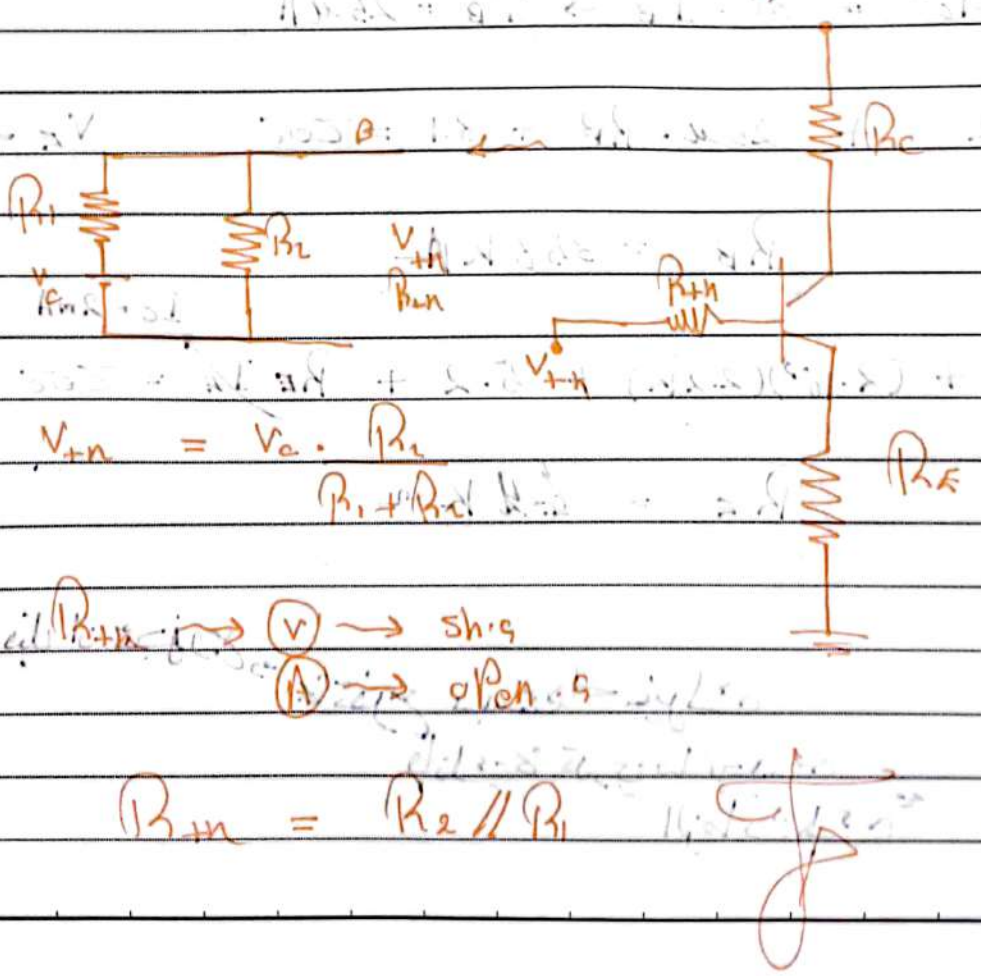
Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

[3] Voltage divider



Exact method

Approximate method



$$V_{th} = V_{cc} \cdot \frac{R_2}{R_1 + R_2}$$

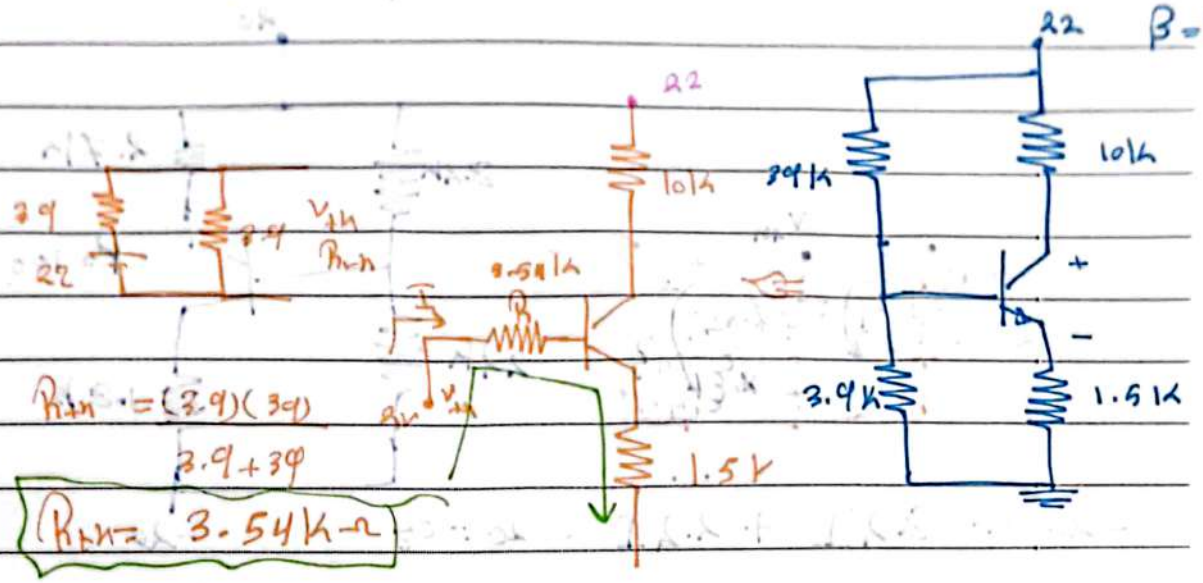
R_{th} → (V) → short

(A) → open

$$R_{th} = R_2 // R_1$$

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CamScanner بالماسح الضوئي

Ex:- find I_B, I_C, V_{CE}, $I_{C(sat)}$



$$V_{th} = 22 \cdot \frac{3.9k}{3.9+39} = 2V$$

$\Rightarrow I_B = ?$

$$-2 + I_B(3.54\cdot10^3) + 0.7 + (1.5\cdot10^3)(140 I_B) = 0$$

$$I_B = 6.52\,\mu A$$

$\Rightarrow I_C = ?$ $\rightarrow$ $I_C = \beta I_B \rightarrow 912.1\,\mu A$

$\Rightarrow V_{CE} = ?$ $\rightarrow$ $-22 + (912\,\mu)(10k) + V_{CE} + (1.5k)(912\,\mu) = 0$

$$V_{CE} = 11.512\,V$$

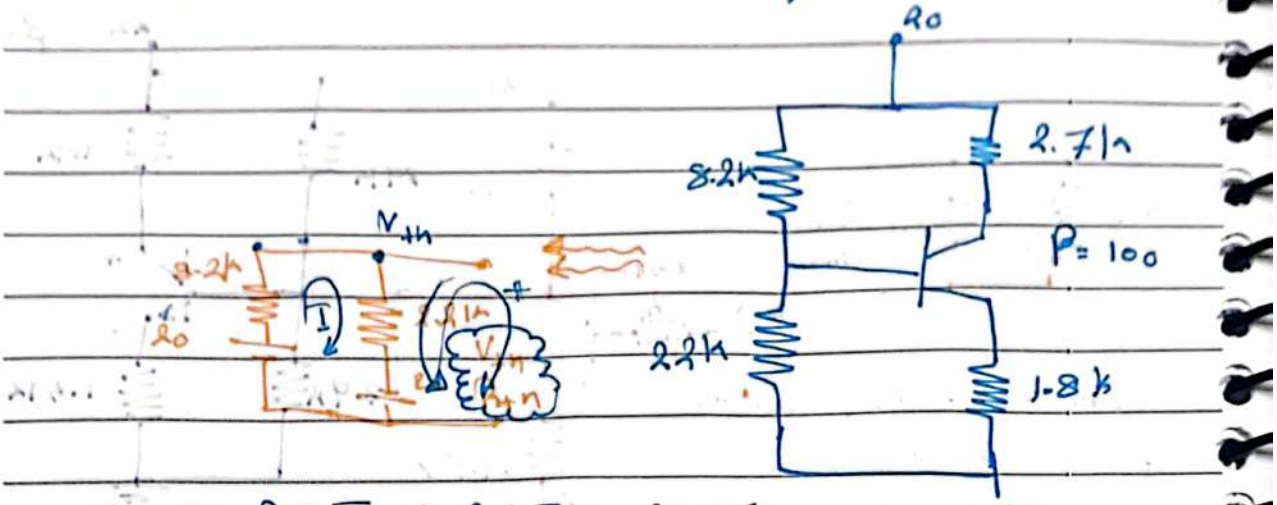
$\Rightarrow V_{sat} = ?$ $\Rightarrow$ $-22 + (10k)I_C + 0 + I_C(1.5k) = 0$

$$I_{C(sat)} = 1.913.1\,\mu A$$

تمت على النبي

Scanned with CamScanner بلغني تحويل

Ex: find I_C, V_C, V_B ?



$-20 + 8.2I + 2.2I - 20 = 0$ $\quad -2\sigma$

$I = 3.84\ mA$

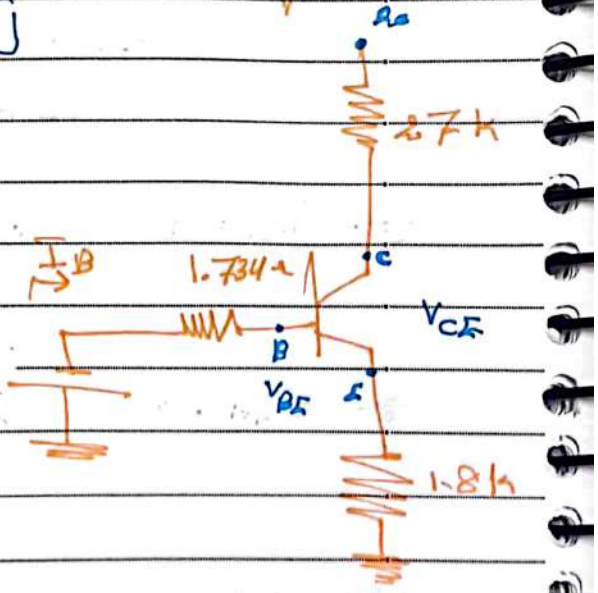
$2.8(3.84m) - 20 - V_{th} = 0$

$V_{th} = -11.57\ V$

$R_{th} = 1.734\ \Omega$

$I_C = \beta I_B$

$11.57 + I_B(1.734) + 0.7 + \beta(1.8k \cdot I_B) = 0$



$I_B = 71.6\ \mu A$

$I_C = (100)(71.6)$

$I_C = 7.16\ mA$

$-20 + 2.7(7.16m) + V_{CB} + 1.8(7.16m) = 0$

$V_{CE} = 7.09\ V$

$V_C = 20 - (2.7k \cdot 7.16 \cdot 10^{-3})$

$V_C = 0.668\ V$

$V_{CE} = V_C - V_E$

$7.09 = 0.668 - V_E \Rightarrow -6.42$

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تم المسح ضوئيا بـ CamScanner

V_{BE} = V_B - V_E

0.7 = V_B - 6.422

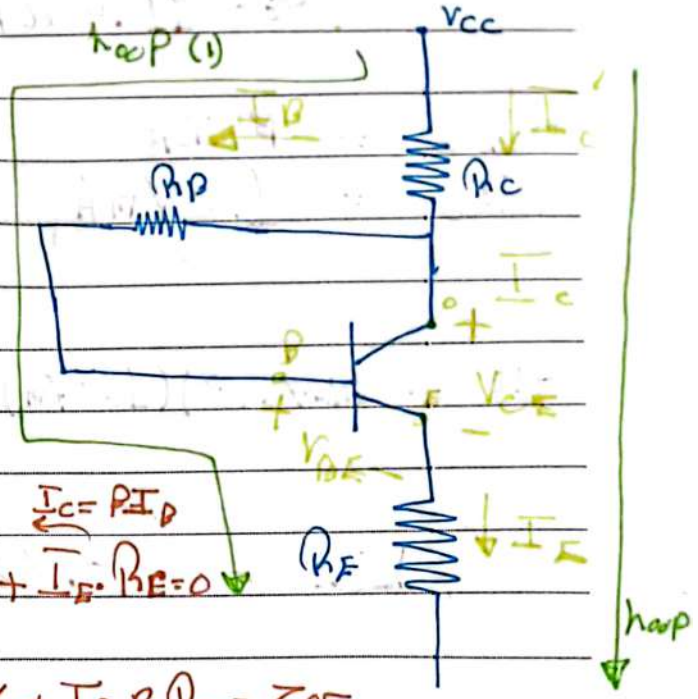
V_B = 7.122

نوكل على الآت

الحمد الآت

*Spectrom

4) collertor feed Back



$I_C' = I_C = I_E$

loop (1) :-

$-V_{CC} + I_B \cdot R_B + 0.7 + I_E \cdot R_E = 0$

$-V_{CC} + I_B R_B + 0.7 + I_B \beta \cdot R_E = 0$

I_B = ?

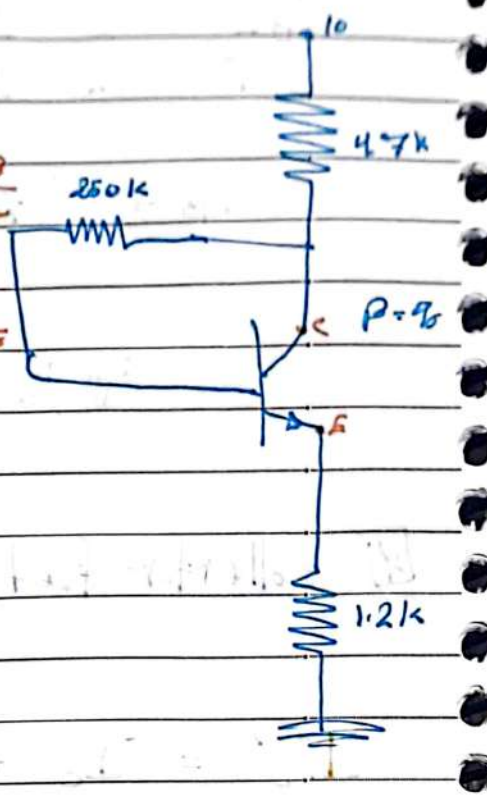
loop (2) :-

$-V_{CC} + I_C' R_C + V_{CE} + I_E R_E = 0$

V_{CE} = ?

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CamScanner بلغني تحويل

Ex/ find the I_C and V_{CE}



$I_B = ?$

$-10 + 4.7k\,I_B + 250k \cdot I_B + 0.7 + (1.2k)I_E = 0$

$-10 + 250 \cdot 10^3 I_B + 0.7 + I_B(91 \cdot 1.2k) = 0$

$I_B \Rightarrow 11.9\,\mu A$

$I_C = \beta I_B$

$I_C = 1.07\,mA$

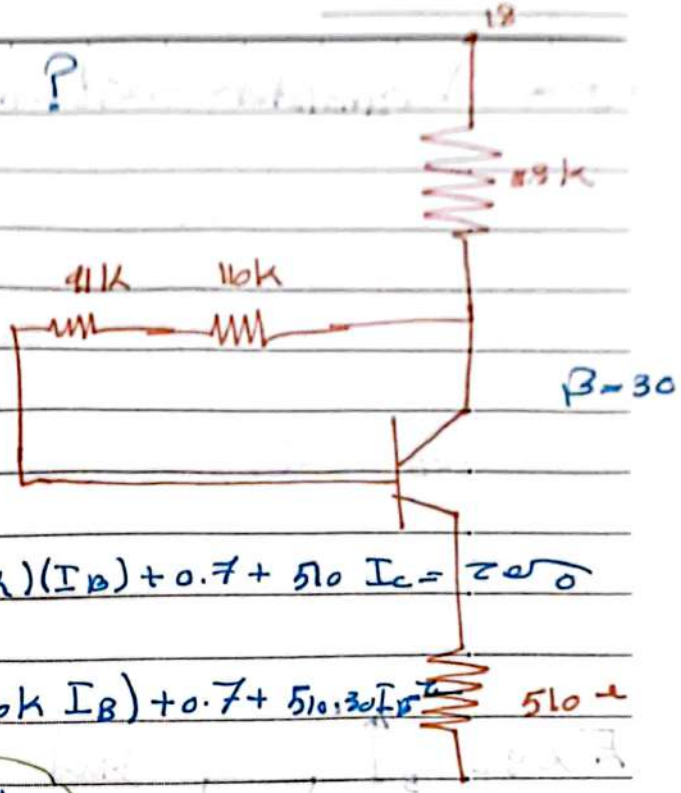
$-10 + (4.7 \cdot 10^3)(1.07m) + V_{CE} + (1.2k)(1.07m) = 0$

$V_{CE} = 3.687\ V$

"كمل ليحي"

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CamScanner بلغني تحويل

Ex: find I_B, I_C, V_{CE} ?



Sol:

$$-18 + (3.3k)(I_C) + (201k)(I_B) + 0.7 + 510\,I_C = 0$$

$$-18 + (3.3k\ 30.\ I_B) + (20k\ I_B) + 0.7 + 510.30I_B = 0$$

$$I_B = 128.8\ \mu A$$

$$I_C = \beta . I_B \Rightarrow I_C = 3.86\ mA$$

$$-18 + (3.86m \cdot 3.3k) + V_{CE} + (510)(3.86m) = 0$$

$$V_{CE} = 3.3\ V$$

"رب اعوذ بك من همزات الشياطين وأعوذ بك رب أن يحضرون"

✻ spectroma

Transistor switching network

on ⇒ saturation

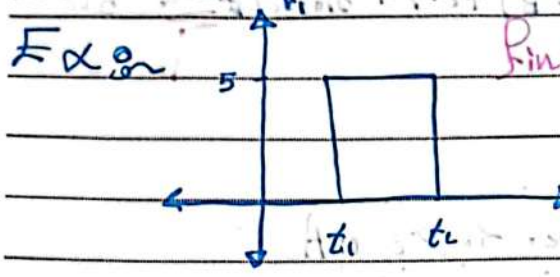
off ⇒ cutoff

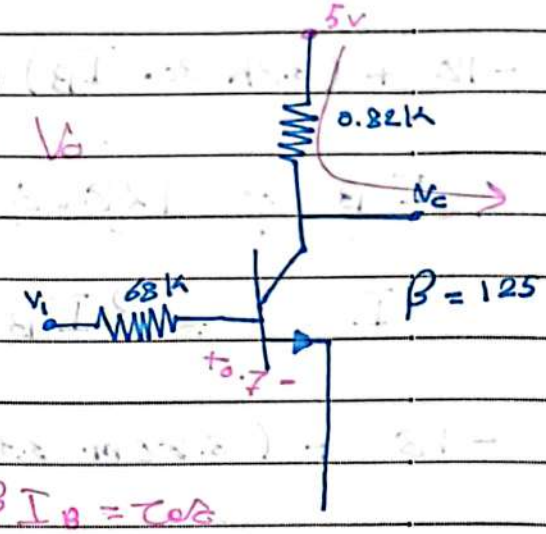
$V_{out} = V_c$

$I_c > I_{c(sat)}$

$I_B > I_{B(sat)}$

Ex/ Find the V_o





$q_1 \rightarrow t_1 \Rightarrow V_i = 5v$

$I_B = 5v$, , $I_C = \beta I_B = 5v$

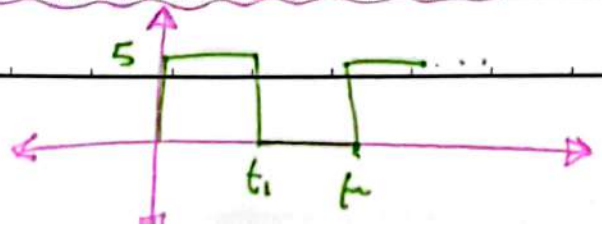
$-5 + I_c R_c + V_c = 5v$ $\quad -5 + (0.82 \cdot 8.6) + V_c = 0$

$V_c = \{V_{out} = 5v\}$ $\quad V_{out} = -2.052$

$t_1 \rightarrow t_2 \rightarrow V_i = 5v$

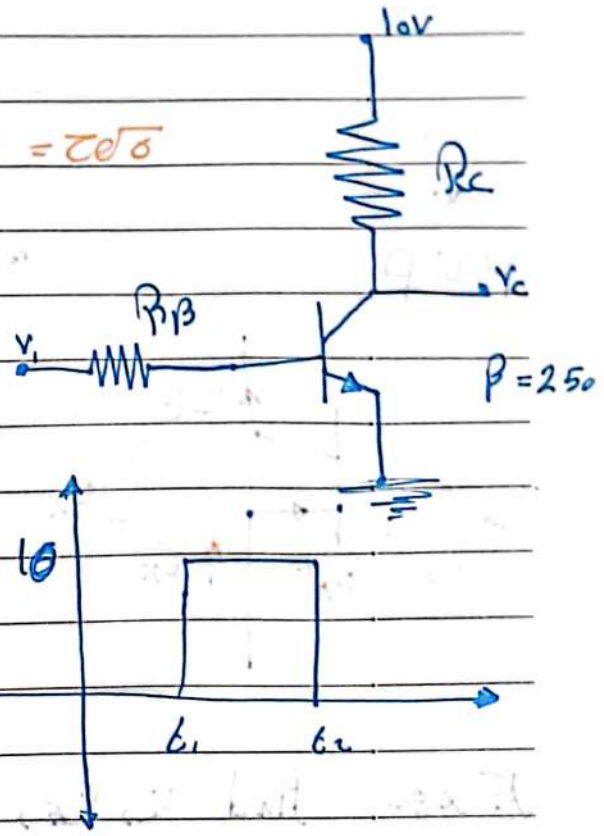
$-5 + 68k \cdot I_B + 0.7 = 0$

$\{I_B = 69.12\ \mu A\}$ $\quad \{I_C = 7.8\ mA\}$



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Find R_B and R_C, if $I_{C(sat)} = 10mA$



sch.

$-10 + R_C(10 \cdot 10^{-3}) + 0 = 0$

$R_C = 1\,k\Omega$

$I_{sat(c)} = \beta I_{B(sat)}$

$10mA = 250\, I_{B(sat)}$

$40\,\mu A = I_{B(sat)}$

$I_B = I_{B(sat)} + 20\mu$

$= 40\mu + 20\mu$

$I_B = 60\,\mu A$

$-V_i + I_B R_B + 0.7 = 0$

$R_B = \frac{10 - 0.7}{60\,\mu A}$

$R_B = 155\,k$

"R" يحمل معنى الحياة

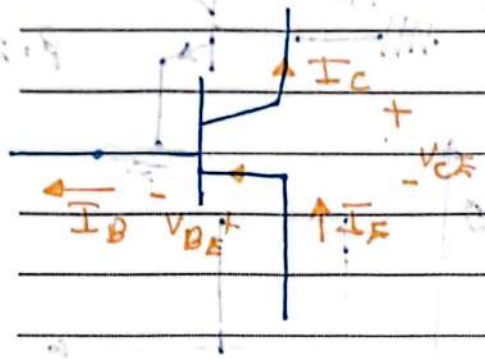
Scanned with CamScanner
CamScanner بـ ممسوحة ضوئيا

تنقسم الـ BJT الى نوعين

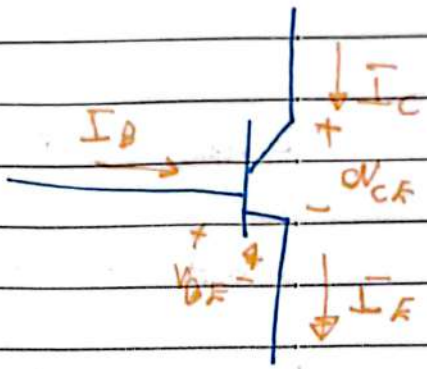
PNP

NPN

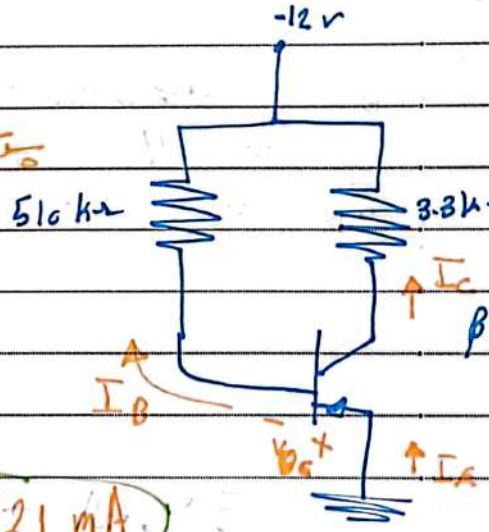
"لا تيأس الفرج قريب"



$I_E \cong I_C$



Example find V_C, V_{CE}, I_C



sol:

$12 - 510k \cdot I_B - 0.7 = 0$

$I_B = \frac{11.3}{510k}$

$I_B = 22.15\ \mu A$

$I_C = \beta I_B = 2.21\ mA$

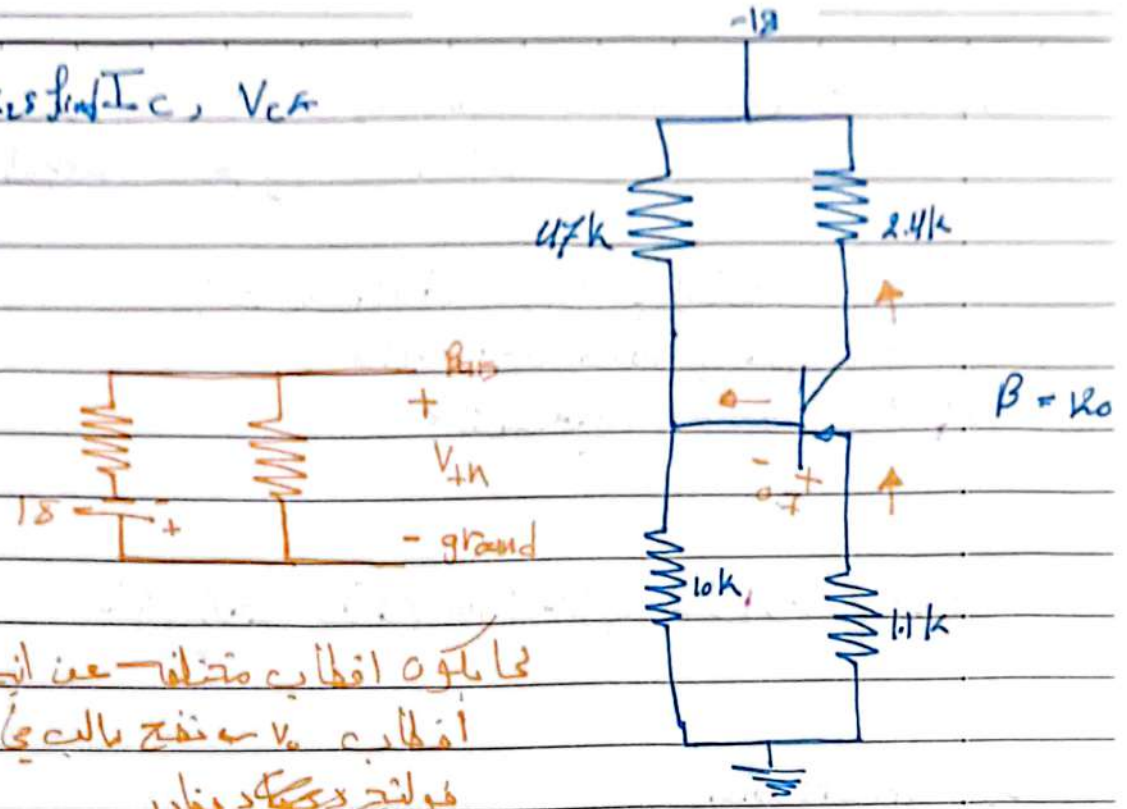
$12 - 3.3k \cdot 2.21m + V_{CE} = 0$

$V_{CE} = -4.707\ V$

$V_C = -4.707\ V$

Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

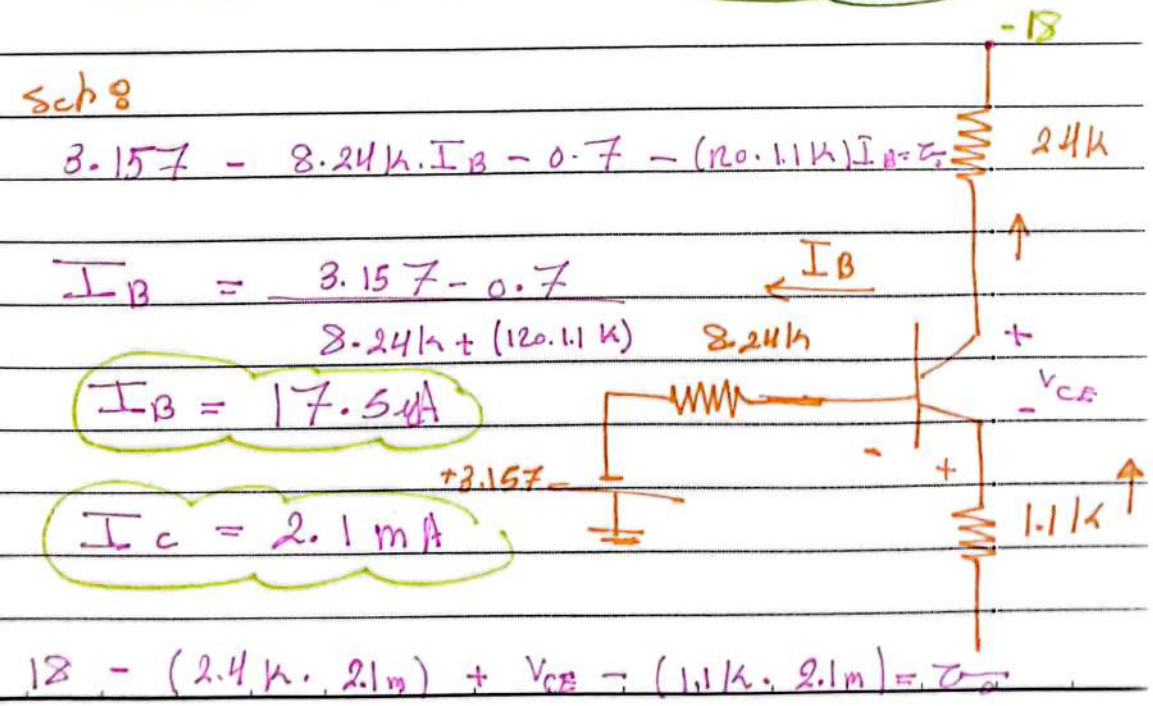
Example find I_C, V_{CE}



لما يكون اقطاب متختلف عن اتجاه
اقطاب V_o سنفتح بالبي
فولتج دايما يقادر

$$V_{th} = \frac{-18 \cdot 10k}{(10+47)k} = -3.157\ V$$

$$R_{th} \simeq 47 // 10\ k = 8.24\ k\Omega$$



sch 8

$$3.157 - 8.24k \cdot I_B - 0.7 - (120 \cdot 1.1k) I_B = 0$$

$$I_B = \frac{3.157 - 0.7}{8.24k + (120 \cdot 1.1k)}$$

$$I_B = 17.5\ \mu A$$

$$I_C = 2.1\ mA$$

$$18 - (2.4k \cdot 2.1m) + V_{CE} - (1.1k \cdot 2.1m) = 0$$

$$V_{CE} = 10.65$$

Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

الحمد لله رب العالمين هيلا بلوك خاص
لترانزستور BJT

الله يرزقني العلم النافع
اذا وجدت فائدة في هذا الملف
لا تنساني من صالح دعائك

هذا العمل متواضع مني خالص لوجه الله

* spectrum

Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

Field effect trasistor :

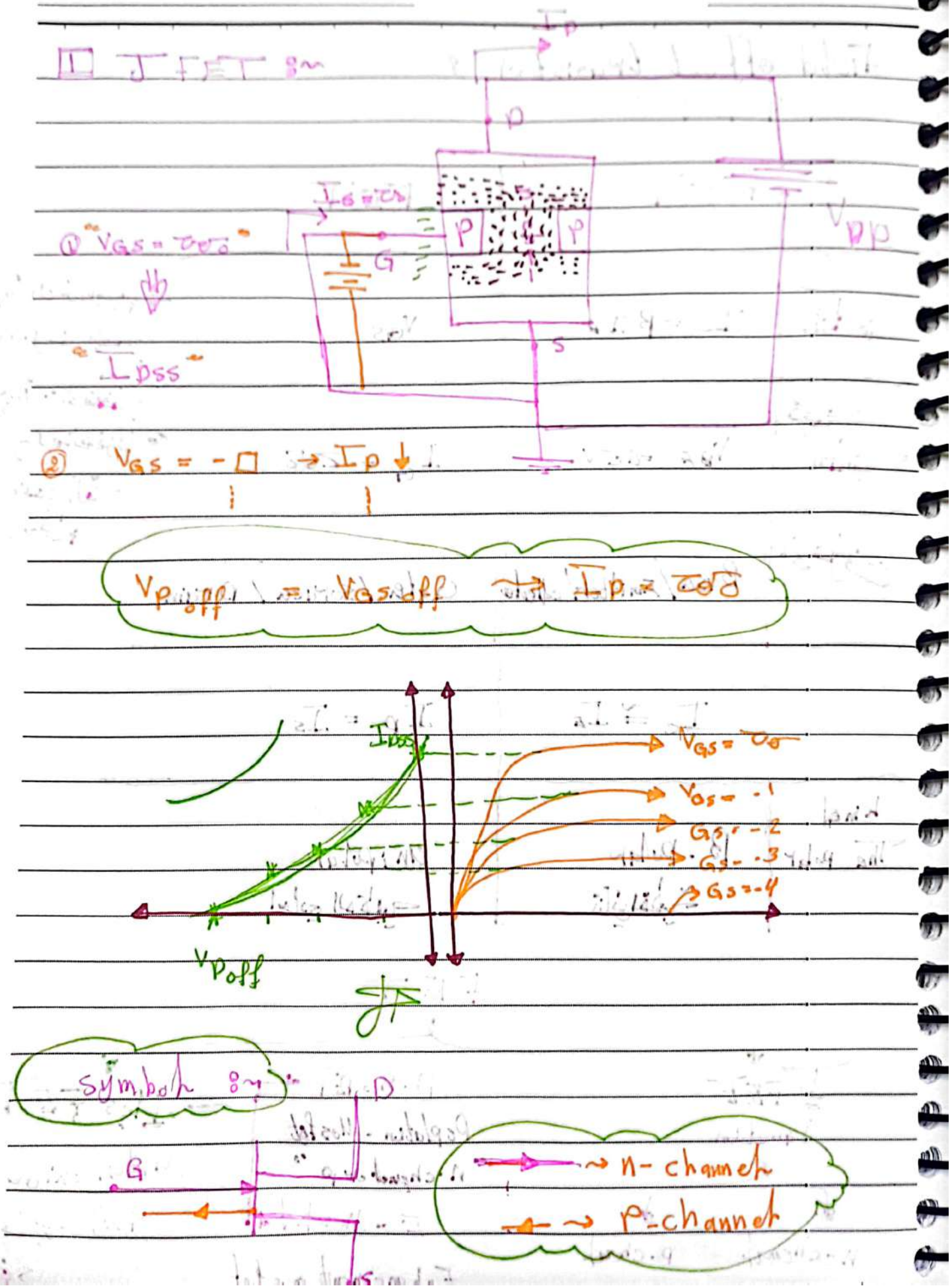
"FET"

	BJT	FET
التحكم	$I_C = \beta I_B$	V_{GS}
قيمة الدخل	$V_{BE} = 0.7V$	$I_G = 0$
الأطراف	Base / Emitter / collector	Gate / source / Drain
	$I_C \cong I_E$	$I_D = I_S$
kind The polar	B. polar ثنائي القطبية	uni polar أحادي القطبية

بالنسبة للمقاومة الداخلية: ذكرنا جعل والتيار الخارج يساوي صفر ما يمشون دون المقاومة الداخلية

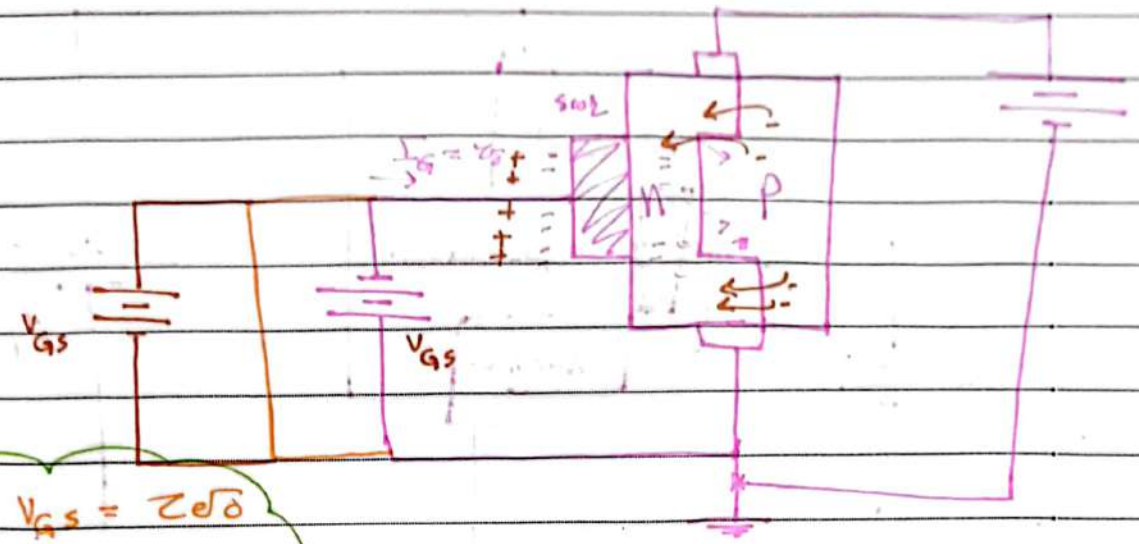
FET

- J FET (junction)
  - n-channel
  - p-channel
- Mos FET (Metal oxide semi FET ... Si SiO_2)
  - D-Mos FET (Depletion - Mosfet "n-channel or p")
  - E-Mos fet (Enhancement mosfet "n-p channel")



Scanned with CamScanner
CamScanner ب ممسوحة ضوئيا

2 Depletion - Mos Fet



① $V_{GS} = zero$

↳ I_{DSS} ✓

② $V_{GS} = -$ □

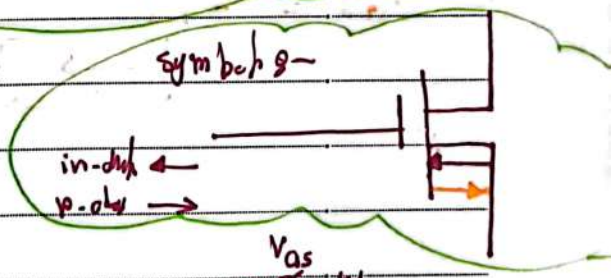
↳ $I_D = \downarrow$

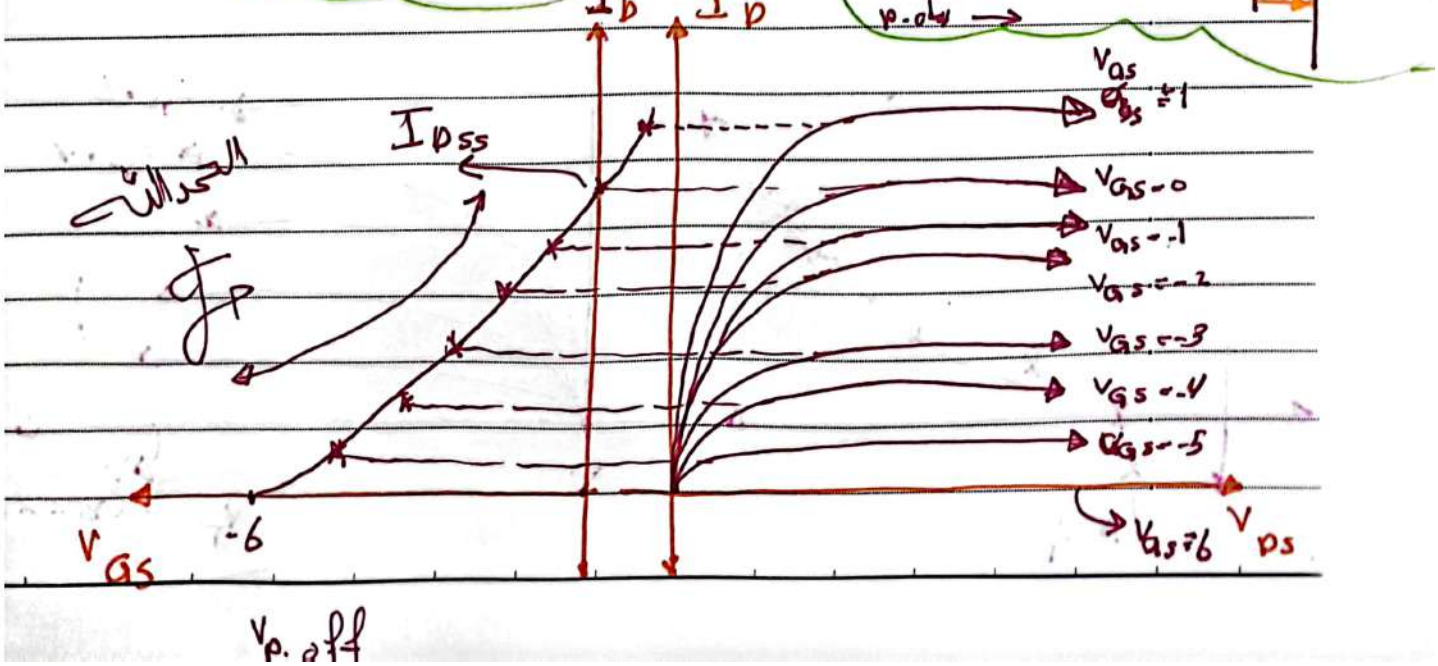
$V_{GS\,off} = V_{P\,off} \Rightarrow I_D = zero$

③ $V_{GS} = +$ □

↳ $I_D \uparrow$

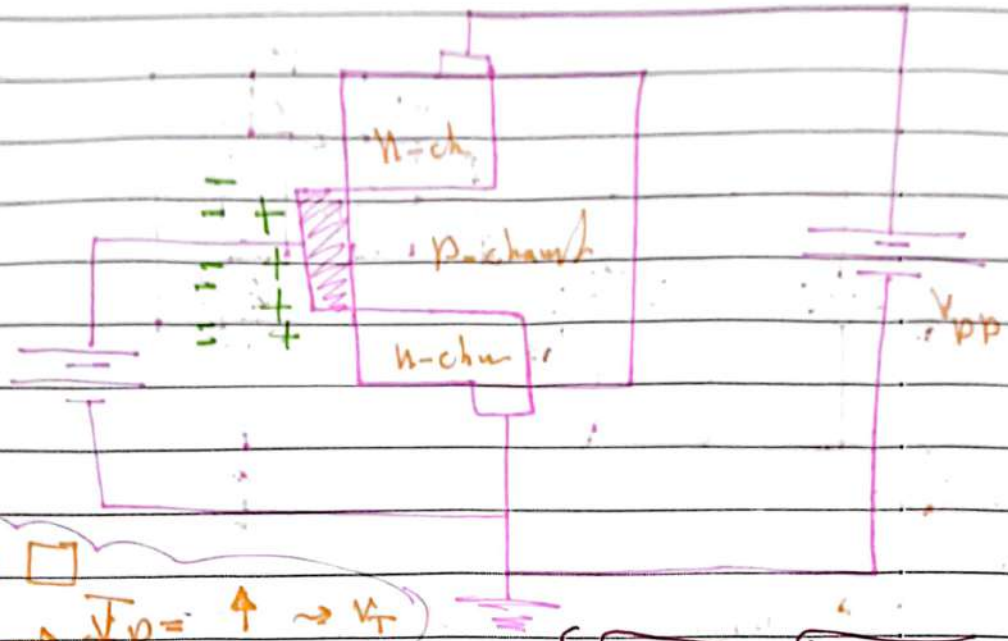
symbol





Scanned with CamScanner
تم المسح ضوئيا بـ CamScanner

3 Enhancement - MOSFET



① $V_{GS} = + \square$

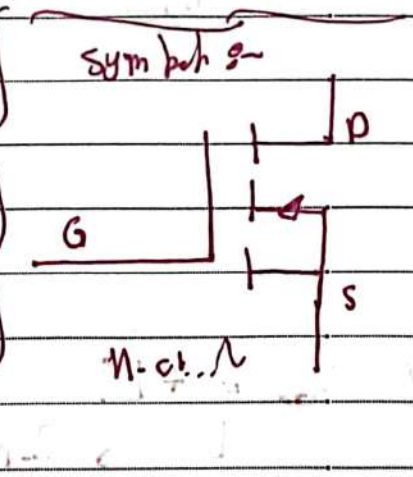
↳ $I_D = \uparrow \rightarrow V_T$

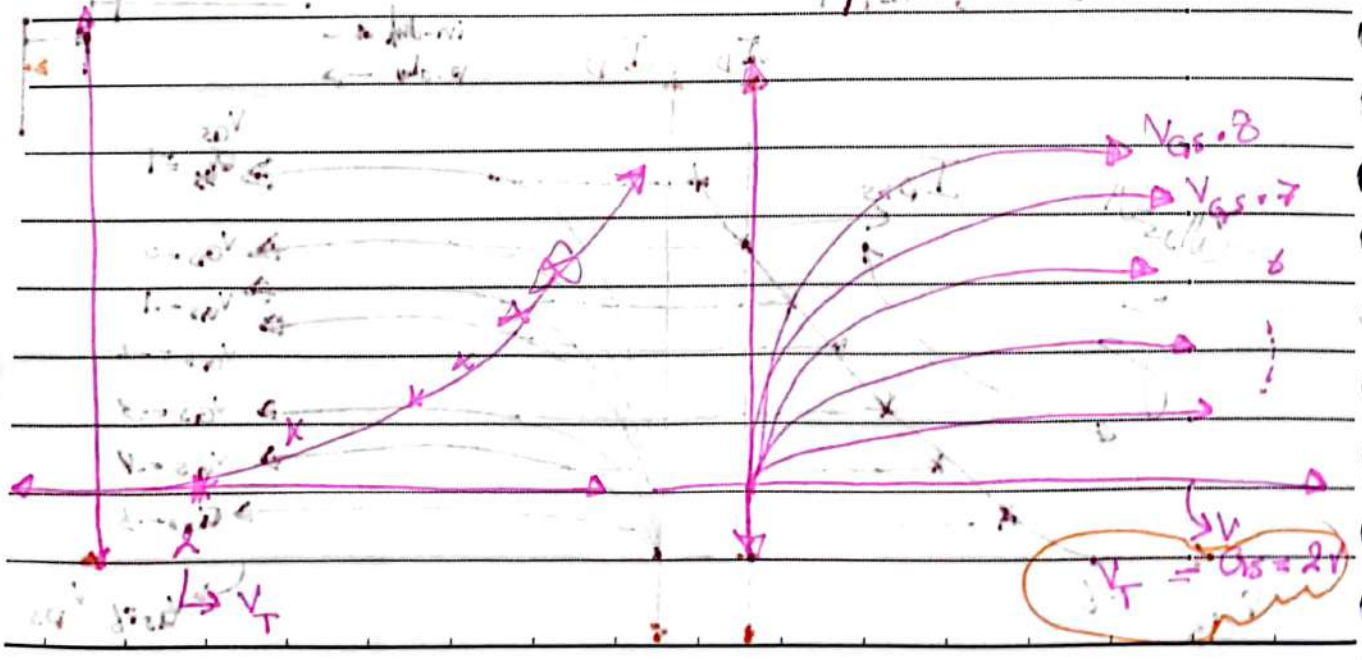
② $V_{GS} = \text{zero}$

↳ $I_D = \text{zero}$

③ $V_{GS} = - \square$

↳ $I_D = \text{zero}$



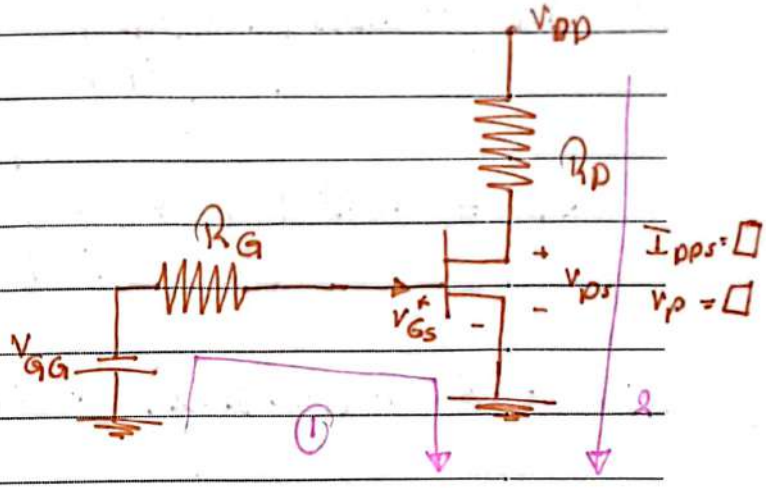


Scanned with CamScanner
CamScanner بلیکن تھوڑا

Shockley's equation

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Fixed Bias



1. $I_G = 0$

2. $V_{GS} = ??$

Loop (1)

$$V_{GG} + R_G I_G + V_{GS} = 0$$ ($R_G I_G = 0$)

$$V_{GS} = -V_{GG}$$

3. $$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

4. $V_{DS} = ??$

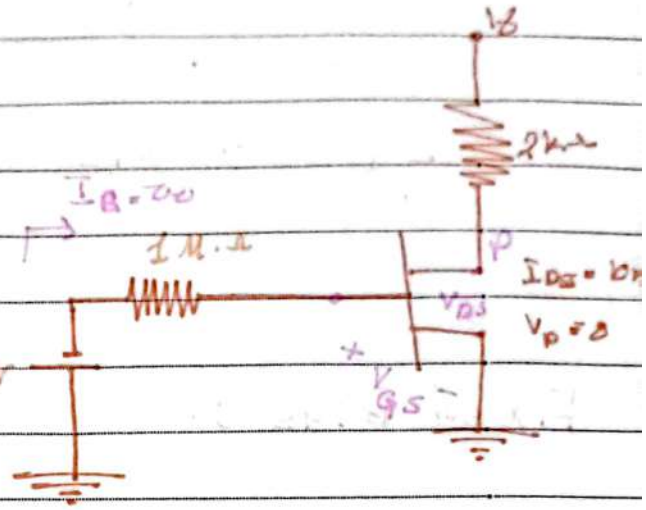
Loop (2)

$$-V_{DD} + R_D I_D + V_{DS} = 0$$

$$V_{DS} = V_{DD} - R_D I_D$$

Scanned with CamScanner

Ex :- find the V_{GS}, I_{DQ}, V_{DS}, V_G ... ?



Sol :-

[1] $2 + 20 + V_{GS} = 20$

$V_{GS} = -2\ V$

[2] $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$

$I_D = 10 \cdot 10^{-3}\left(1 + \frac{2}{-8}\right)^2 = 5.62\ mA$

[3] $-16 + (2 \cdot 5.62) + V_{DS} = 20$

$V_{DS} = 4.76\ V$

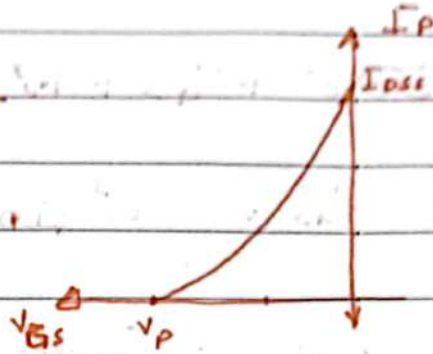
[4] $V_{GS} = V_G - V_S$ ($V_S = 20$)

$-2 = V_G$

* $I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$ بالرسم:

(A)

V_{GS}	I_D
zero	I_{DSS}
V_P	zero
$V_P/2$	$I_{DSS}/4$
$0.3\,V_P$	$I_{DSS}/2$

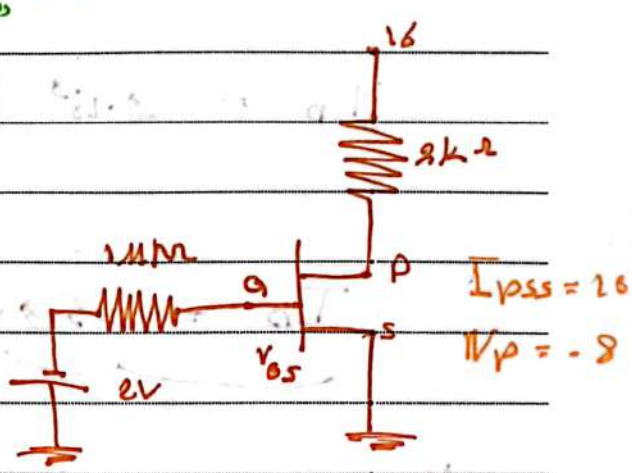


(B) Load line: أوجد معادلة الـ Load line (I) ثم رسمها

Ex 8-

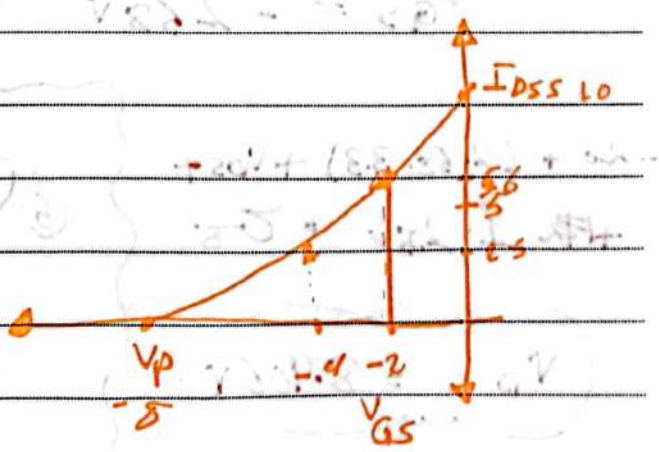
(1)

$V_P = -4$	2.5
$V_P = -2$	5.62



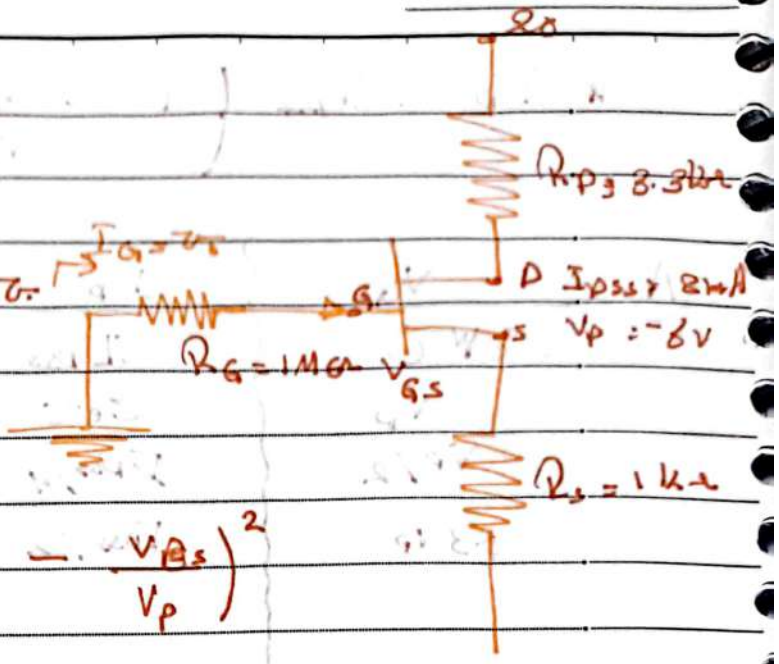
$V_{GSQ} = -2$

$I_{DSSQ} = 5.62$



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2] Self Bias



$$0 + V_{GS} + 10^3 I_D = 0$$

$$V_{GS} = -10^3 (I_D)$$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = .8m\left(1 + \frac{10^3 I_D}{-6}\right)^2$$

$$I_D = 8\cdot10^{-3}\left(1 - 2\cdot\frac{10^3 I_D}{-6} + \left(\frac{10^3 I_D}{-6}\right)^2\right)$$

$$I_D = 2.58\,mA$$

$$V_{GS} = -2.58\,V$$

$$-20 + (2.58)(3.3) + V_{DS} + (1.2)(5) = 0$$

$$V_{DS} = 8.906\,V$$

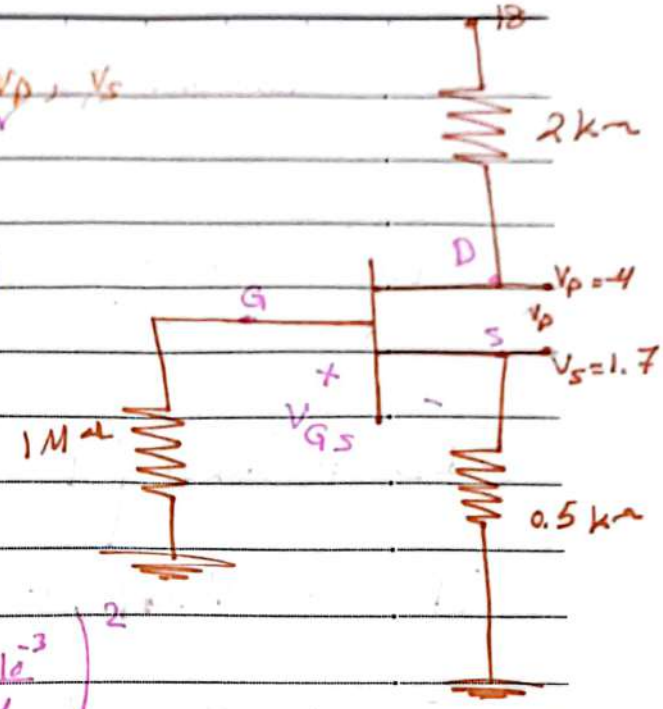
خطوات الحل على الحاسبة

① shift ⑨ ② 2 ③ = ④

نعيد المعادلة نفس ما هي ⑤ shift ⑥ calc ⑦ =

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Ex: find I_D, V_{GS}, I_{DSS}, V_p, V_S



⇒ sol:

KVL $+V_{GS} = -0.5 I_D$

$I_D = 3.4mA$

$V_{GS} = -1.7V$

$3.4 \cdot 10^{-3} = I_{DSS}\left(1 + \frac{1.7 \cdot 10^{-3}}{-4}\right)^2$

$V_S = I_D \cdot R_S$

$1.7 = I_D \cdot 0.5k$

$I_D = 3.4mA$

$I_{DSS} = 10.2mA$

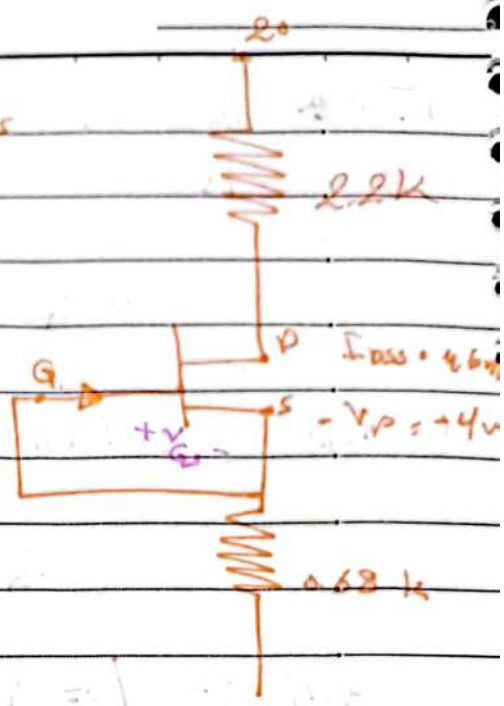
"الحمد لله"

$-18 + 2(3.4) + V_D = 0$

$V_D = 11.2V$

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تم المسح ضوئيا بـ CamScanner

Ex 8- find the V_{GS}, I_D, V_{DS}, V_D, V_S



$V_{GS} = 0V$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I = 4.5m(1-0)^2$$

$I_D = 4.5\,mA$

$-20 + (2.2 \times 4.5) + V_{DS} + (0.68 \cdot 4.5) = 0$

$V_{DS} = 7.04V$

$-20 + (2.2 \cdot 4.5) + V_D = 0 \rightarrow V_D = 10.1V$

$V_{DS} = V_D - V_S$

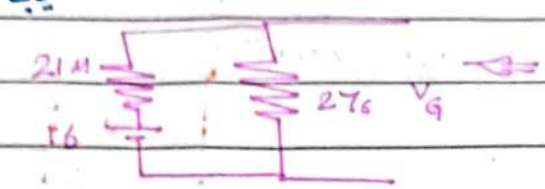
$7.04 = 10.1 - V_S \rightarrow V_S = 3.06V$

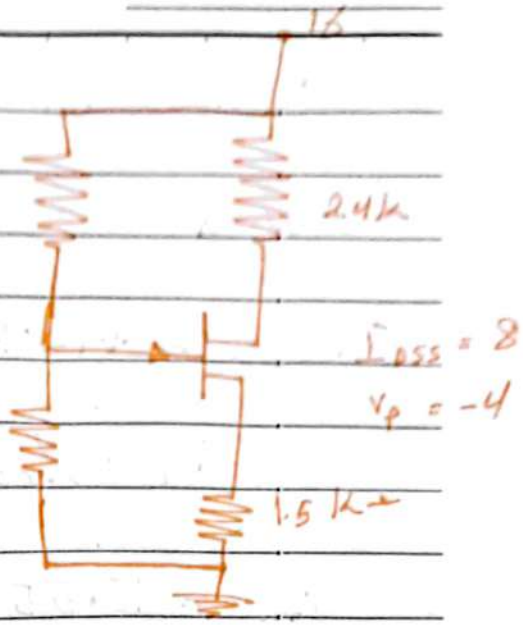
الحمد لله رب العالمين

Scanned with CamScanner

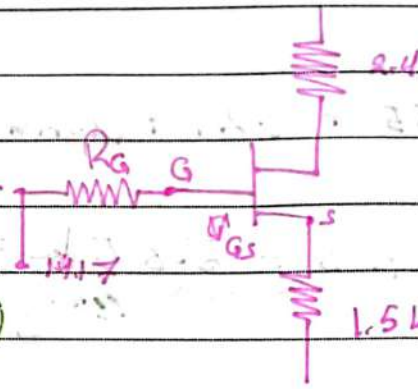
Ex: Find $I_D, V_{GS}, V_D, V_S, V_{DS}$

هون في خطأ عند ايجاد V_{TH}





$$V_{TH} = V_G = 14.17\,V$$



$$-14.17 + 0 + V_{GS} + 1.5k I_D = 0$$

$$V_{GS} = 14.17 - 1.5k I_D$$

$$I_D = 8\cdot10^{-3}\left(1 + \frac{14.17 - 1.5\,I_D}{4}\right)^2$$

$I_D = 9.24\,mA$

$V_{GS} = 0.31\,V$

$$-16 + (2.4)(9.24) + V_{DS} + (1.5)(9.24) = 0$$

$V_{DS} = -20.036\,V$

$V_S = (1.5)(9.24)$

$V_S = 13.86\,V$

$V_{DS} = V_D - V_S$

$-20.036 = V_D - 13.86$

$V_D = -6.176\,V$

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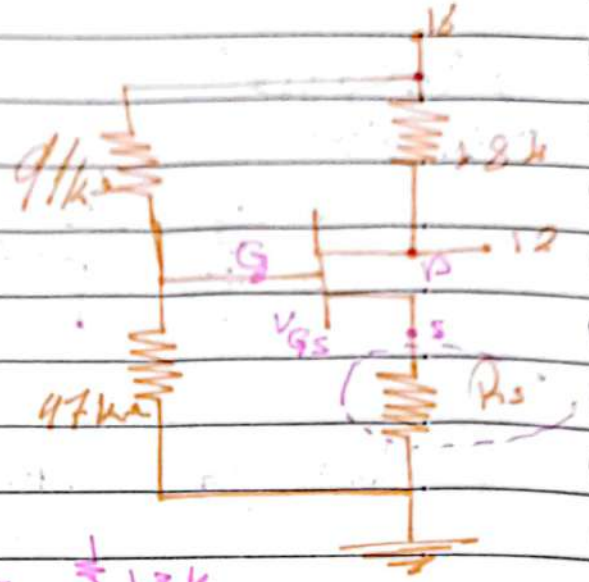
Ex: find the R_s if $V_D = 12V$, $V_{GS} = -2$?

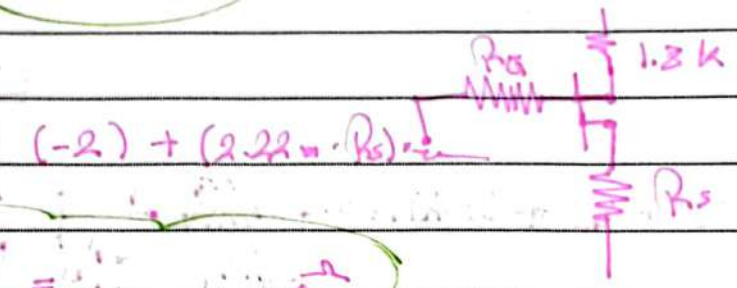
Sol:

$-16 + 12 + 1.8k\, I_D = 0$

$I_D = 2.22\,mA$

$V_{th} = \quad V$



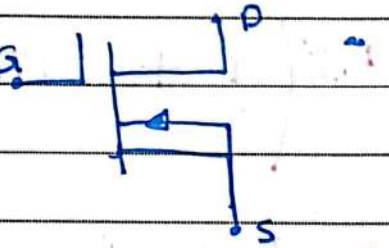


$- \quad + (-2) + (2.22m \cdot R_s) =$

$R_s =$

FET هيك تحلينا

D-MOSFET "



"

"سوديا"

$(0, I_{DSS})$

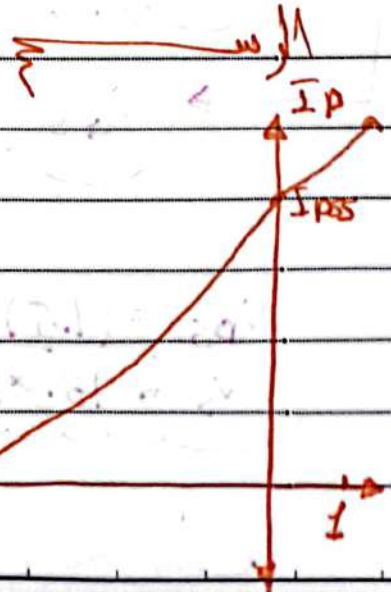
$(V_P, 0)$

$(\frac{1}{2}V_P, \frac{I_{DSS}}{4})$

$(0.3V_P, \frac{I_{DSS}}{2})$

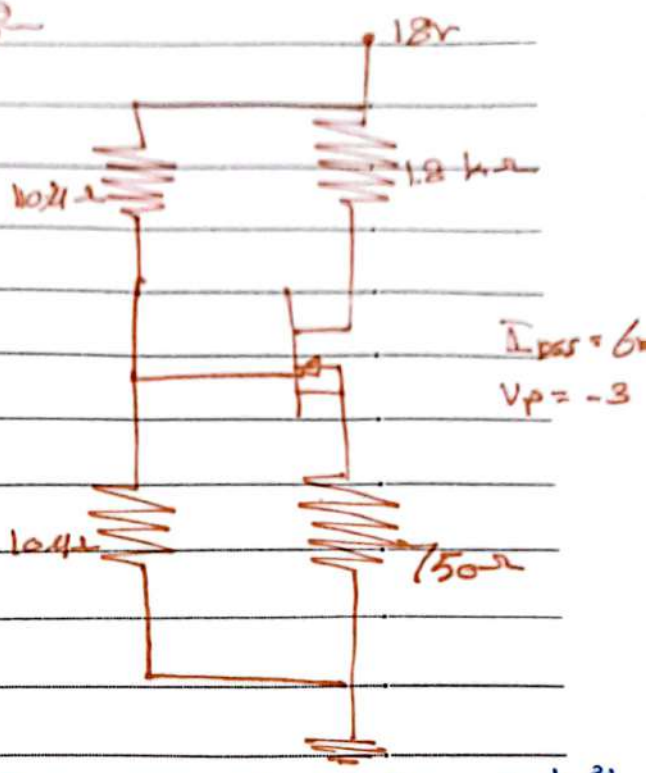
$V_{GS} = 1$

$(1, I_D = ?)$



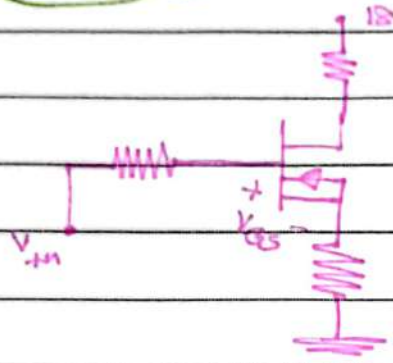
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CamScanner بلغني تعرف

Ex/ Find I_D, V_{GS} ?



$$V_{th} = \frac{18 \cdot 10 \cdot 10^{-6}}{110\mu + 10\mu}$$

$$V_{th} = 1.5\,V$$



$$-1.5 + 750\,I_D + V_{GS} + 750\,I_D = 0$$

$$V_{GS} = 1.5 - 750\,I_D$$

$$I_D = 6 \cdot 10^{-3}\left(1 + \frac{(1.5 - 750\,I_D)}{3}\right)^2$$

$$I_D = 8.12\,mA$$

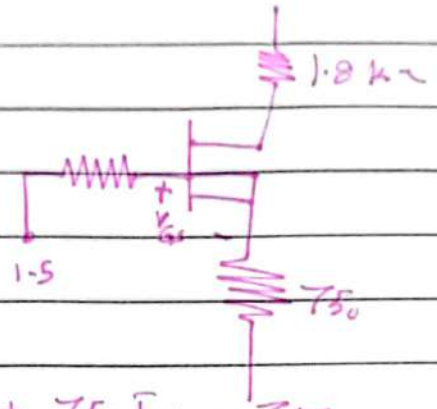
$$V_{GS} = -0.84$$

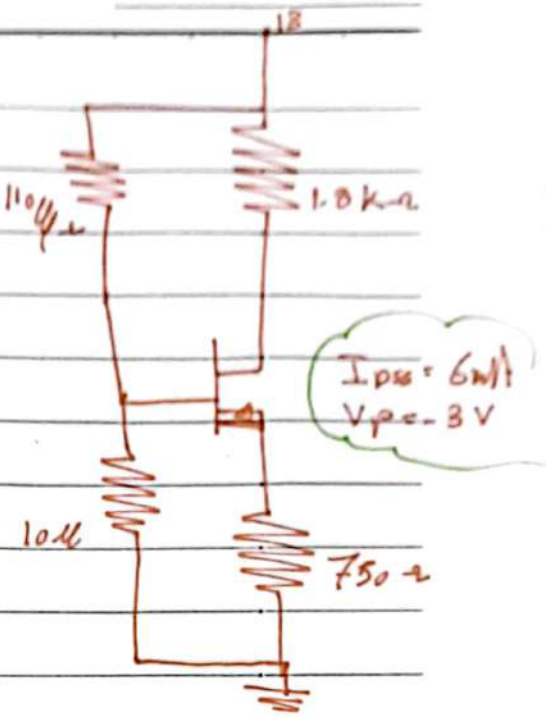
"ارفع رأسك انت سوري حر"
"عاشت سورية حرية أبية"

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Ex 8 find I_D, V_{GS}

$V_{th} = 1.5\,V$





$-1.5 + V_{GS} + 750 I_D = 0$

$$V_{GS} = 1.5 - 750 I_D$$

قانون

$$I_D = 6m \left(1 + \frac{(1.5 - 750 I_D)}{3}\right)^2$$

حل رياضي

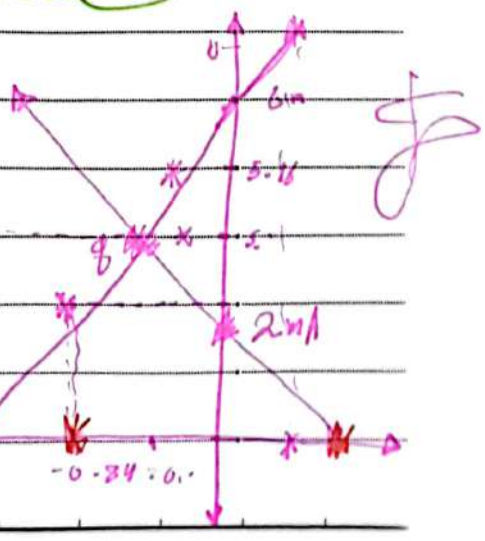
$$I_D = 3.12\,mA$$

$$V_{GS} = -0.84\,V$$

① $(0, 6mA)$

② $(-3, 0)$

③ $(-0.42, 5.16m)$



بما أن $V_{GS} = 1.5 - 750 I_D$

$\rightarrow V_{GS} = 0 \rightarrow I_D = 2m$

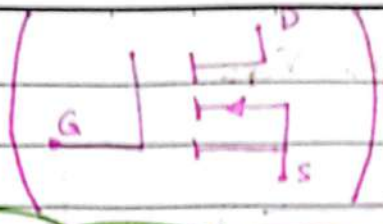
$\rightarrow I_D = 0 \rightarrow V_{GS} = 1.5$

$(1.5, 0)$

$(0, 2m)$

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تم المسح ضوئيا بـ CamScanner

E-MOSFET

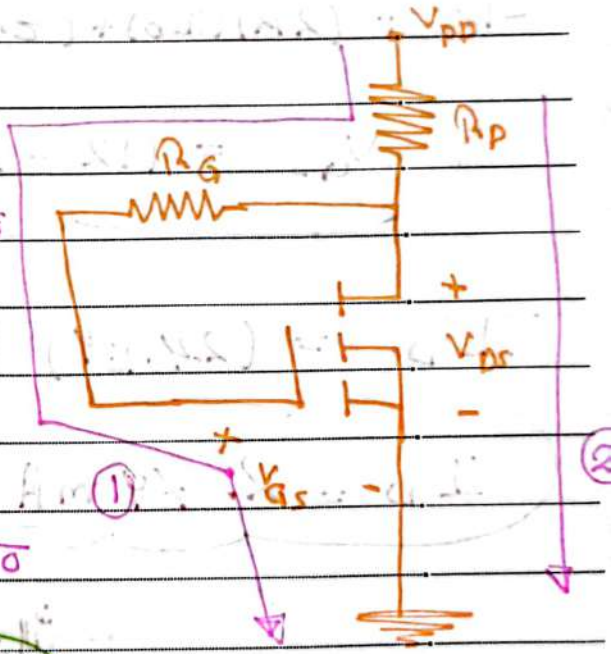


$$I_D = k\left(V_{GS} - V_{GS(TH)}\right)^2$$

أهم القوانين لـ E-Mos

$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{GS(TH)}\right)^2}$$

Feed Back :-



Loop (1)

[1] $-V_{DD} + R_D I_D + R_G(I_G) + V_{GS} = zero$

$$V_{GS} = V_{DD} - I_D R_D$$

[2] Loop

$$-V_{DD} + R_D I_D + V_{DS} = zero$$

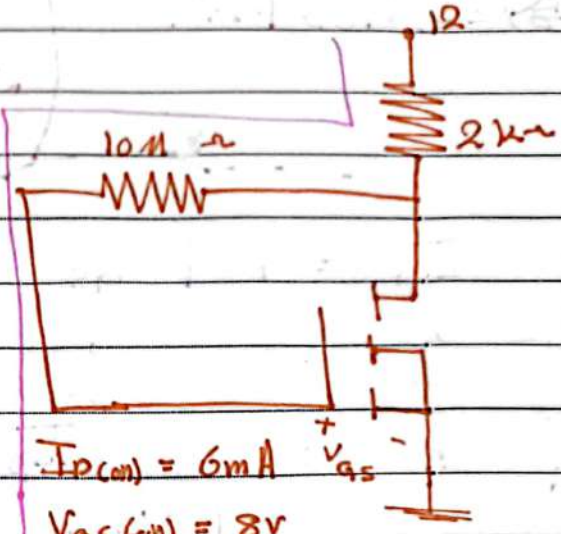
$$V_{DS} = V_{DD} - I_D R_D$$

* spectrum

* بسواعدنا نبنيها

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Ex: find I_D, V_{GS}



$I_{D(on)} = 6mA$

$V_{GS(on)} = 8V$

$V_{GS(th)} = 3V$

① $$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(th)})^2}$$

$$k = \frac{6 \cdot 10^{-3}}{(8-3)^2}$$

$$k = 2.4 \cdot 10^{-4} \ A/V^2$$

$$-12 + (2k)(I_D) + (\text{zero}) + V_{GS} = \text{zero}$$

$$V_{GS} = (12 - 2kI_D)$$

$$I_D = (2.4 \cdot 10^{-4})\left((12 - 2kI_D) - 3\right)^2$$

$$I_D = 2.79 \ mA$$

$$V_{GS} = 6.42V$$

الحمد لله

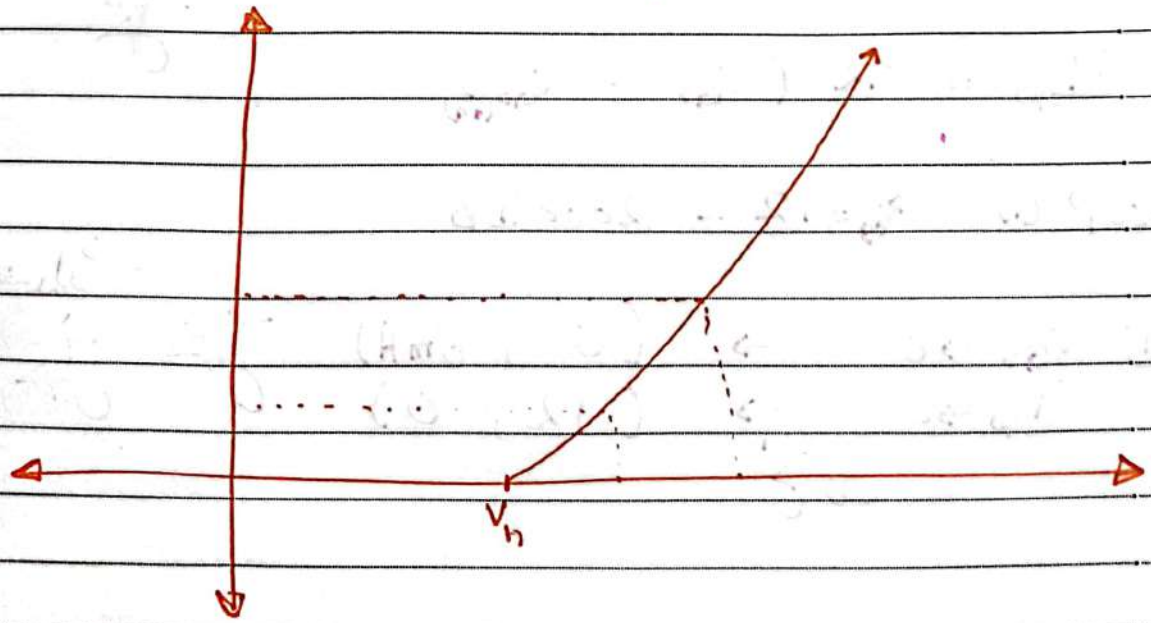
"جنة جنة جنة ولدنا"

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تم المسح ضوئيا بـ CamScanner

$I_D = k(V_{GS} - V_{GS(th)})$ بالرسم

[1] V_{GS}	I_D	
V_{th} ..	صفر	$(V_{th}, 0)$
$V_{GS(on)}$	$I_{D(on)}$	$(V_{GS(on)}, I_{D(on)})$
$V_{th} < V_{GS} < V_{GS(on)}$	$I_D = \square$	(,)
$V_{GS} > V_{GS(on)}$	$I_D = \square$	(,)

[2] .. loop ① .. معادلة الدخل



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تم المسح ضوئيا بـ CamScanner

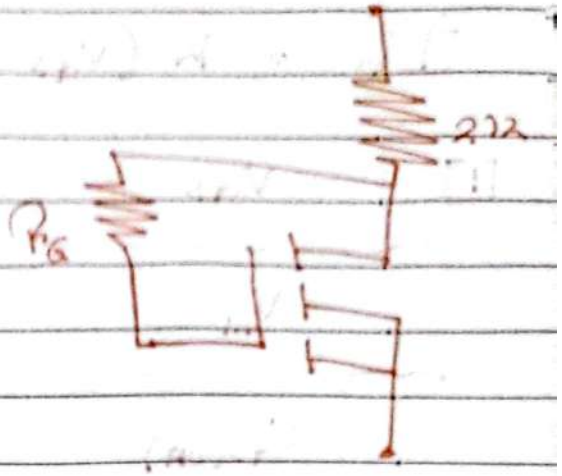
Ex. 8~

$k = 0.24\ mA/V^2$

$I_{D(on)} = 6mA$

$V_{GS(on)} = 8V$

$V_{GS(th)} = 3V$



① (3 , 0)

② (8 , 6) ③ (6 , 2.16m)

④ (9 , 8.64m)

$I_D = k\,(V_{GS} - V_{GS(th)})^2$

loop (1) $\rightarrow V_{GS} = 12 - 2000\,I_D$

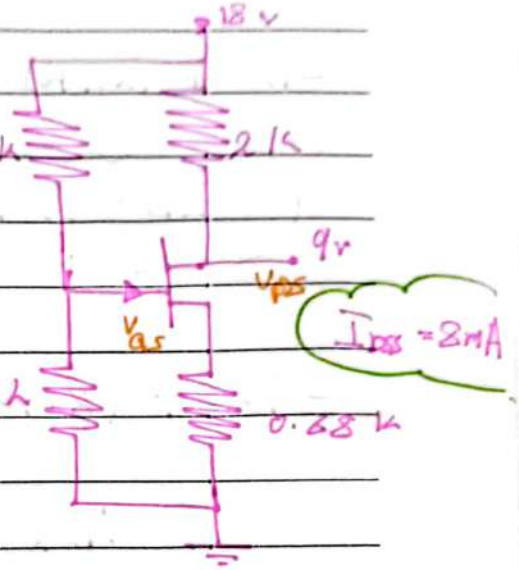
① $V_{GS} \rightarrow 0 \rightarrow (0 , 6mA)$

$I_D \rightarrow 0 \rightarrow (12 , 0)$

②

"Problem" مثال

find I_D, V_S, V_{DS}, V_G, V_{GS}, V_P



$$V_{TH} = \frac{18 \cdot 91}{91 + 750} = 1.94V$$

$$-18 + 2k I_D + 9 = 0$$

$$I_D = \frac{9}{2k} \rightarrow 4.5\ mA$$

$$V_G = V_{TH} = 1.94V$$

$$V_S = 0.88k \cdot 4.5m$$

$$V_S = 3.96V$$

$$-1.94 + V_{GS} + 3.96 = 0$$

$$V_{GS} = -1.12V$$

$$4.5m = 8m \left(1 + \frac{1.12}{V_P}\right)^2$$

شرط: $|V_P| > |V_{GS}|$

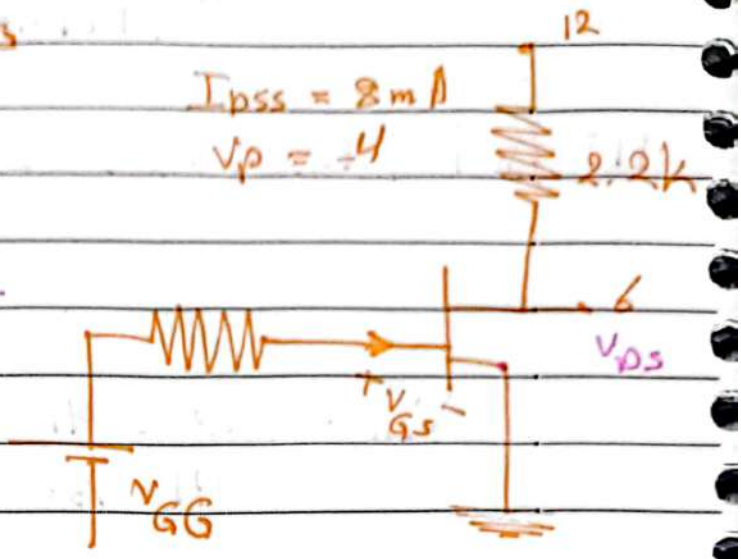
$V_P = \pm 0.64\ V$ ✗

$$V_P = \frac{V_{GS}}{1 \pm \sqrt{\frac{I_D}{I_{DSS}}}}$$

$= -4.452V$ ✓

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تم المسح ضوئيا بـ CamScanner

[2] find I_D, V_{DS}, V_{GG}



Sol 8~

$-12 + (2.2k)\,I_D + 6 = 0$

$I_D = 2.72\,mA$

$-12 + (2.72)(2.2) + V_{DS} = 0$

$V_{DS} = 6.016\,V$

$-V_{GG} + Zero + V_{GS} = Zero$

$V_{GS} = V_{GG}$ → $V_{GG} = -1.66\,V$

$2.72m = 8m\left(1 + \frac{V_{GS}}{4}\right)^2$

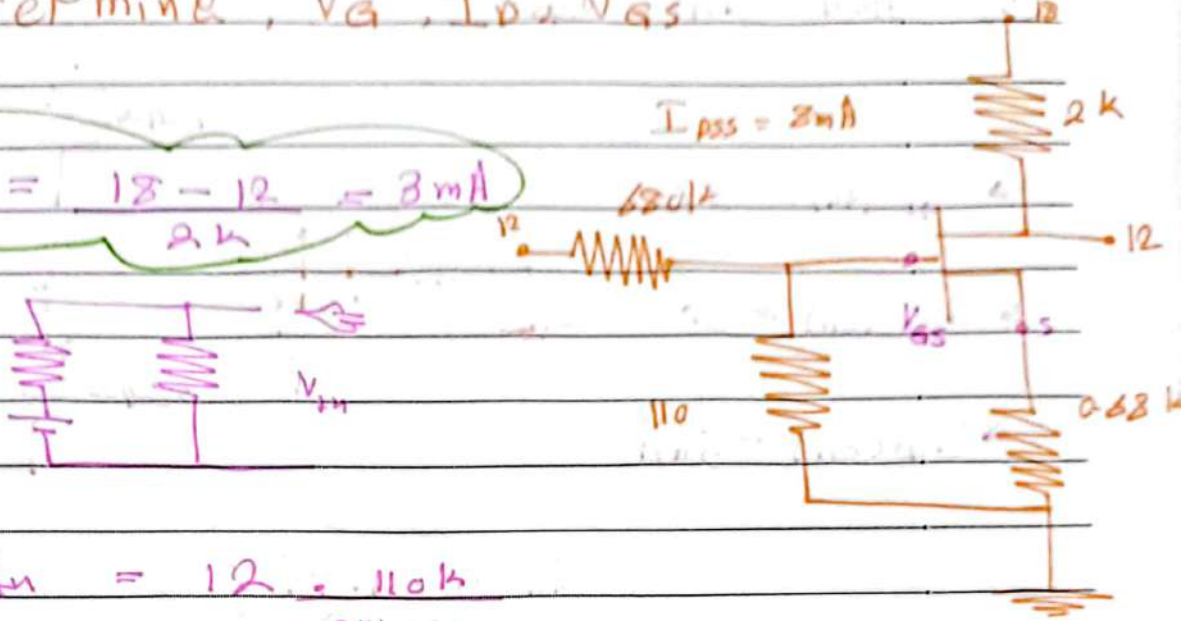
$V_{GS} = -1.66\,v$

"الى الله نسعى" . "الى الله نصير"

Scanned with CamScanner بلغني تحويلها

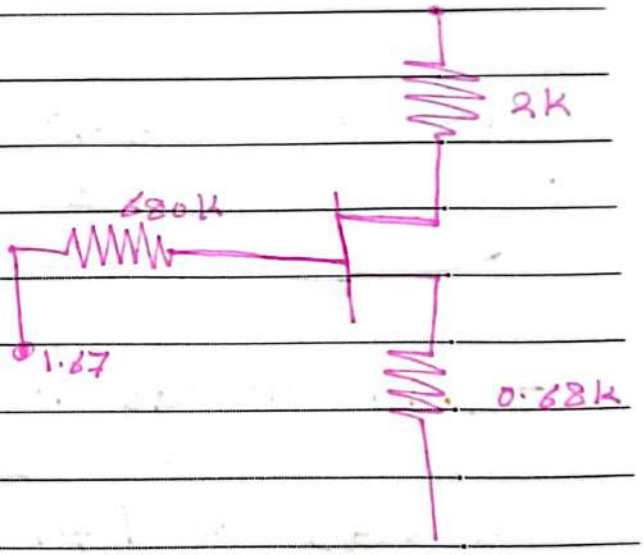
[3] Determine V_G, I_D, V_{GS}.

$$I_D = \frac{18 - 12}{2k} = 3mA$$



$$V_{th} = 12 \cdot \frac{110k}{680k + 110k}$$

$$V_{th} = 1.67V$$



$$V_{th} = 1.67$$

$$-1.67 + V_G = 0$$

$$V_G = 1.67V$$

$$-18 + (2k + 0.68k)\,I_D + V_{DS} = 0$$ (I_D: 3m)

$$V_{DS} = 9.96V$$

$$V_{DS} = V_D - V_S$$

$$9.96 = 12 - V_S$$

$$V_S = 2.04$$

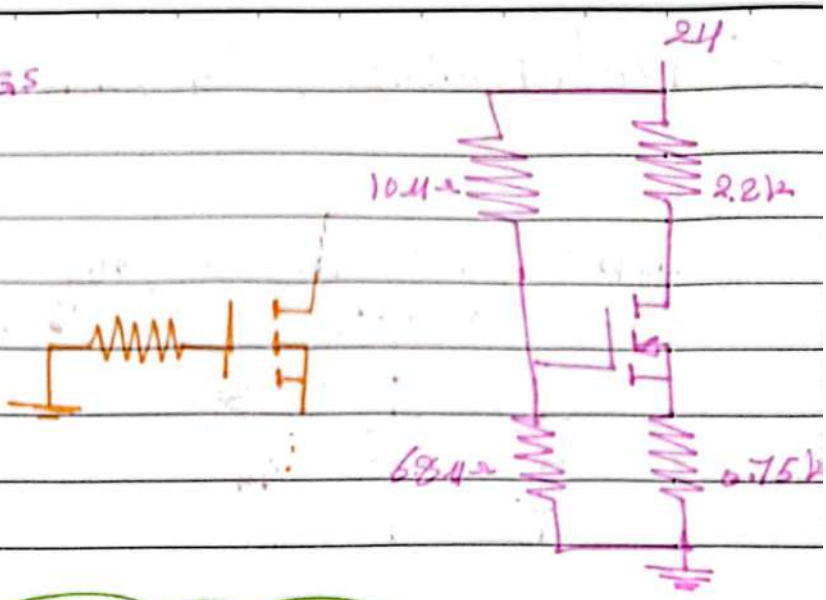
$$V_{GS} = V_G - V_S$$

$$= 1.67 - 2.04$$

$$V_{GS} = -0.37V$$

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[4] find I_D, V_{GS}



→ $V_{GS(th)} = 3V$

→ $V_{GS(on)} = 6V$

→ $I_{D(on)} = 5mA$

$$V_{th} = 9.71\ V$$

$$k = \frac{5 \cdot 10^{-3}}{6-3} = 5.557 \times 10^{-4}\ A/V^2$$

$$-9.71 + V_{GS} + 0.75k I_D = 0$$

$$V_{GS} = 9.71 - 0.75 I_D$$

$$I_D = k (V_{GS(on)} - V_{th})^2$$

$$I_D = (5.557 \cdot 10^{-3})(9.71 - 0.75 \cdot 10^3 I_D - 3)$$

$$I_D = 4.967\ mA$$

$$V_{GS} = 5.984\ V$$

Scanned with CamScanner

انتهى الشهر الحمد لله

هيك بنكون ان شاء الله خلصنا

مادة الالكترونيات - كاملة

بإذن الله

اللهم بارك لنا في علمنا

واستعملنا ولا تستبدلنا

هذا العمل صدقة جارية على

روح "والدي نحميه" الله يرحمه

ويغفر له

9/7/2016 ؟؟

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